Surveying Wi-Fi HaLow

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wirelessLAN PROFESSIONALS

Instructor

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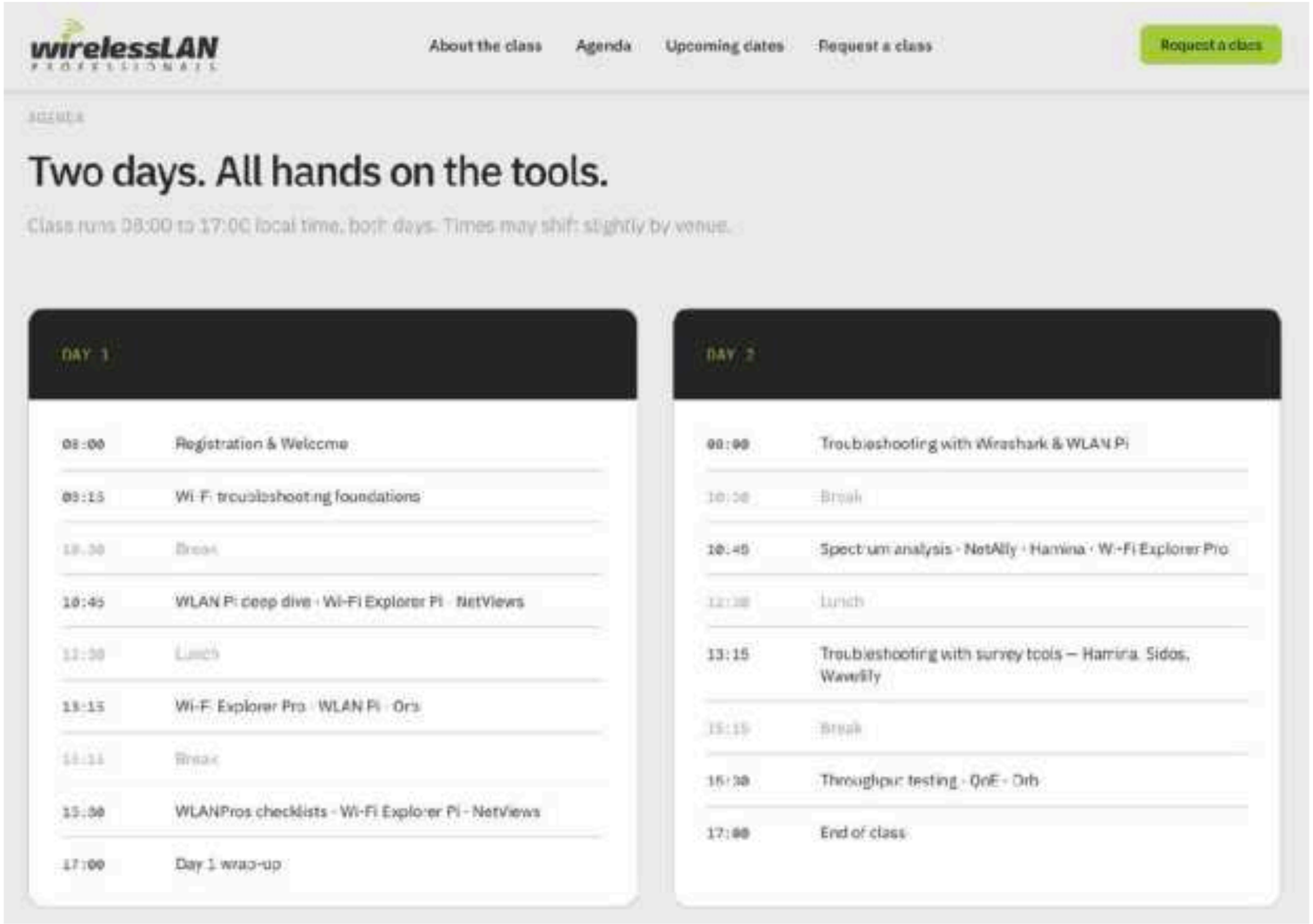


Director

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#WLPC Prague 2026

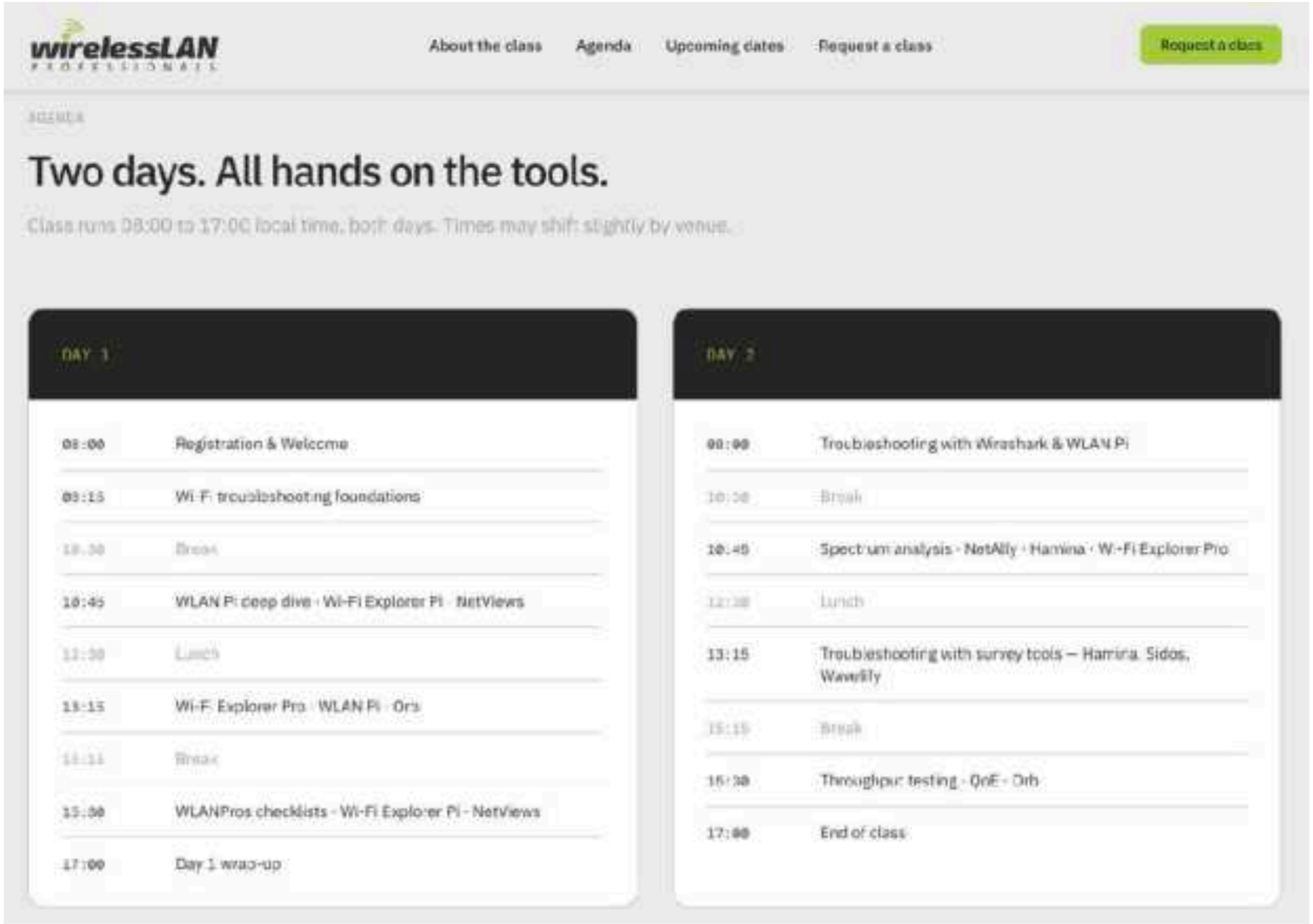


https://www.wifitroubleshooting.com/



https://www.wifitroubleshooting.com/

Belgium [Oct 19-20]



https://www.wifitroubleshooting.com/


ORIGINALS
MFG
MAISON
KITSUNÉ
PARIS, FRANCE
BASKETBALL



100% Free WLANPros Wi-Fi Design

Hamina

sidos

Wavelify

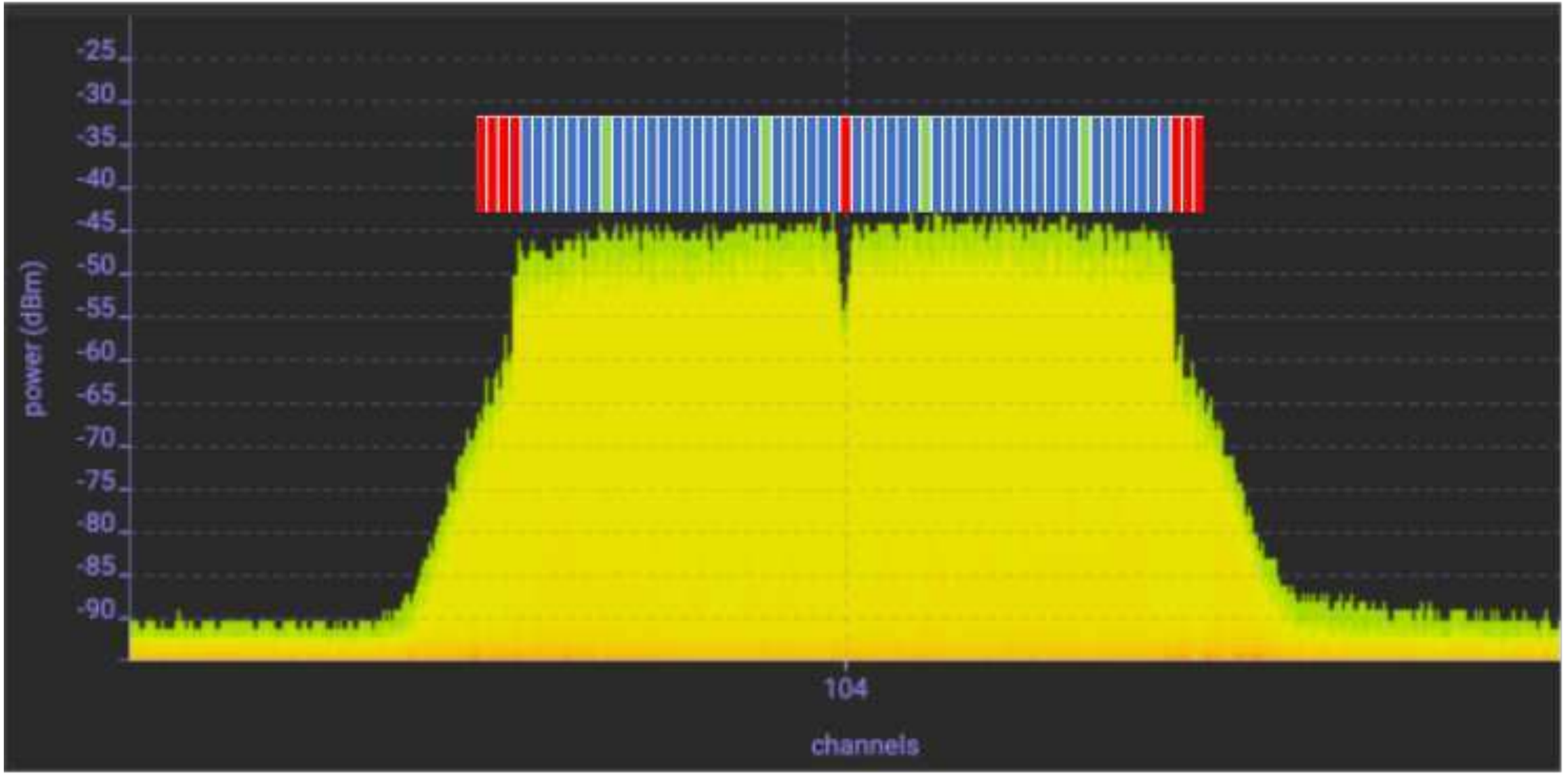
ekahau

https://thewifiofthings.com/

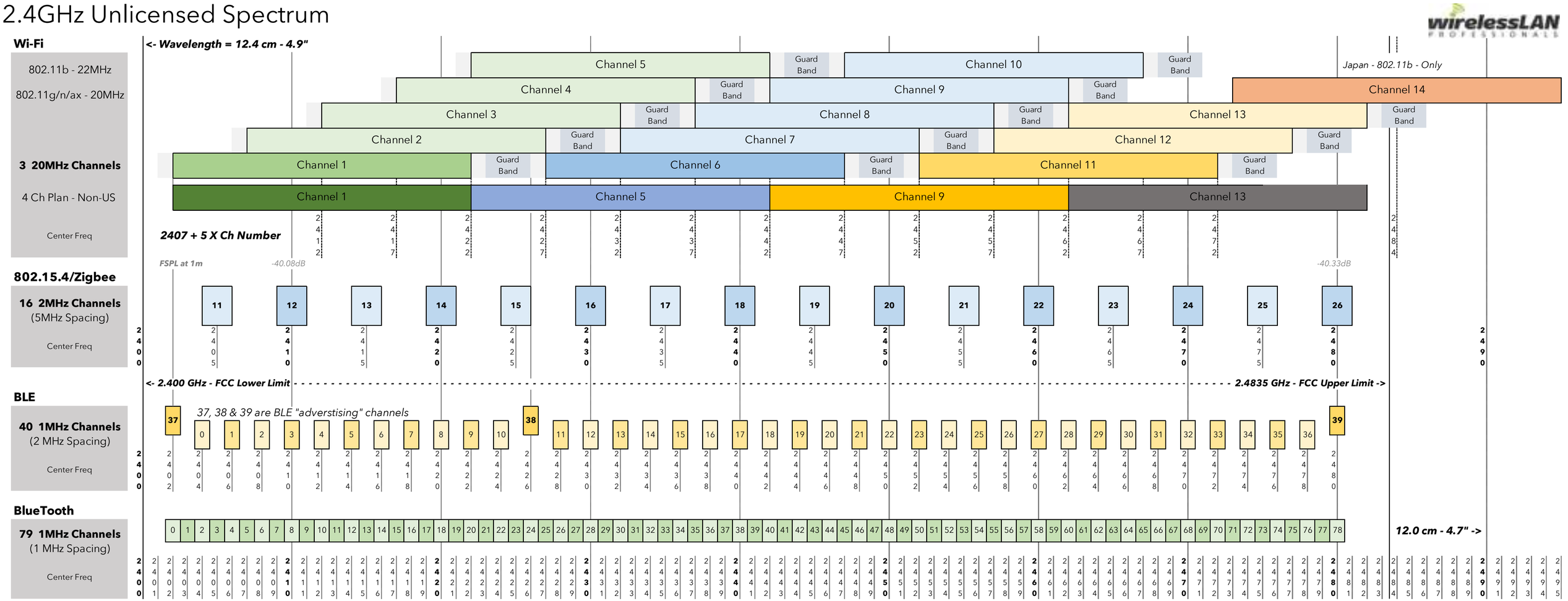
How it Works

Channels & Spectrum

64 Subcarriers [312.5 kHz]



2.4GHz Unlicensed Spectrum



5 GHz Channel Allocations 500 MHz

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Frequency	5000 + 5 X Ch. Number				DFS Channels				DFS Channels																								
Radio Band	U-NII-1				U-NII-2a				U-NII-2c (Extended)					TDWR								165 was ISM, now U-NII-3					UNII-4						
Center Freq	5.180	5.200	5.220	5.240	5.260	5.280	5.300	5.320	5.500	5.520	5.540	5.560	5.580	5.600	5.620	5.640	5.660	5.680	5.700	5.720	5.745	5.765	5.785	5.805	5.825	5.845	5.865	5.885					
20 MHz	36	40	44	48	52	56	60	64	100	104	108	112	116	120	124	128	132	136	140	144	149	153	157	161	165	169	173	177					
40 MHz	38		46		54		62		102		110		118		126		134		142		151		159		167		175						
80 MHz	42				58				106				122				138				155				171								
160 MHz	50								114												163												
FCC - US	1,000 mW Tx Power Indoor & Outdoor No DFS needed				250 mw w/6dBi Indoor & Outdoor DFS Required				250 mw w/6dBi Indoor & Outdoor DFS Required					120, 124, 128 US - Allowed						*144* *Now* *Allowed*	1,000 mW Tx Power Indoor & Outdoor No DFS needed											
ISED - Canada	FCC - Except Outdoor License Req. >200 mW				Same as FCC				Same as FCC					TDWR Not Allowed			Same as FCC				Canada PtP allows Higher EIRP											
ACMA - Australia	200 mW EIRP Indoor				200 mW EIRP - DFS & TPC 100 mW EIRP - DFS-Only Indoor				1,000 mW - DFS & TPC 500 mW - DFS-Only - No TPC Indoor/Outdoor					TDWR Not Allowed			1,000 mW - DFS & TPC 500 mW - DFS-Only Indoor/Outdoor				4,000 mW Tx Power Indoor & Outdoor No DFS needed											
ETSI - EU	100 mW No DFS/TPC Indoor				200 mW EIRP DFS/TPC Indoor				1,000 mW EIRP DFS/TPC Indoor/Outdoor											UK No 144	4,000 mW EIRP DFS/TPC - Outdoor Fixed Wireless Access											
	200 mW EIRP DFS/TPC - Indoor													10-min TWDR Scan Time						25mW SRD	25mW - SRD - No DFS											
20 MHz	36	40	44	48	52	56	60	64	100	104	108	112	116	120	124	128	132	136	140	144	149	153	157	161	165	169	173	177					
Center Freq	5.180	5.200	5.220	5.240	5.260	5.280	5.300	5.320	5.500	5.520	5.540	5.560	5.580	5.600	5.620	5.640	5.660	5.680	5.700	5.720	5.745	5.765	5.785	5.805	5.825	5.845	5.865	5.885					

<- Wavelength 5.8cm - 2.3" ***Wavelength 5.1cm - 2.0" ->***

Free Space Path Loss 1m	-45.74	-45.77	-45.80	-45.84	-45.87	-45.90	-45.94	-45.97	-46.26	-46.29	-46.32	-46.35	-46.38	-46.41	-46.44	-46.48	-46.51	-46.54	-46.57	-46.60	-46.64	-46.67	-46.70	-46.73	-46.76	-46.79	-46.82	-46.84

6 GHz Channel Allocations

FCC - USA

<- Wavelength 5.1cm - 2.0" — *1.2 Gigahertz of Spectrum* — *<- Wavelength 4.6cm - 1.8"* — *Wavelength 4.2cm - 1.6" ->*

Low Power Indoor

5dBm/MHz - Net EIRP 18dBM = LPI

Qty	Radio Band	20 MHz	UNII-5	UNII-6	UNII-7	7|8	UNII-8
59	20 MHz		1 **5** 9 13 17 **21** 25 29 33 **37** 41 45 49 **53** 57 61 65 **69** 73 77 81 **85** 89 93	97 **101** 105 109 113	**117** 121 125 129 **133** 137 141 145 **149** 153 157 161 **165** 169 173 177 **181** 185	189	193 **197** 201 205 209 **213** 217 221 225 **229** 233
29	40 MHz		3 11 19 27 35 43 51 59 67 75 83 91	99 107	115 123 131 139 147 155 163 171 179 187		195 203 211 219 227
14	80 MHz		7 23 39 55 71 87	103	119 135 151 167 183		199 215
7	160 MHz		15 47 79	111	143 175		207
3	320 MHz		31	95	159		

Preferred Scanning Channels In **BOLD** (PSC)

	5				**21**				**37**				**53**				**69**				**85**				**101**				**117**				**133**				**149**				**165**				**181**				**197**				**213**				**229**	
Free Space Path Loss at 1m	-46.95	-46.98	-47.01	-47.03	-47.06	-47.09	-47.12	-47.15	-47.18	-47.21	-47.23	-47.26	-47.29	-47.32	-47.35	-47.37	-47.40	-47.43	-47.46	-47.48	-47.51	-47.54	-47.57	-47.59	-47.62	-47.65	-47.67	-47.70	-47.73	-47.75	-47.78	-47.81	-47.83	-47.86	-47.89	-47.91	-47.94	-47.97	-47.99	-48.02	-48.04	-48.07	-48.09	-48.12	-48.14	-48.17	-48.20	-48.22	-48.25	-48.27	-48.30	-48.32	-48.35	-48.37	-48.40	-48.42	-48.44	-48.47	-48.49

Standard Power AP

36dBm with Automated Frequency Coordination (AFC)

Qty	Radio Band	20 MHz	UNII-5	UNII-6	UNII-7	7|8	UNII-8
59	20 MHz		1 **5** 9 13 17 **21** 25 29 33 **37** 41 45 49 **53** 57 61 65 **69** 73 77 81 **85** 89 93	~~97~~ **~~101~~** ~~105~~ ~~109~~ ~~113~~	**117** 121 125 129 **133** 137 141 145 **149** 153 157 161 **165** 169 173 177 **181** 185	~~189~~	~~193~~ **~~197~~** ~~201~~ ~~205~~ ~~209~~ **~~213~~** ~~217~~ ~~221~~ ~~225~~ **~~229~~** ~~233~~
29	40 MHz		3 11 19 27 35 43 51 59 67 75 83 91	~~99~~ ~~107~~	~~115~~ 123 131 139 147 155 163 171 179 ~~187~~		~~195~~ ~~203~~ ~~211~~ ~~219~~ ~~227~~
14	80 MHz		7 23 39 55 71 87	~~103~~	~~119~~ 135 151 167 ~~183~~		~~199~~ ~~215~~
7	160 MHz		15 47 79	~~111~~	143 ~~175~~		~~207~~
3	320 MHz		31	~~95~~	~~159~~		

Very Low Power AP

25mW/14dBm — *Still under review by the FCC for inclusion in the United States*

Qty	Radio Band	20 MHz	UNII-5	UNII-6	UNII-7	7|8	UNII-8
59	20 MHz		1 **5** 9 13 17 **21** 25 29 33 **37** 41 45 49 **53** 57 61 65 **69** 73 77 81 **85** 89 93	97 **101** 105 109 113	**117** 121 125 129 **133** 137 141 145 **149** 153 157 161 **165** 169 173 177 **181** 185	189	193 **197** 201 205 209 **213** 217 221 225 **229** 233
29	40 MHz		3 11 19 27 35 43 51 59 67 75 83 91	~~99~~ ~~107~~	~~115~~ 123 131 139 147 155 163 171 179 187		195 ~~203~~ ~~211~~ ~~219~~ ~~227~~
14	80 MHz		7 23 39 55 71 87	~~103~~	~~119~~ 135 151 167 183		~~199~~ ~~215~~
7	160 MHz		15 47 79	~~111~~	143 175		~~207~~
3	320 MHz		31	~~95~~	159		

Client Devices

6dB Below AP Peak

ETSI - EU

500 Megahertz of Spectrum

Radio Band

5dBm/MHz - Net EIRP 18dBM = LPI — 25mW/14dBm = VLP

UNII-5

Low Power Indoor (LPI) & Very Low Power (VLP)

Qty	Center Freq	20 MHz	
24	20 MHz	1 **5** 9 13 17 **21** 25 29 33 **37** 41 45 49 **53** 57 61 65 **69** 73 77 81 **85** 89 93	24
12	40 MHz	3 11 19 27 35 43 51 59 67 75 83 91	12
6	80 MHz	7 23 39 55 71 87	6
3	160 MHz	15 47 79	3

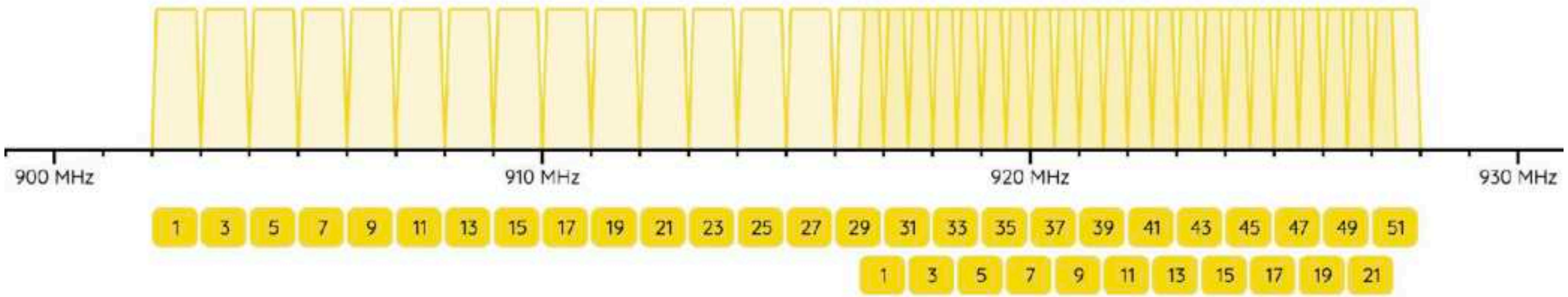
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Sub 1 Gig [S1G]

1 Mhz

32 Subcarriers 31.25 KHz [24D, 2P, 6N]

Mandatory



https://spectrum.potatofi.com/

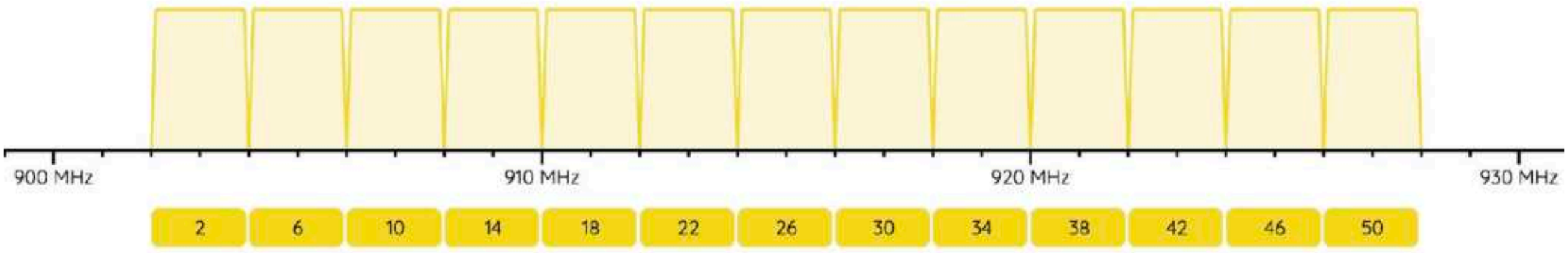
wirelessLAN PROFESSIONALS

Sub 1 Gig [S1G]

2 Mhz

64 Subcarriers 31.25 KHz [52D, 4P, 8N]

Mandatory



https://spectrum.potatofi.com/

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Sub 1 Gig [S1G]

4 Mhz

128 Subcarriers 31.25 KHz [108D, 6P, 14N]

Optional



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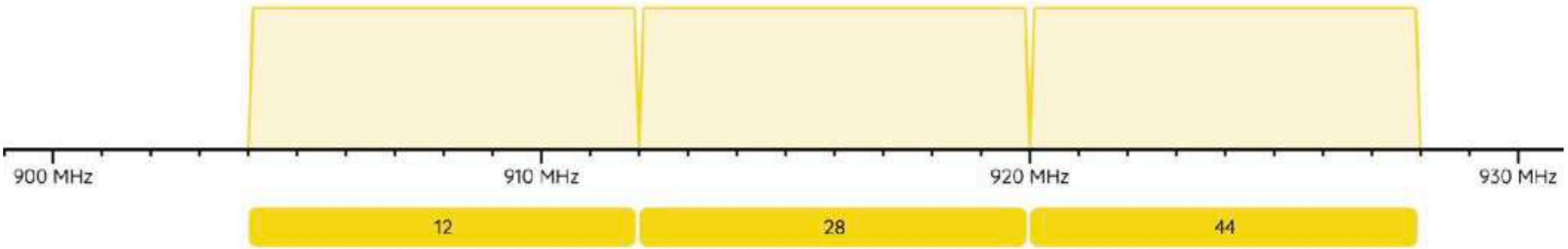
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Sub 1 Gig [S1G]

8 Mhz

256 Subcarriers 31.25 KHz [234D, 8P, 14N]

Optional



https://spectrum.potatofi.com/

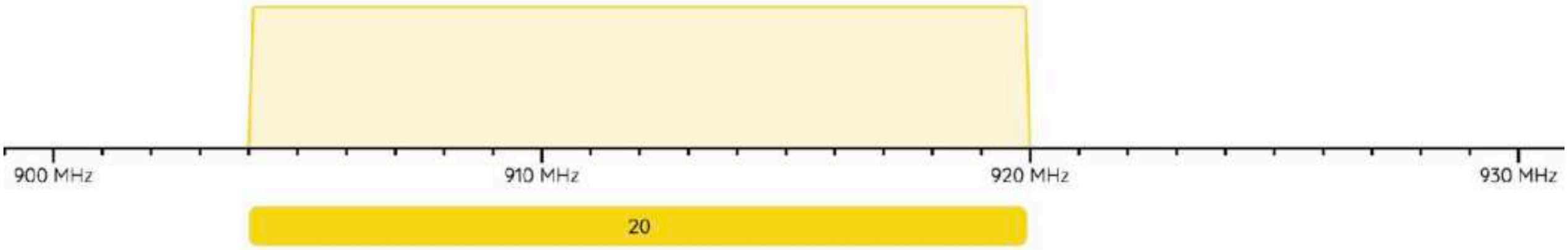
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Sub 1 Gig [S1G]

16 Mhz

512 Subcarriers 31.25 KHz [468D, 16P, 28N]

Optional



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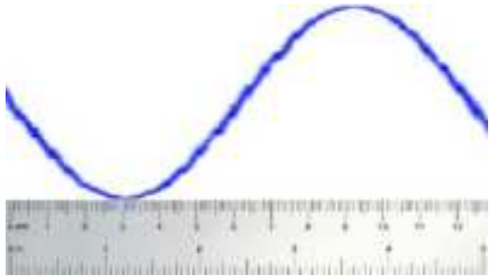
How it Works

RF

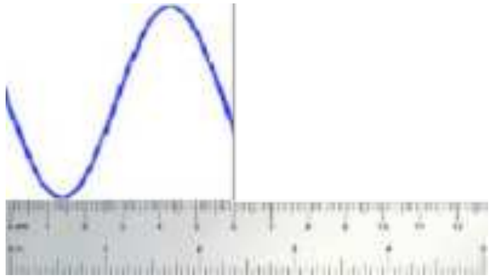
Sine Waves

Wavelength

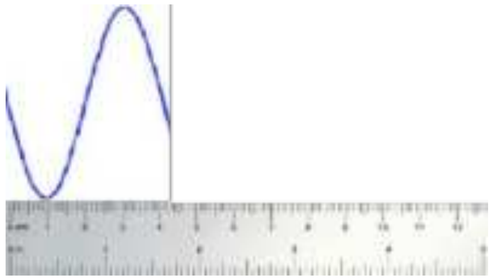
$\lambda = c / f$



2.4 GHz



5 GHz

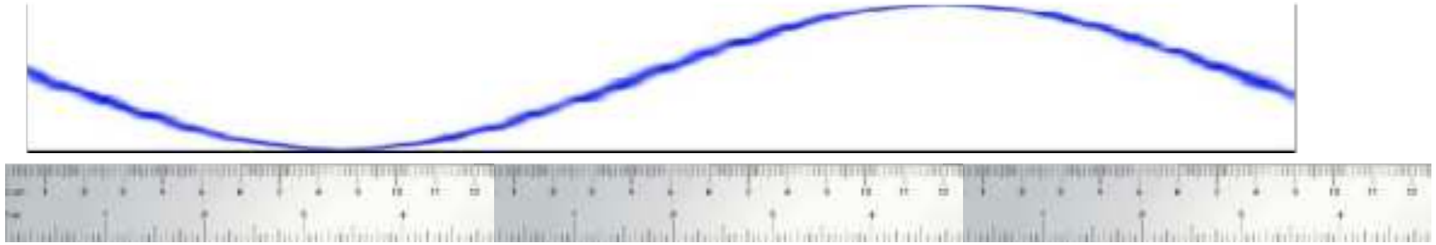


6 GHz

$\lambda = c / f$

915MHz = 32.8 cm
2.4 GHz = 12.5 cm
5 GHz = 6 cm
6 GHz = 4.2 cm
60GHz = 0.42 cm

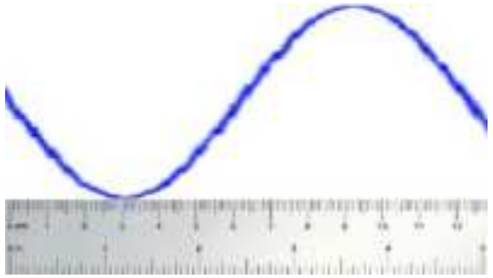
Sine Waves



915 MHz

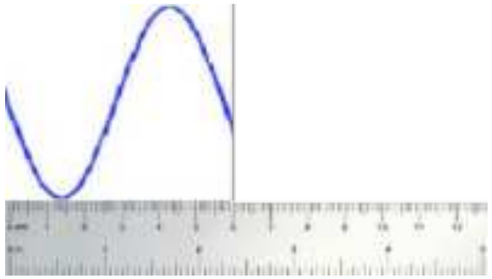
Wavelength

$\lambda = c / f$



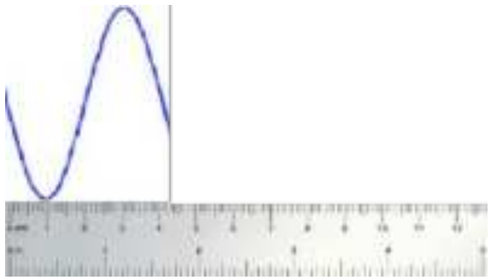
2.4 GHz

$\lambda = c / f$



5 GHz

915MHz = 32.8 cm
2.4 GHz = 12.5 cm
5 GHz = 6 cm
6 GHz = 4.2 cm
60GHz = 0.42 cm



6 GHz

Distance

1 Km [1 MHz]

Narrow Channels and Tx Power

10 dB advantage over 2.4 GHz 20 MHz

S1G Relay Node and Serve Clients

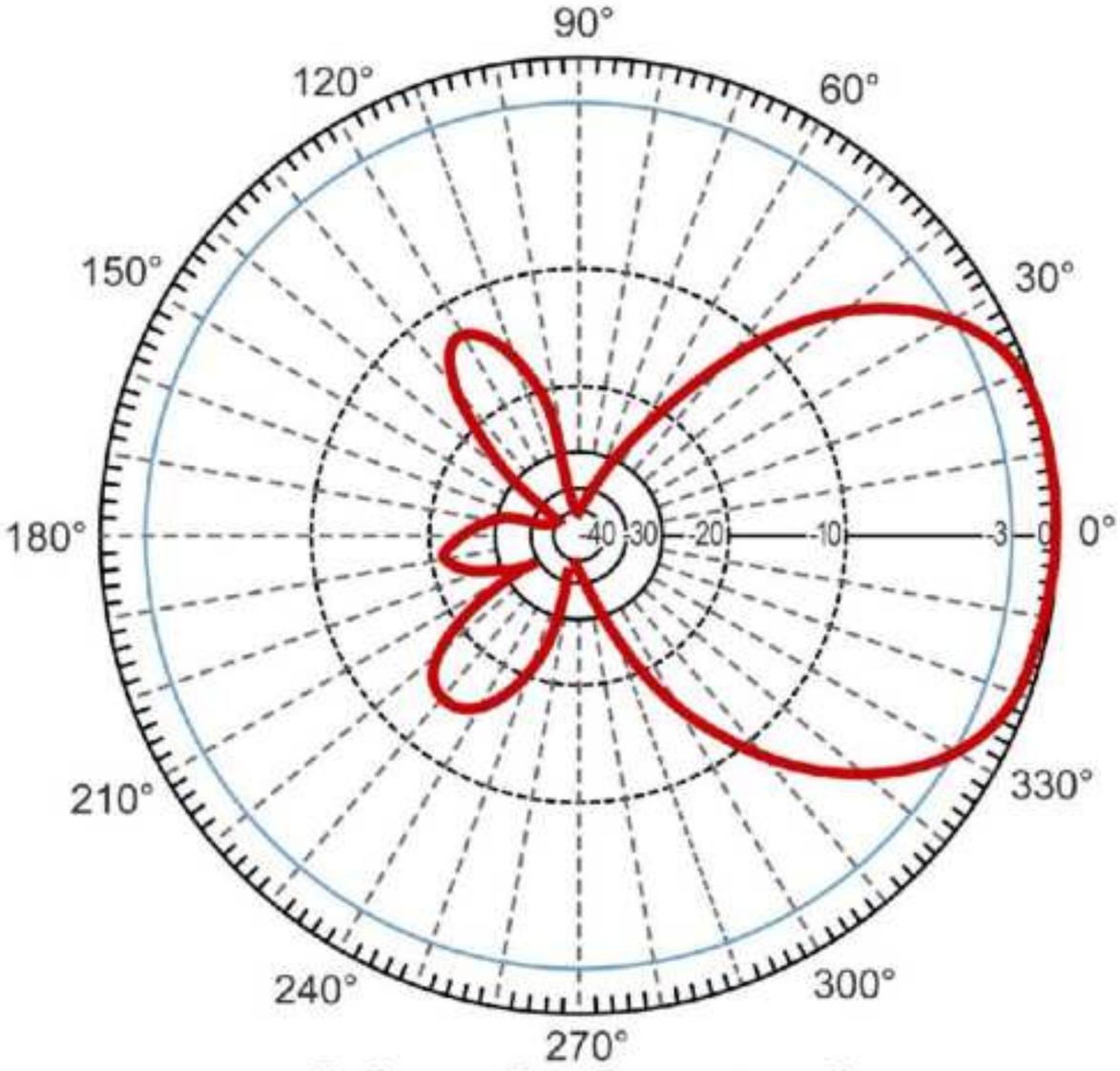
-109 dBm [1 MHz]

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How it Works

Antennas

Antennas





Reflections and Multipath



Reflections and Multipath




ALTO
PROHIBIDO
RAMPA DEL
MTR 65




UNIVERSITY
RED ZONE
RED ZONE
UTAH
10
20
30
40
50


SCANIA
SCANIA
SCANIA
SCANIA
SCANIA



How it Works

NICs

How NICs Send and Receive Information

- Convert electromagnetic energy into bits (ones and zeros)
- Modulation schemes

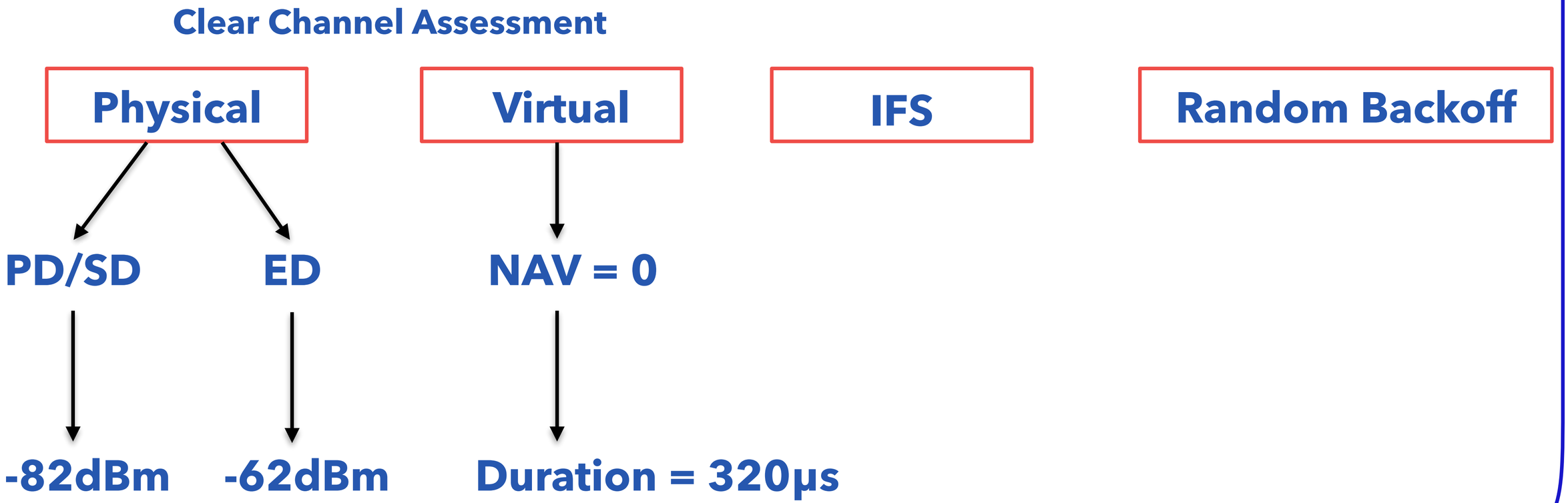


How it Works

Contention

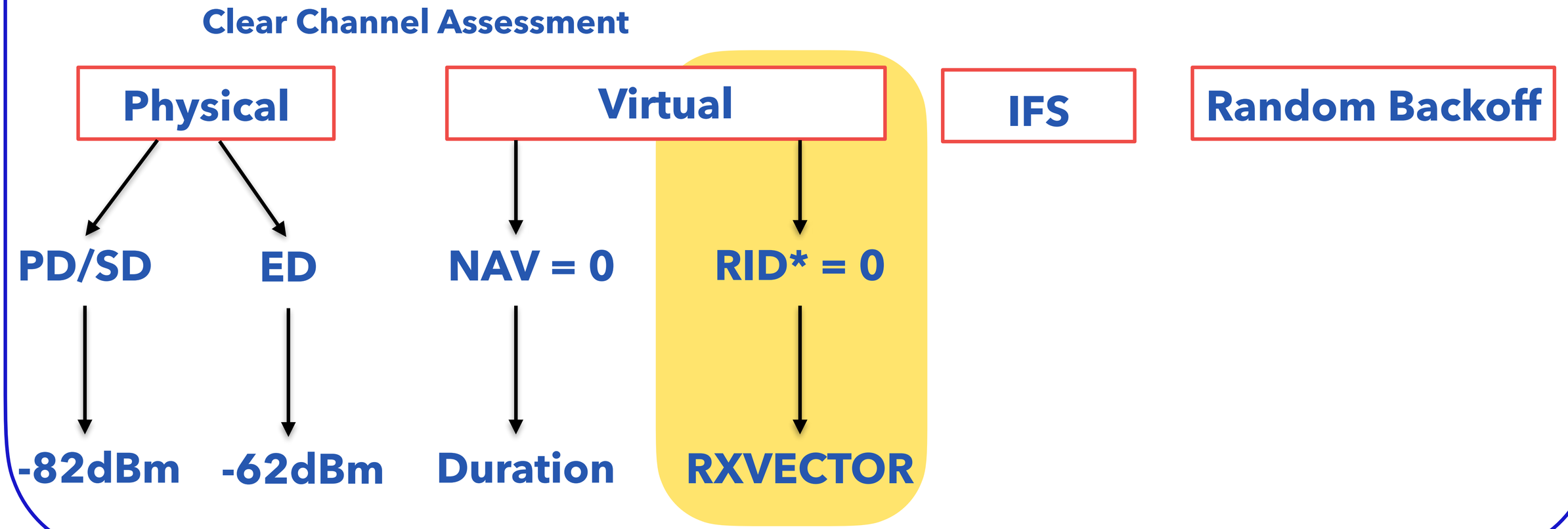
Distributed Coordination Function

Carrier Sense Multiple Access / Collision Avoidance



Distributed Coordination Function

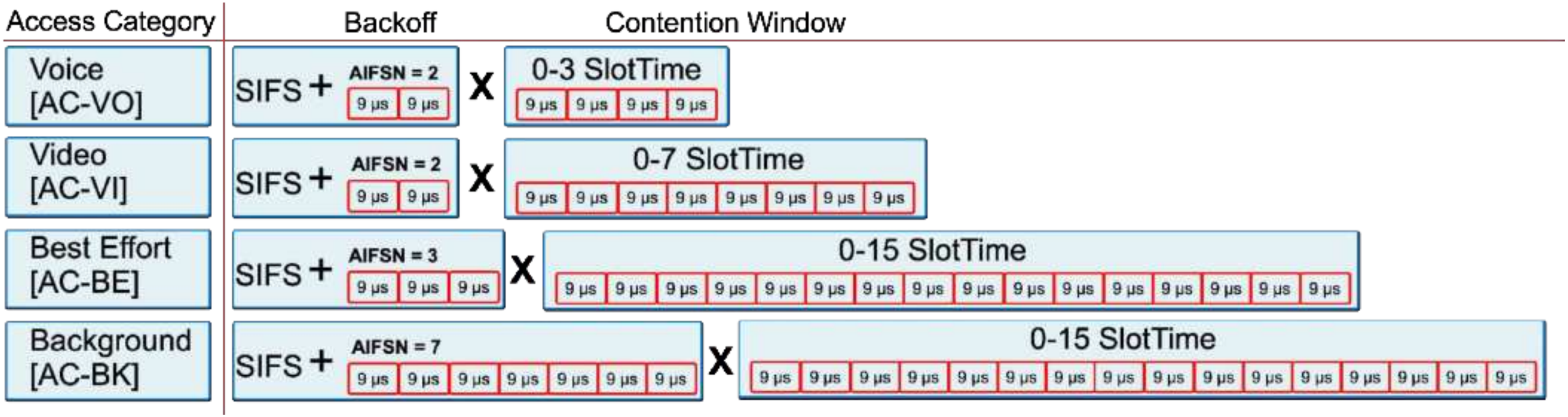
Carrier Sense Multiple Access / Collision Avoidance



* Response Indication Deferral

Enhanced Distributed Channel Access

EDCA

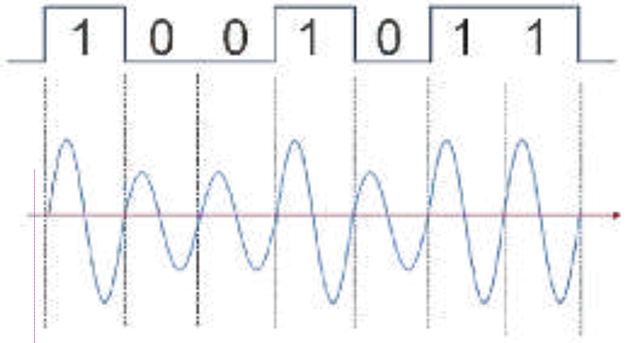


How it Works

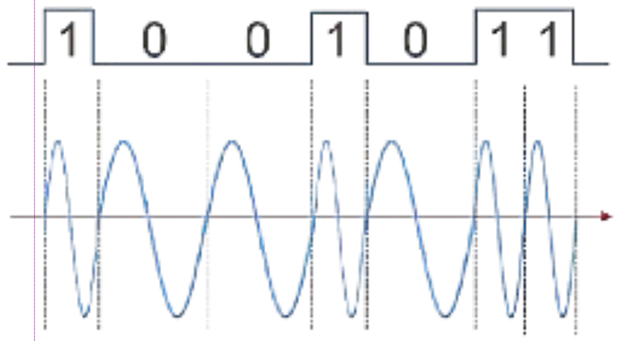
Modulation

Modulation Types

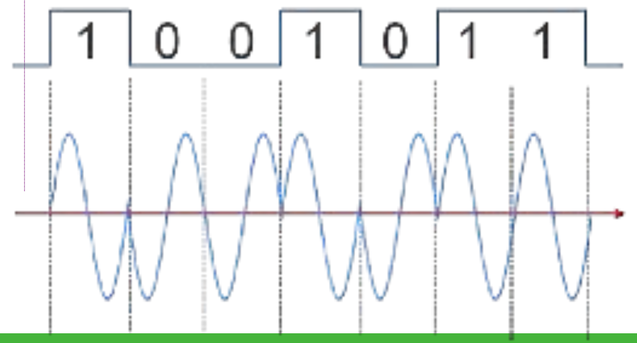
Amplitude Modulation



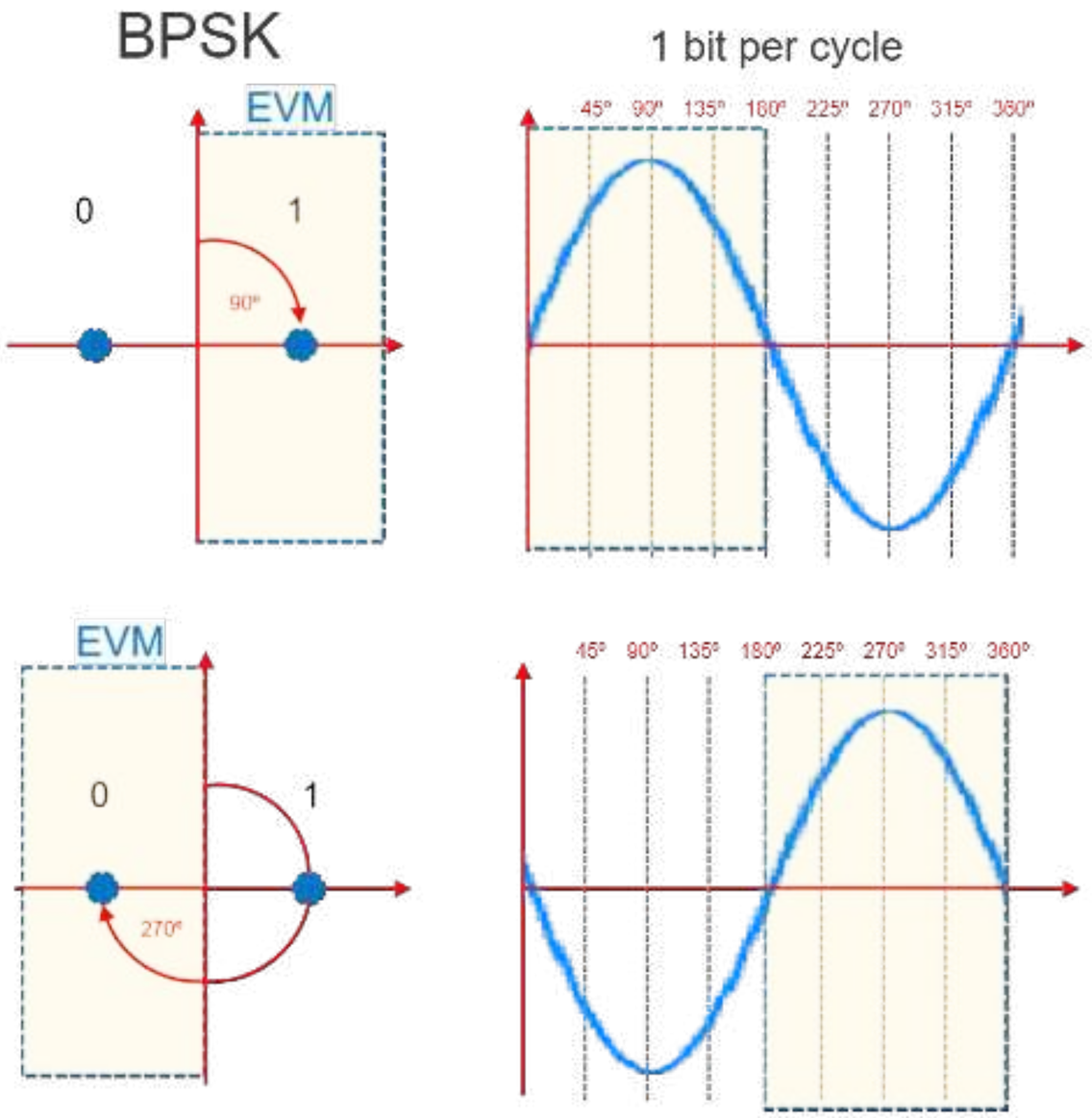
Frequency Modulation



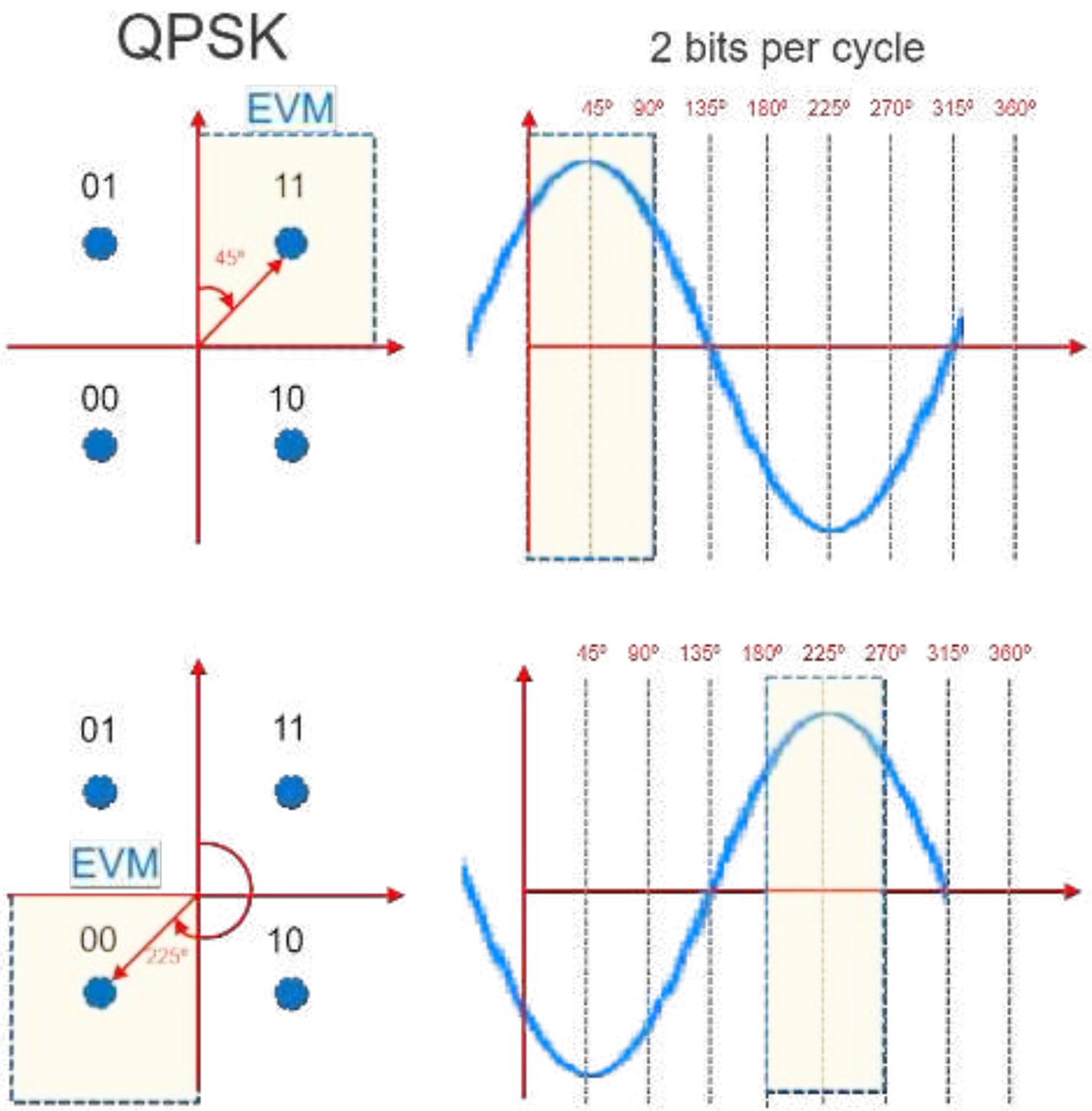
Phase Modulation



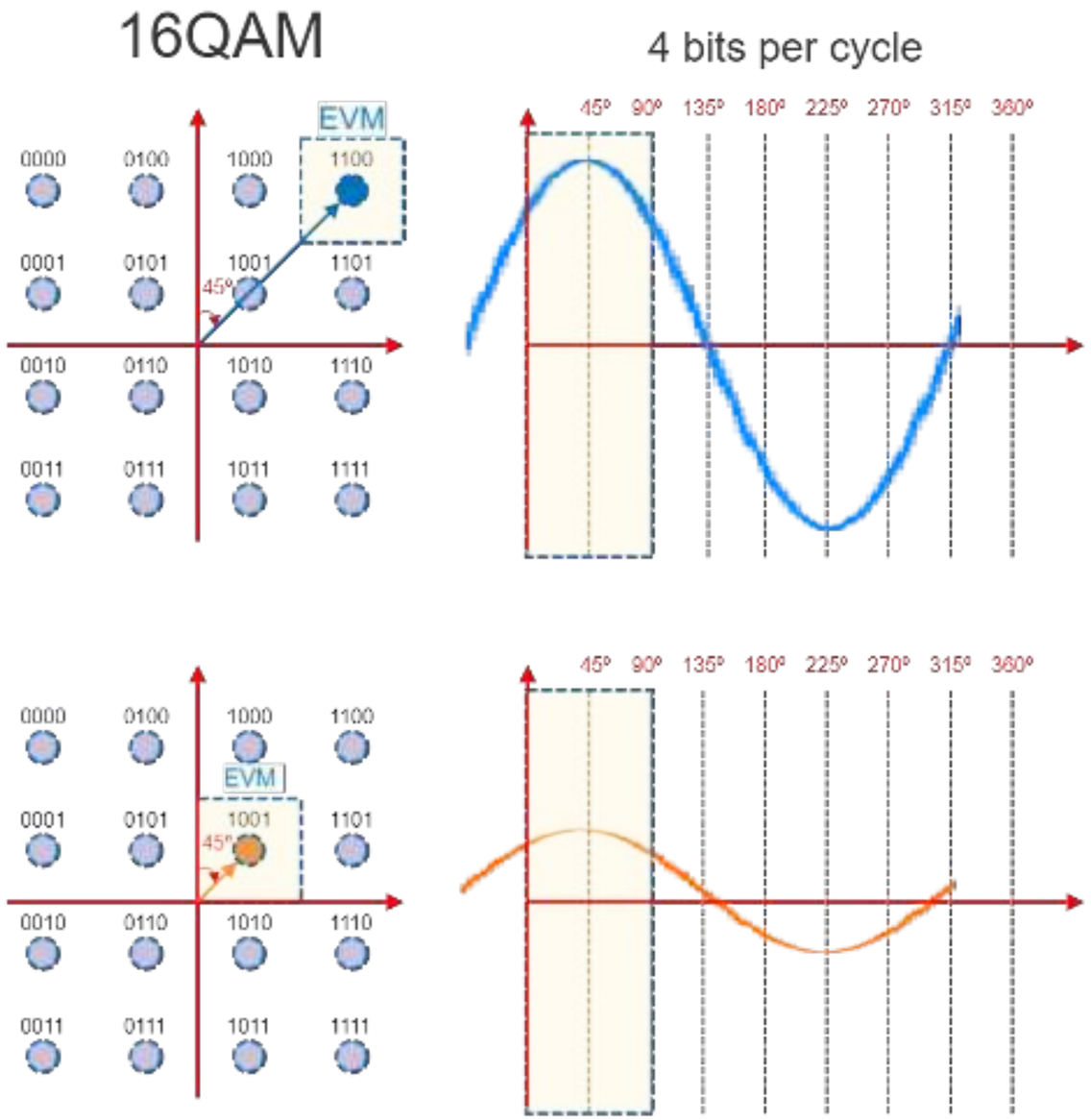
Binary Phase Shift Keying



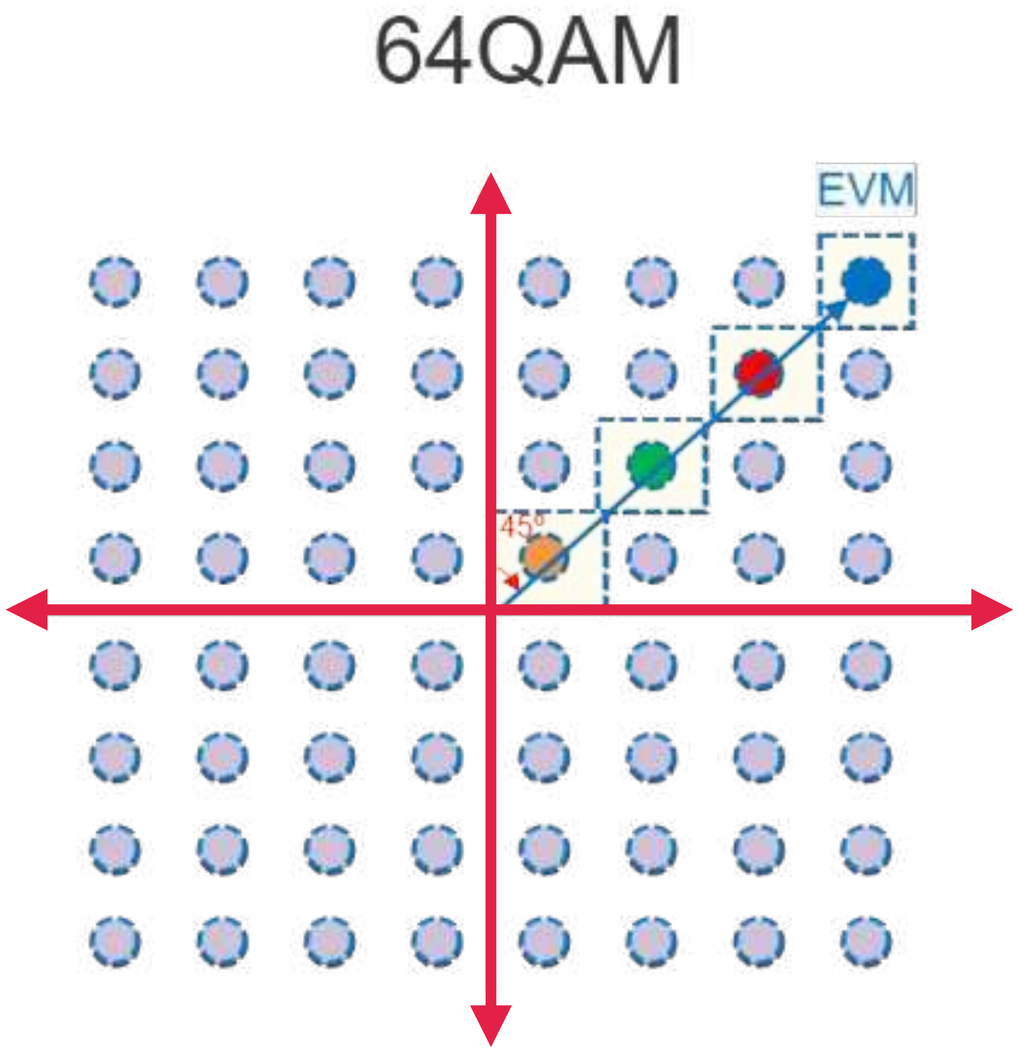
Quadrature Phase Shift Keying

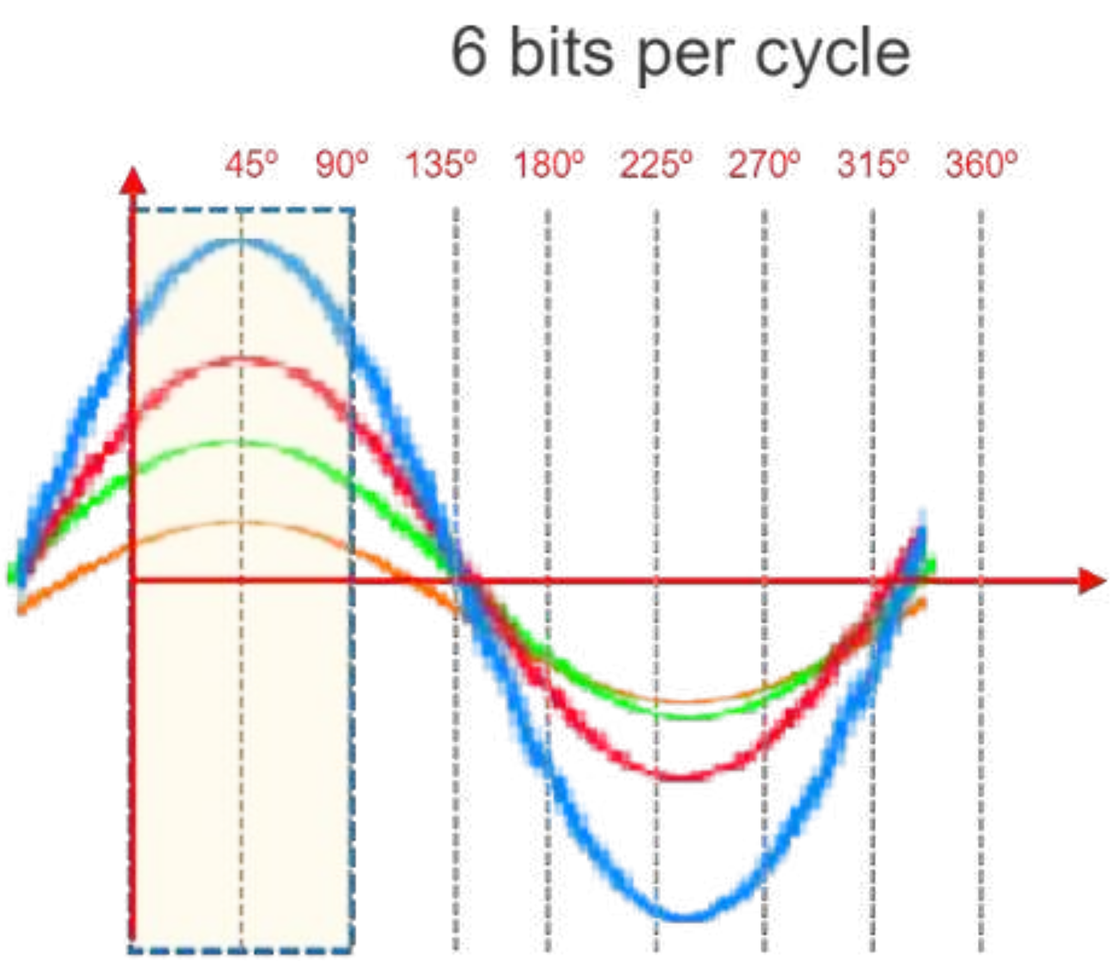


16 Quadrature Amplitude Modulation

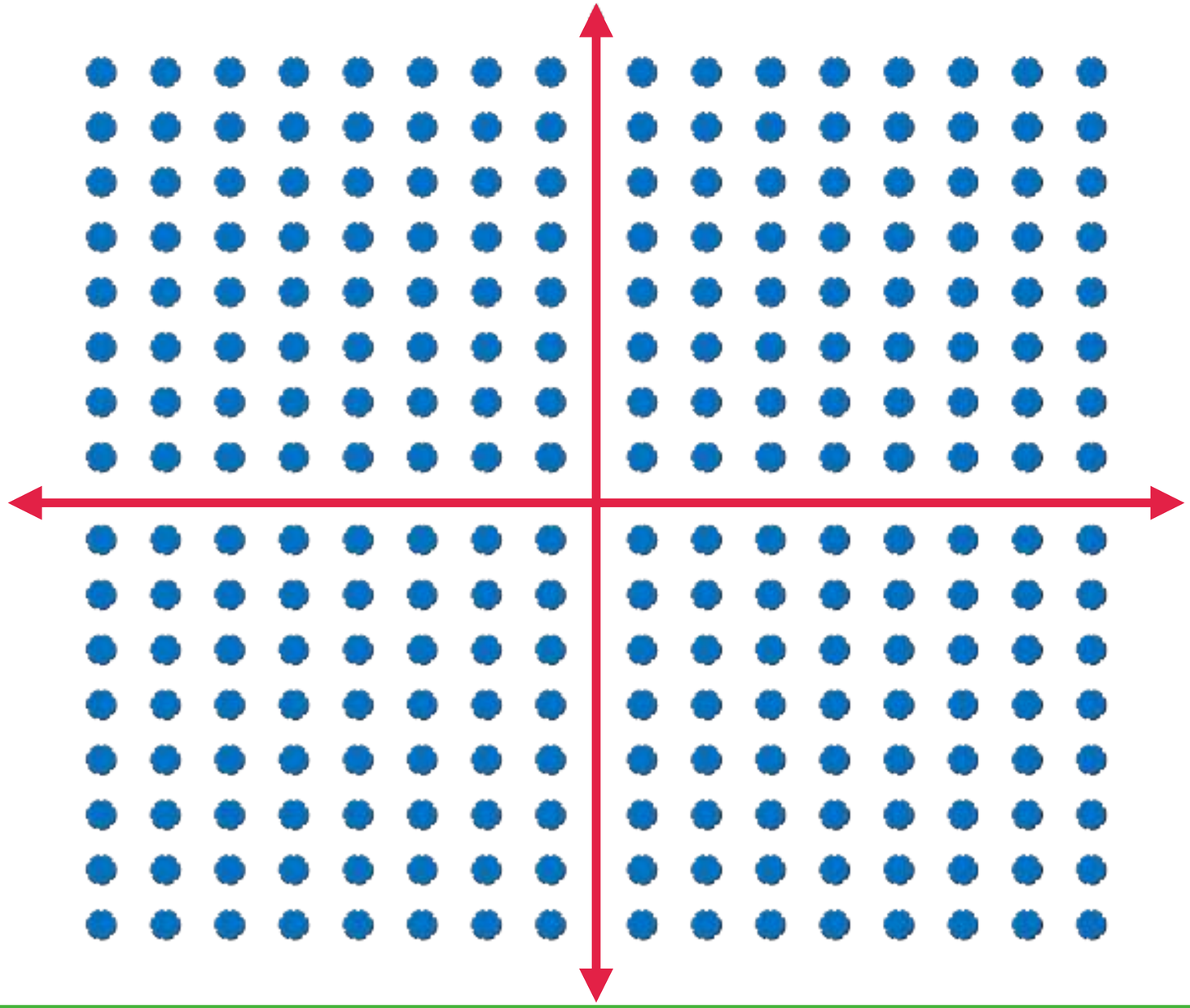


64 Quadrature Amplitude Modulation

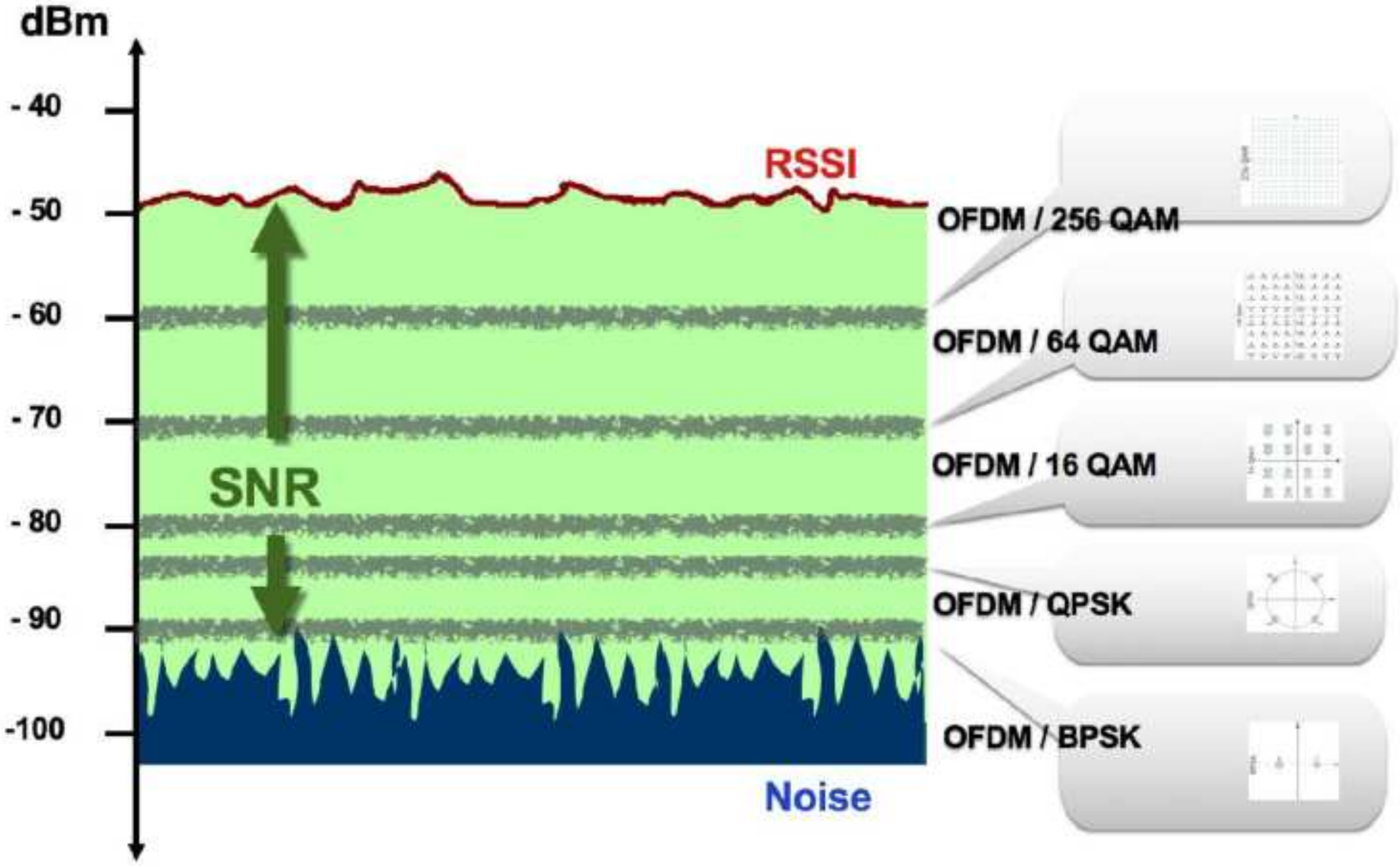




256 Quadrature Amplitude Modulation



RSSI, SNR, Noise, Modulation



Modulation and Coding Schemes

802.11n and 802.11ac — MCS, SNR and RSSI

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HT MCS	VHT MCS	Modulation	Coding	20MHz				40MHz				80MHz				160MHz			
				Data Rate		Min. SNR	RSSI	Data Rate		Min. SNR	RSSI	Data Rate		Min. SNR	RSSI	Data Rate		Min. SNR	RSSI
				800ns	400ns			800ns	400ns			800ns	400ns			800ns	400ns		
1 Spatial Stream																			
0	0	BPSK	1/2	6.5	7.2	2	-82	13.5	15	5	-79	29.3	32.5	8	-76	58.5	65	11	-73
1	1	QPSK	1/2	13	14.4	5	-79	27	30	8	-76	58.5	65	11	-73	117	130	14	-70
2	2	QPSK	3/4	19.5	21.7	9	-77	40.5	45	12	-74	87.8	97.5	15	-71	175.5	195	18	-68
3	3	16-QAM	1/2	26	28.9	11	-74	54	60	14	-71	117	130	17	-68	234	260	20	-65
4	4	16-QAM	3/4	39	43.3	15	-70	81	90	18	-67	175.5	195	21	-64	351	390	24	-61
5	5	64-QAM	2/3	52	57.8	18	-66	108	120	21	-63	234	260	24	-60	468	520	27	-57
6	6	64-QAM	3/4	58.5	65	20	-65	121.5	135	23	-62	263.3	292.5	26	-59	526.5	585	29	-56
7	7	64-QAM	5/6	65	72.2	25	-64	135	150	28	-61	292.5	325	31	-58	585	650	34	-55
	8	256-QAM	3/4	78	86.7	29	-59	162	180	32	-56	351	390	35	-53	702	780	38	-50
	9	256-QAM	5/6			31	-57	180	200	34	-54	390	433.3	37	-51	780	866.7	40	-48
2 Spatial Streams																			
8	0	BPSK	1/2	13	14.4	2	-82	27	30	5	-79	58.5	65	8	-76	117	130	11	-73
9	1	QPSK	1/2	26	28.9	5	-79	54	60	8	-76	117	130	11	-73	234	260	14	-70
10	2	QPSK	3/4	39	43.3	9	-77	81	90	12	-74	175.5	195	15	-71	351	390	18	-68
11	3	16-QAM	1/2	52	57.8	11	-74	108	120	14	-71	234	260	17	-68	468	520	20	-65
12	4	16-QAM	3/4	78	86.7	15	-70	162	180	18	-67	351	390	21	-64	702	780	24	-61
13	5	64-QAM	2/3	104	115.6	18	-66	216	240	21	-63	468	520	24	-60	936	1040	27	-57
14	6	64-QAM	3/4	117	130.3	20	-65	243	270	23	-62	526.5	585	26	-59	1053	1170	29	-56
15	7	64-QAM	5/6	130	144.4	25	-64	270	300	28	-61	585	650	31	-58	1170	1300	34	-55
	8	256-QAM	3/4	156	173.3	29	-59	324	360	32	-56	702	780	35	-53	1404	1560	38	-50
	9	256-QAM	5/6			31	-57	360	400	34	-54	780	866.7	37	-51	1560	1733	40	-48
3 Spatial Streams																			
16	0	BPSK	1/2	19.5	21.7	2	-82	40.5	45	5	-79	87.8	97.5	8	-76	175.5	195	11	-73
17	1	QPSK	1/2	39	43.3	5	-79	81	90	8	-76	175.5	195	11	-73	351	390	14	-70
18	2	QPSK	3/4	58.5	65	9	-77	121.5	135	12	-74	263.3	292.5	15	-71	526.5	585	18	-68
19	3	16-QAM	1/2	78	86.7	11	-74	162	180	14	-71	351	390	17	-68	702	780	20	-65
20	4	16-QAM	3/4	117	130	15	-70	243	270	18	-67	526.5	585	21	-64	1053	1170	24	-61
21	5	64-QAM	2/3	156	173.3	18	-66	324	360	21	-63	702	780	24	-60	1404	1560	27	-57
22	6	64-QAM	3/4	175.5	195	20	-65	364.5	405	23	-62			26	-59	1580	1755	29	-56
23	7	64-QAM	5/6	195	216.7	25	-64	405	450	28	-61	877.5	975	31	-58	1755	1950	34	-55
	8	256-QAM	3/4	234	260	29	-59	486	540	32	-56	1053	1170	35	-53	2106	2340	38	-50
	9	256-QAM	5/6	260	288.9	31	-57	540	600	34	-54	1170	1300	37	-51			40	-48

Wi-Fi MCS index

PHY data rates · 802.11n / ac / ax / be

Copy CSV | CSV | PDF | Light

28 rows · **39** rate columns

802.11n · Wi-Fi 4 | 802.11ac · Wi-Fi 5 | 802.11ax · Wi-Fi 6 / 6E | 802.11be · Wi-Fi 7

Q 1024-QAM, 5/6, 2,882.4

SELECTED RATE

Click a rate in the table to find out how it is calculated.

WI-FI STANDARD — None

802.11n Wi-Fi 4 | 802.11ac Wi-Fi 5 | 802.11ax Wi-Fi 6 / 6E | 802.11be Wi-Fi 7

SPATIAL STREAMS — All

1 2 3 4 5 6 7 8

CHANNEL WIDTH (MHZ) — None

20 40 80 160 320

GUARD INTERVAL — None

400 ns | 800 ns | 1.6 µs | 3.2 µs

				802.11n · Wi-Fi 4				802.11ac · Wi-Fi 5						802.11ax · Wi-					
				20 MHz		40 MHz		20 MHz		40 MHz		80 MHz		160 MHz		20 MHz · 242-tone RU			40 MHz · 484-tone RU		
MCS	SS	MODULATION	CODING	800 ns	400 ns	800 ns	400 ns	800 ns	400 ns	800 ns	400 ns	800 ns	400 ns	800 ns	400 ns	3.2 µs	1.6 µs	800 ns	3.2 µs	1.6 µs	800 ns
1 SPATIAL STREAM																					
0	1	BPSK	1/2	6.5	7.2	13.5	15.0	6.5	7.2	13.5	15.0	29.3	32.5	58.5	65.0	7.3	8.1	8.6	14.6	16.3	17.2
1	1	QPSK	1/2	13.0	14.4	27.0	30.0	13.0	14.4	27.0	30.0	58.5	65.0	117.0	130.0	14.6	16.3	17.2	29.3	32.5	34.4
2	1	QPSK	3/4	19.5	21.7	40.5	45.0	19.5	21.7	40.5	45.0	87.8	97.5	175.5	195.0	21.9	24.4	25.8	43.9	48.8	51.6
3	1	16-QAM	1/2	26.0	28.9	54.0	60.0	26.0	28.9	54.0	60.0	117.0	130.0	234.0	260.0	29.3	32.5	34.4	58.5	65.0	68.8
4	1	16-QAM	3/4	39.0	43.3	81.0	90.0	39.0	43.3	81.0	90.0	175.5	195.0	351.0	390.0	43.9	48.8	51.6	87.8	97.5	103.2
5	1	64-QAM	2/3	52.0	57.8	108.0	120.0	52.0	57.8	108.0	120.0	234.0	260.0	468.0	520.0	58.5	65.0	68.8	117.0	130.0	137.6
6	1	64-QAM	3/4	58.5	65.0	121.5	135.0	58.5	65.0	121.5	135.0	263.3	292.5	526.5	585.0	65.8	73.1	77.4	131.6	146.3	154.9
7	1	64-QAM	5/6	65.0	72.2	135.0	150.0	65.0	72.2	135.0	150.0	292.5	325.0	585.0	650.0	73.1	81.3	86.0	146.3	162.5	172.1
8	1	256-QAM	3/4	—	—	—	—	78.0	86.7	162.0	180.0	351.0	390.0	702.0	780.0	87.8	97.5	103.2	175.5	195.0	206.5
9	1	256-QAM	5/6	—	—	—	—	—	—	180.0	200.0	390.0	433.3	780.0	866.7	97.5	108.3	114.7	195.0	216.7	229.4
10	1	1024-QAM	3/4	—	—	—	—	—	—	—	—	—	—	—	—	109.7	121.9	129.0	219.4	243.8	258.1
11	1	1024-QAM	5/6	—	—	—	—	—	—	—	—	—	—	—	—	121.9	135.4	143.4	243.8	270.8	286.8
12	1	4096-QAM	3/4	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
13	1	4096-QAM	5/6	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2 SPATIAL STREAMS																					
0	2	BPSK	1/2	13.0	14.4	27.0	30.0	13.0	14.4	27.0	30.0	58.5	65.0	117.0	130.0	14.6	16.3	17.2	29.3	32.5	34.4
1	2	QPSK	1/2	26.0	28.9	54.0	60.0	26.0	28.9	54.0	60.0	117.0	130.0	234.0	260.0	29.3	32.5	34.4	58.5	65.0	68.8
2	2	QPSK	3/4	39.0	43.3	81.0	90.0	39.0	43.3	81.0	90.0	175.5	195.0	351.0	390.0	43.9	48.8	51.6	87.8	97.5	103.2
3	2	16-QAM	1/2	52.0	57.8	108.0	120.0	52.0	57.8	108.0	120.0	234.0	260.0	468.0	520.0	58.5	65.0	68.8	117.0	130.0	137.6
4	2	16-QAM	3/4	78.0	86.7	162.0	180.0	78.0	86.7	162.0	180.0	351.0	390.0	702.0	780.0	87.8	97.5	103.2	175.5	195.0	206.5
5	2	64-QAM	2/3	104.0	115.6	216.0	240.0	104.0	115.6	216.0	240.0	468.0	520.0	936.0	1,040.0	117.0	130.0	137.6	234.0	260.0	275.3
6	2	64-QAM	3/4	117.0	130.0	243.0	270.0	117.0	130.0	243.0	270.0	526.5	585.0	1,053.0	1,170.0	131.6	146.3	154.9	263.3	292.5	309.7

MCS Table (HT/VHT/HE) - MCSINDEX.NET

MCS Index	Spatial Stream	Modulation	Coding	OFDM (802.11ah) HaLow									
				1MHz		2MHz		4MHz		8MHz		16MHz	
				0.8µs GI	0.4µs GI	0.8µs GI	0.4µs GI	0.8µs GI	0.4µs GI	0.8µs GI	0.4µs GI	0.8µs GI	0.4µs GI
0	1	BPSK	1/2	0.3	0.33	0.65	0.7	1.35	1.5	2.925	3.3	5.85	6.5
1	1	QPSK	1/2	0.6	0.67	1.3	1.44	2.7	3.0	5.85	6.5	11.7	13.0
2	1	QPSK	3/4	0.9	1.00	1.95	2.17	4.05	4.5	8.78	9.8	17.6	19.5
3	1	16-QAM	1/2	1.2	1.33	2.6	2.89	5.4	6.0	11.7	13.0	23.4	26.0
4	1	16-QAM	3/4	1.8	2.00	3.9	4.33	8.1	9.0	17.55	19.5	35.1	39.0
5	1	64-QAM	2/3	2.4	2.67	5.2	5.78	10.8	12.0	23.4	26.0	46.8	52.0
6	1	64-QAM	3/4	2.7	3.00	5.85	6.5	12.15	13.5	26.33	29.3	52.65	58.5
7	1	64-QAM	5/6	3.0	3.33	6.5	7.22	13.5	15.0	29.25	32.5	58.5	65.0
8	1	256-QAM	3/4	3.6	4.00	7.8	8.67	16.2	18.0	35.1	39.0	70.2	78.0
9	1	256-QAM	5/6	4.0	4.44	-	-	18	20.0	39	43.3	78	86.7
10	1	BPSK	1/2 x 2	0.15	0.17	-	-	-	-	-	-	-	-
0	2	BPSK	1/2	0.6	0.7	1.3	1.4	2.7	3.0	5.85	6.5	11.7	13.0
1	2	QPSK	1/2	1.2	1.3	2.6	2.9	5.4	6.0	11.7	13.0	23.4	26.0
2	2	QPSK	3/4	1.8	2.0	3.9	4.3	8.1	9.0	17.55	19.5	35.1	39.0
3	2	16-QAM	1/2	2.4	2.7	5.2	5.8	10.8	12.0	23.4	26.0	46.8	52.0
4	2	16-QAM	3/4	3.6	4.0	7.8	8.7	16.2	18.0	35.1	39.0	70.2	78.0
5	2	64-QAM	2/3	4.8	5.33	10.4	11.6	21.6	24.0	46.8	52.0	93.6	104
6	2	64-QAM	3/4	5.4	6.0	11.7	13.0	24.3	27.0	52.65	58.5	105.3	117
7	2	64-QAM	5/6	6.0	6.67	13	14.4	27	30.0	58.5	65.0	117	130
8	2	256-QAM	3/4	7.2	8.0	15.6	17.3	32.4	36.0	70.2	78.0	140.4	156
9	2	256-QAM	5/6	8.0	8.85	-	-	35.9	39.8	77.7	86.3	155.4	173
0	3	BPSK	1/2	0.9	1.0	2.0	2.2	4.05	4.5	8.78	9.8	17.55	19.5
1	3	QPSK	1/2	1.8	2.0	1.95	4.3	8.1	9.0	17.55	19.5	35.1	39.0
2	3	QPSK	3/4	2.7	3.0	3.90	6.5	12.15	13.5	26.33	29.3	52.65	58.5
3	3	16-QAM	1/2	3.6	4.0	5.85	8.7	16.2	18.0	35.1	39.0	70.2	78
4	3	16-QAM	3/4	5.4	6.0	7.80	13.0	24.3	27.0	52.65	58.5	105.3	117
5	3	64-QAM	2/3	7.2	8.0	11.70	17.3	32.4	36.0	70.2	78.0	140.4	156
6	3	64-QAM	3/4	8.1	9.0	15.90	19.5	36.45	40.5	-	-	157.95	176
7	3	64-QAM	5/6	9.0	10.0	17.55	21.7	40.5	45.0	87.75	97.5	175.5	195
8	3	256-QAM	3/4	10.8	12.0	19.50	26.0	48.6	54.0	105.3	117.0	211	234
9	3	256-QAM	5/6	12.0	13.3	23.40	28.8	53.78	59.8	117	129	-	-
0	4	BPSK	1/2	1.2	1.3	2.6	2.9	5.4	6.0	11.7	13.0	23.4	26.0
1	4	QPSK	1/2	2.4	2.7	5.2	5.8	10.8	12.0	23.4	26.0	46.8	52.0
2	4	QPSK	3/4	3.6	4.0	7.8	8.7	16.2	18.0	35.1	39.0	70.2	78.0
3	4	16-QAM	1/2	4.8	5.3	10.4	11.6	21.6	24.0	46.8	52.0	93.6	104.0
4	4	16-QAM	3/4	7.2	8.0	15.6	17.3	32.4	36.0	70.2	78.0	140.4	156.0
5	4	64-QAM	2/3	9.6	10.7	20.8	23.1	43.2	48.0	93.6	104.0	187.2	208.0
6	4	64-QAM	3/4	10.8	12.0	23.4	26.0	48.6	54.0	105.3	117.0	210.6	234.0
7	4	64-QAM	5/6	12.0	13.3	26	28.8	54	60.0	117	130.0	234	260.0
8	4	256-QAM	3/4	14.4	16.0	31.2	34.7	64.8	72.0	140.4	156.0	280.8	312.0
9	4	256-QAM	5/6	15.9	17.7	-	-	71.71	79.7	155.4	172.9	310.8	345.3

by Troy Martin

https://mcsindex.net/

MCS Table (HT/VHT/HE) - MCSINDEX.NET

MCS Index	Spatial Stream	Modulation	Coding	OFD				
				1MHz		2MHz		
				0.8µs GI	0.4µs GI	0.8µs GI	0.4µs GI	0.
0	1	BPSK	1/2	0.3	0.33	0.65	0.7	
1	1	QPSK	1/2	0.6	0.67	1.3	1.44	
2	1	QPSK	3/4	0.9	1.00	1.95	2.17	
3	1	16-QAM	1/2	1.2	1.33	2.6	2.89	
4	1	16-QAM	3/4	1.8	2.00	3.9	4.33	
5	1	64-QAM	2/3	2.4	2.67	5.2	5.78	
6	1	64-QAM	3/4	2.7	3.00	5.85	6.5	
7	1	64-QAM	5/6	3.0	3.33	6.5	7.22	
8	1	256-QAM	3/4	3.6	4.00	7.8	8.67	
9	1	256-QAM	5/6	4.0	4.44	-	-	
10	1	BPSK	1/2 x 2	0.15	0.17	-	-	
0	2	BPSK	1/2	0.6	0.7	1.3	1.4	
1	2	QPSK	1/2	1.2	1.3	2.6	2.9	

Modulation and Coding Schemes

MCS Index	Spatial Streams	Modulation Type	Data Rate (Mbps)									
			1 MHz Channels		2 MHz Channels		4 MHz Channels		8 MHz Channels		16 MHz Channels	
			Long GI[3]	Short GI	Long GI	Short GI	Long GI	Short GI	Long GI	Short GI	Long GI	Short GI
0	1	BPSK	0.30	0.33	0.65	0.72	1.4	1.5	2.9	3.3	5.9	6.5
1	1	QPSK	0.60	0.67	1.3	1.4	2.7	3.0	5.9	6.5	11.7	13.0
2	1	QPSK	0.90	1.00	2.0	2.2	4.1	4.5	8.8	9.8	17.6	19.5
3	1	16-QAM	1.2	1.3	2.6	2.9	5.4	6.0	17.6	19.5	35.1	39.0
4	1	16-QAM	1.8	2.0	3.9	4.3	8.1	9.0	17.6	19.5	35.1	39.0
5	1	64-QAM	2.4	2.7	5.2	5.8	10.8	12.0	22.3	23.6	48.6	52.0
6	1	64-QAM	2.7	3.0	5.9	6.5	12.2	13.5	26.3	29.3	52.7	58.5
7	1	64-QAM	3.0	3.3	6.5	7.2	13.5	15.0	29.3	32.5	58.5	65.0
8	1	256-QAM	3.6	4.0	7.8	8.7	16.2	18.0	35.0	39.0	70.2	78.0
9	1	256-QAM	4.0	4.4	N/A	N/A	18.0	20.0	43.3	43.3	78.0	86.7[4]
10	1	BPSK	0.15	0.17	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

https://wi-fi.org/

How it Works

dB Math

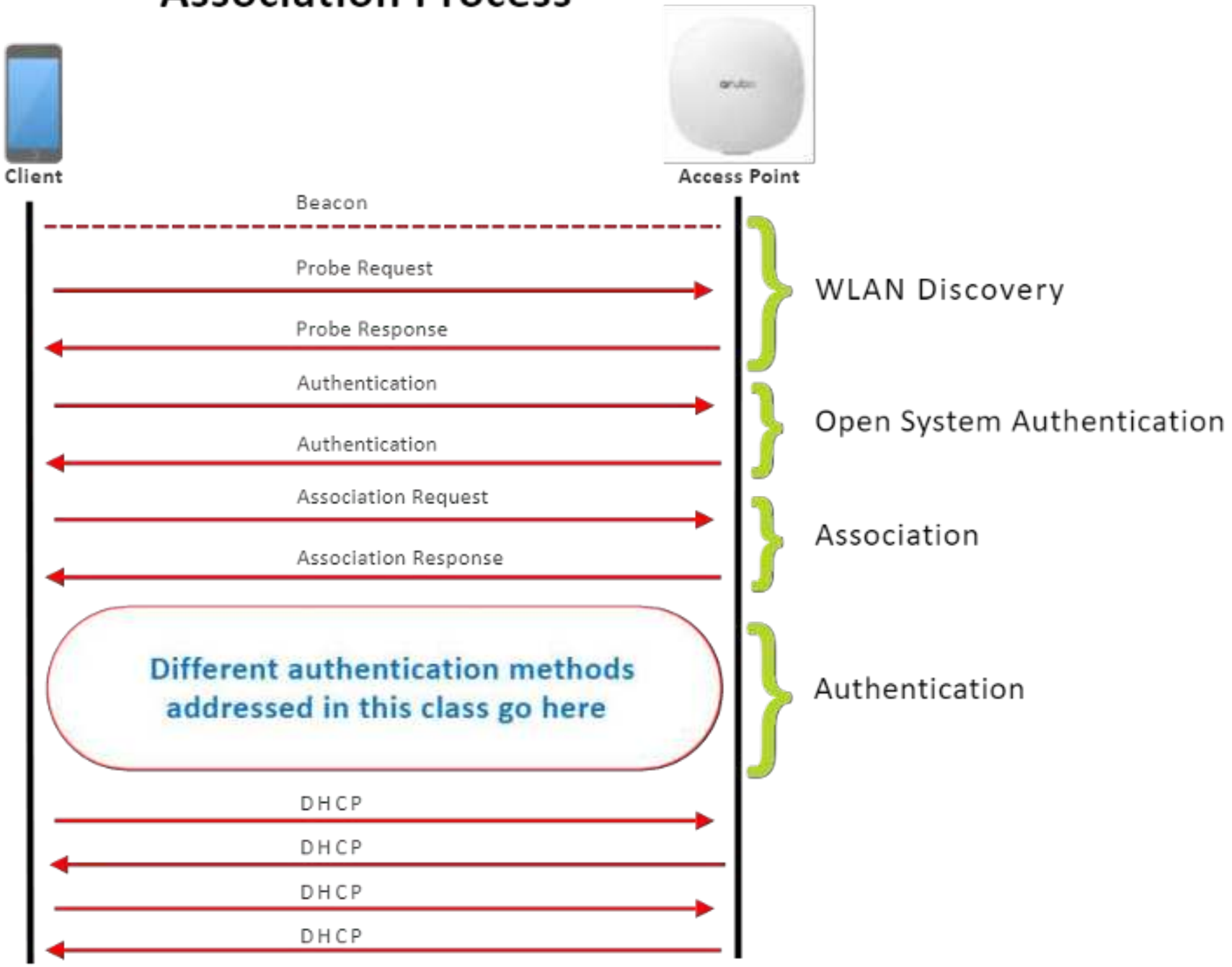
Conversion Chart dBm/mW

0 dBm	**1 mW**	**11 dBm**	12.5 mW	**22 dBm**	160 mW
1 dBm	1.25 mW	**12 dBm**	16 mW	**23 dBm**	200 mW
2 dBm	1.56 mW	**13 dBm**	20 mW	**24 dBm**	256 mW
3 dBm	2 mW	**14 dBm**	25 mW	**25 dBm**	320 mW
4 dBm	2.5 mW	**15 dBm**	32 mW	**26 dBm**	400 mW
5 dBm	3.12 mW	**16 dBm**	40 mW	**27 dBm**	512 mW
6 dBm	4 mW	**17 dBm**	50 mW	**28 dBm**	640 mW
7 dBm	5 mW	**18 dBm**	64 mW	**29 dBm**	800 mW
8 dBm	6.25 mW	**19 dBm**	80 mW	**30 dBm**	**1000 mW**
9 dBm	8 mW	**20 dBm**	**100 mW**	**40 dBm**	10000 mW
10 dBm	**10 mW**	**21 dBm**	128 mW	**43 dBm**	20000 mW

How it Works

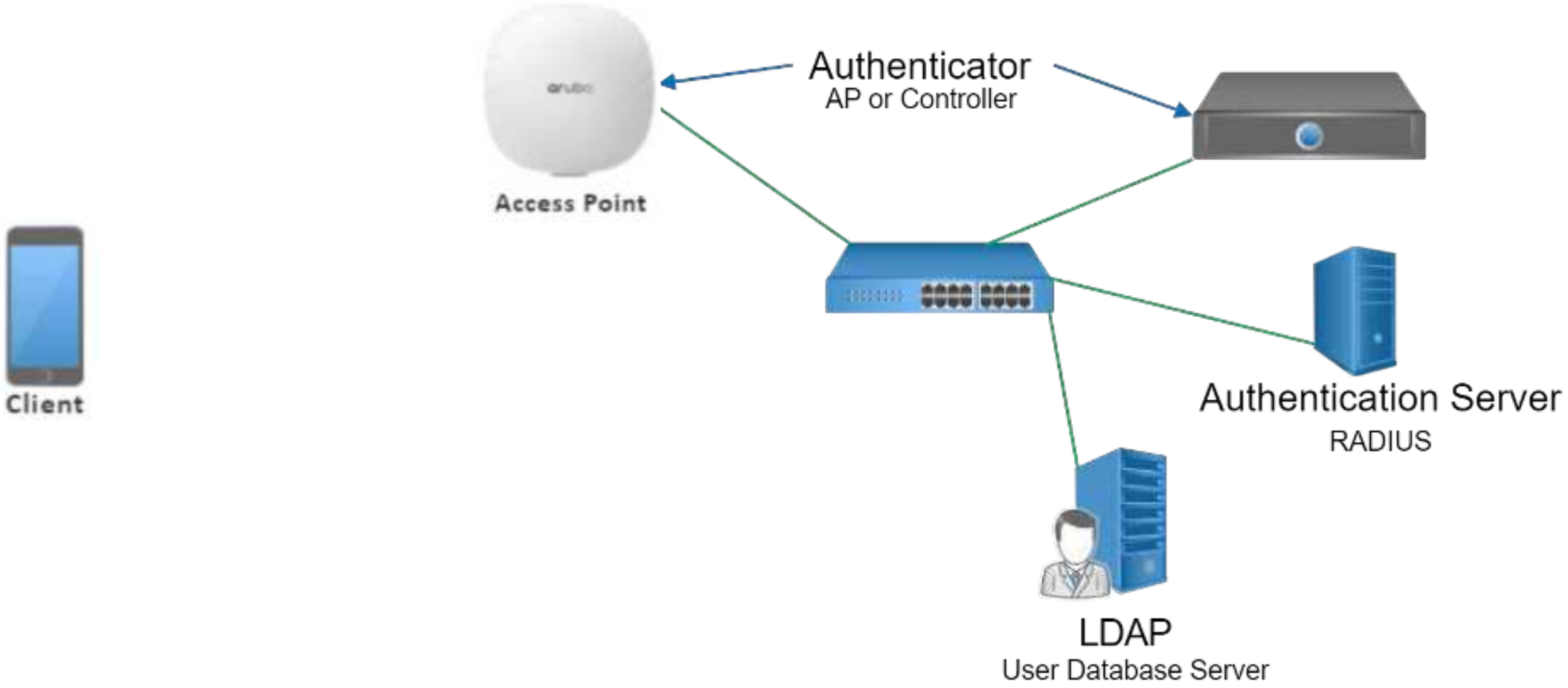
Association

Association Process



Authentication

802.1X/EAP Authentication



Authentication

PSK Authentication



Client

No.	Time	SA	DA	Transmitter	Receiver	Channel	Info Long	Protocol
235	0.208718	Student01_AP	My_iPhone	Student01_AP	My_iPhone	48	Key (Message 1 of 4)	EAPOL
236	0.208753		Student01_AP		Student01_AP	48	Acknowledgement, Flags=....	802.11
237	0.213656	My_iPhone	Student01_AP	My_iPhone	Student01_AP	48	Key (Message 2 of 4)	EAPOL
238	0.213704		My_iPhone		My_iPhone	48	Acknowledgement, Flags=...	802.11
239	0.216035	Student01_AP	My_iPhone	Student01_AP	My_iPhone	48	Key (Message 3 of 4)	EAPOL
240	0.216082		Student01_AP		Student01_AP	48	Acknowledgement, Flags=....	802.11
241	0.217128	My_iPhone	Student01_AP	My_iPhone	Student01_AP	48	Key (Message 4 of 4)	EAPOL
242	0.217174		My_iPhone		My_iPhone	48	Acknowledgement, Flags=....	802.11



Access Point

DHCP

Discovery, Offer, Request, Acknowledge



dhcp

No.	Time	SA	DA	Transmitter	Receiver	Channel	Info Long	Protocol	Frame Type	SSID
219	*REF*	Student01_AP	Broadcast	Student01_AP	Broadcast	48	Beacon frame, SN=1275, FN...	802.11	Beacon frame	"Student01_VLAN10_5GHz"
305	0.272760	0.0.0.0	255.255.255.255	Student01_AP	Broadcast	48	DHCP Discover - Transaction ...	DHCP	Data	
364	0.792296	10.10.10.1	10.10.10.249	Student01_AP	My_iPhone	48	DHCP Offer - Transaction I...	DHCP	QoS Data	
396	1.798519	0.0.0.0	255.255.255.255	My_iPhone	Student01_AP	48	DHCP Request - Transaction ...	DHCP	QoS Data	
398	1.800006	0.0.0.0	255.255.255.255	Student01_AP	Broadcast	48	DHCP Request - Transaction ...	DHCP	Data	
399	1.802408	10.10.10.1	10.10.10.249	Student01_AP	My_iPhone	48	DHCP ACK - Transaction I...	DHCP	QoS Data	

Security

WPA2-Enterprise
WPA3
OWE

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Power Save

Target Wake Time [TWT]
Non-TIM [TIM Segmentation]
Restricted Access Windows [RAW]
Null Data Packet [NDP]
Short Beacons [10/sec]
One Beacon every second
AKM Optional in Beacon included in Probe Responses

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Other Advantages

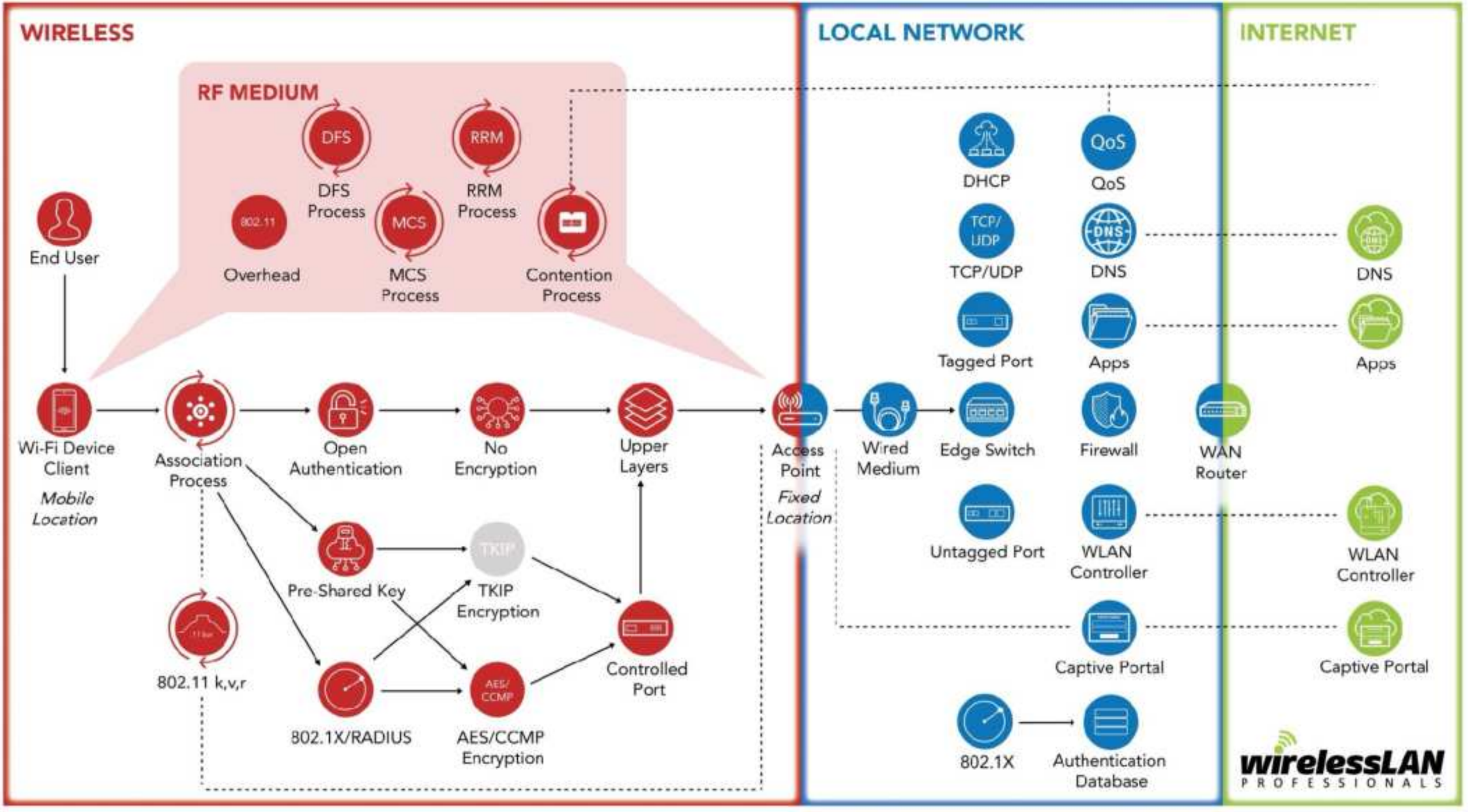
OTA firmware upgrades
Native IP support
8191 Clients [per BSS]
FILS

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Use Cases

Doorlocks
Motorola Door locks
Cameras
PtP or network extension
Signage
Home automation
Lighting
Amazon 802.11ah
Safety pit mining
Japan is opening 850 MHz [no duty cycle limitations]

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WIRELESS
LOCAL NETWORK
INTERNET
RF MEDIUM
DFS
DFS Process
RRM
RRM Process
802.11
Overhead
MCS
MCS Process
Contention Process
End User
Wi-Fi Device Client
Mobile Location
Association Process
Open Authentication
No Encryption
Upper Layers
Access Point
Fixed Location
Wired Medium
Edge Switch
Firewall
WAN Router
DHCP
QoS
TCP/UDP
DNS
Tagged Port
Apps
Untagged Port
WLAN Controller
Captive Portal
802.1X
Authentication Database
Pre-Shared Key
TKIP
TKIP Encryption
802.11 k,v,r
802.1X/RADIUS
AES/CCMP
AES/CCMP Encryption
Controlled Port
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Potential Wireless LAN Troubleshooting Causes

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	Wired/Wireless	Location	Potential Issues
1	**Wireless**	End User	Skills, Knowledge Perceptions, Device on/off, Understanding of Concepts & Device capabilities, Wi-Fi vs Cellular
2	*Mobile*	Wi-Fi Client Device	Drivers, Radio Capabilities, Profiles, Supported PHY, QoS, Power Save, Applications, Location, Vendor IE Support, Chipset Behavior, Roaming Algorithms, Auto-Negotiated MCS, MDM, Protection
3		RF Media	RSSI, SNR, SNiR, Primary & Secondary Coverage, CCI/ACI, Retry Rates, Average MCS, Jitter, Latency, Consistency, Regulatory Domains, Non-Wi-Fi Interference, Spectrum Analysis
4	Per Frame Tx	Contention Process	Preamble Detect, Energy Detect, Triggers, NAV Timers, TxOP, AIFS, Random Slots, QoS, WMM, Duration ID, Ch Capacity, Non-Wi-Fi Inteference
5	Per Frame Tx	MCS Process	Per Frame Decisions - Modulation Technique, Coding Technique, Ch Width, Guard Interval, Spatial Streams, Tx Power, ACK vs No ACK, TX decides
6	Per Time	Radio Resource Mgmt	Per Period Decisions - Channel, Tx Power, CCI, ACI, Noise, Duty Cycle, Retry Rates, CRC's, Load, Ch Width, DFS, User Traffic, Reg Domain, KPI's, Thresholds, Neighbor Discovery, Interference, Timing
7	Per Time	DFS Process	802.11 is NOT primary User - AP Scans for 60-Seconds, AP Enabled, Continuos Scanning, If RADAR detected, send CSA, Change to new CH, After 30-min can return, after 60-second scan
8	Per Frame Tx	Single Frame on RF	Overhead to delivery IP Payload - AIFS, CW, BPSK Preamble, RTS, SIFS, Preamble BPSK, CTS, SIFS, Preamble, Preamble VHT, Header MBR, Payload PHY rate, CRC, SIFS, Preamble, ACK
9	Per Timers	Association Process	Beacon, Probe Request, Probe Reponse, Authentication Request, Authentication Response, Association Request, Association Response, Decide on which AP by: RSSI, SNR, Auth Method, Encrypt Method, Channel Switch Announcement, Error Ratios, MCS/Data Rates Supported, Heuristics, Internal Lists, De-Authentication, Dis-Associate, 802.11 k, v, r, MBR, Proprietary Methods!
10		802.11 k, v, r	AP's try to influence the roaming decisions via 'standard' modes
11	Per Changes	Authentication Process	Open, Pre-Shared Key, 802.1X RADIUS, PSK includes Exchange of 4-Way Handshake to trigger Encryption Keys, 802.1X EAP Exchange, ending in 4-Way Handshake
12		Encrytion Process	None, TKIP, AES/CCMP, Punishment for using TKIP, Confusion with Wi-Fi Alliance naming - WPA2 PSK... is PSK-WPA2
13	From LAN	Upper Layers	DHCP, IP , DNS, VLAN, Subnet Mask, Default Gateway, Captive Portal
14		Controlled Port	AP Controls which 802.11 Frames can cross Wireless to Wired Boundary
15	*Fixed*	Access Point	Configurations, SSIDs, Minimum Basic Rates, Supported PHY Rates, Band Steering, Client Control, Radio Capabilities, Tx Rates, Client Isolation, Roaming, QOS PoE, Antenna Pattern, Mounting, 1GB backhaul limit, AP Locations, Physcial Layer Issues, Firmware Revisions, Custom Configurations, RRM/ARM, Proprietary
16	**Local Network**	Wired Medium	EIA/TIA 568A/B, Category Mismatch, Validation Tests, Grounding, other issues
17		Edge Switch	VLANs, Port Speeds, PoE, Configurations, QoS, End-to-End?, COS vs DSCP
18		Local Network	Distributed vs Centralized Forwarding, ACLs, VLANs, QoS, Tunnels, Layers, NAT
19		TCP/UDP	Following all TCP issues as well as UDP reasons for using each
20		Quality of Service	Tagged Port vs Untagged Port, DSCP, WMM Categories, End-to-End QoS
21		Applicatons	MTU, TCP Window, Round Trip Time, Processing Time, TCP Retransmission times
22		DHCP Server	Lease Durations, Configurations, Broadcast Storms, Latency, Performance, Address Pool Scopes, Scalability, DHCP Options, Auto Renew
23		DNS	Configuration, Scalability, Security, Accuracy, Customization, Control, Blacklists
24		802.1X/RADIUS	Configuration, Ports, Ranges, Licensing Issues, EAP types, Custom VSA, Scalability, Resouces, Certificate Issues, Fast/Secure Roaming types
25		Active Directory	Accounts, Credentials, EAP Compatibility, Custom RADIUS Attributes
26		Controller Functions	Code Versions, Bugs, Configurations, Local vs Cloud, Licensing Issues, Distributed vs Centralized Forwarding, VLAN choices
27		Firewall	Firewall Rules, Capacity, Compatibility, Rate Limiting, Bandwidth Shaping
28		WAN Router	Size of Internet Pipe, Inernet Destination Issues, Costs, Availability, Consistency
29	**Internet**	Internet Connection	Bandwidth Throttling, Jitter, Latency
30		Captive Portal	Security, Client Issues, Privacy, Friction, Triggers, Certificates, DNS, Captive Portal Location, Control, Monetization, Legal, MiFi

	Items to Review and Report	2.4GHz		5GHz		6GHz	
1	Captive Web Portal Use	Use only if required		Use only if required		Use only if required	
2	1, 6, 11 Only	Ch 1, 6, 11 Only		Not Applicable in 5GHz		Not Applicable in 6GHz	
3	20 MHz Channels	20MHz Channels Only		Use		Use	
4	40 MHz Channels	No 40MHz Ch		Check CCC		Use	
5	80 MHz Channels	Not Available in 2.4GHz		Not Recommended		Check CCC	
6	Interference - Co-Channel	>-85dBm		>-85dBm		>-85dBm	
7	OBSS - Consistent Channel Bonding	Do NOT bond channels!		High Side Consistent		High Side Consistent	
8	Channels - DFS in Use	Not Applicable		Use & Review		Not Applicable in 6GHz	
9	Minimum Basic Rates - Trim	MBR>=12Mbps, No b		MBR>=12Mbps		MBR>=12Mbps	
10	SSID's - Across Bands	Have 2.4GHz-only SSIDs		Use w/RNR		Use w/RNR	
11	Signal - Primary RSSI	-67dBm		-65dBm		-65dBm	
12	Signal - Secondary RSSI *^*	-67dBm		-65dBm		-65dBm	
13	AP Transmit Power *^*	7dBm Target		13dBm Target		16dBm Target	
14	WPA 2/3 Transition *^*	Not Recommended		Not Recommended		Not Allowed	
15	WPA 3	Recommended		Recommended		Required	
16	802.11 k - Neighbor Report	Recommended		Recommended		Recommended	
17	802.11 r - Fast Transition *^*	Recommended		Recommended		Recommended	
18	802.11 v - Transition Management	Recommended		Recommended		Recommended	
19	Channels - Utilization/BSS Load *^*	Target <40%		Target <20%		Target <20%	
20	Clients per Radio - BSS Load	<40		<40		<40	
		2.4 GHz Results	0%	5 GHz Results	0%	6 GHz Results	0%

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How it Works

Survey


Treklean




FISHERMAN FD1+





Morse Micro

wavelify.com/halow/site-maps/?mode=survey&folder=Nq6X5zJHCnxR5YTZTtAx&name=E...

887 days left in your Business trial

Choose a Plan

HaLow Site Maps(900 MHz)

Maps | Surveys

2 Maps

Search maps...

Date Last Opened

Import | Archived | + Create Map



Enteranet - HaLow Drone

1 floor · 0 APs

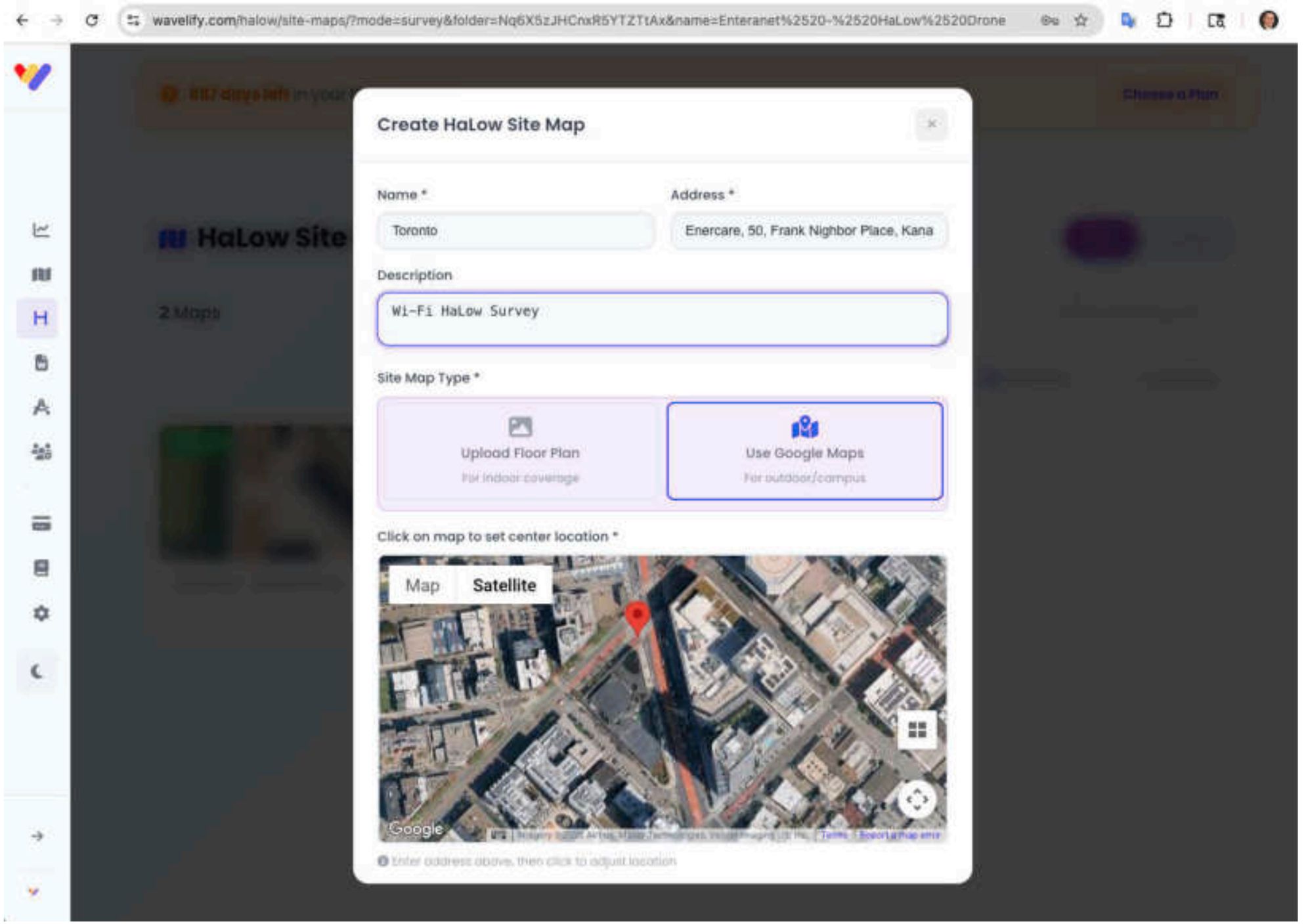


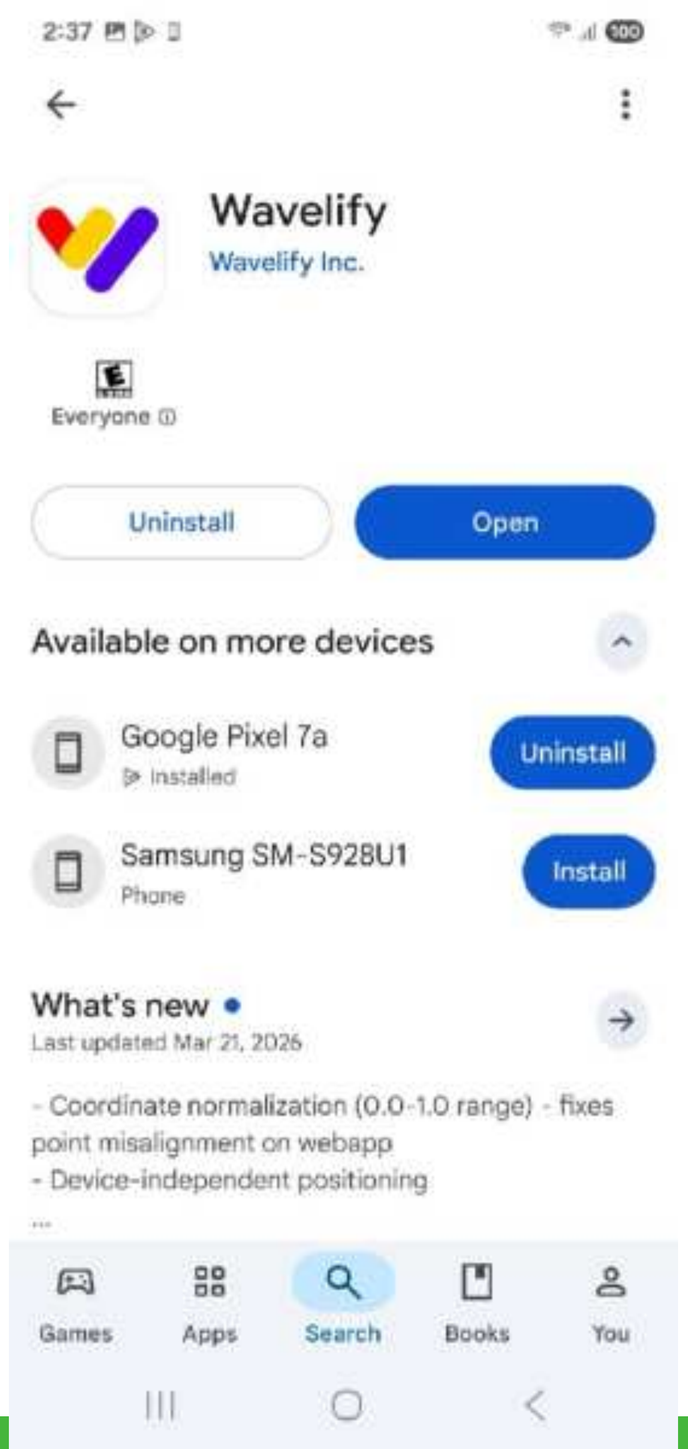
Enteranet HaLow

1 floor · 0 APs

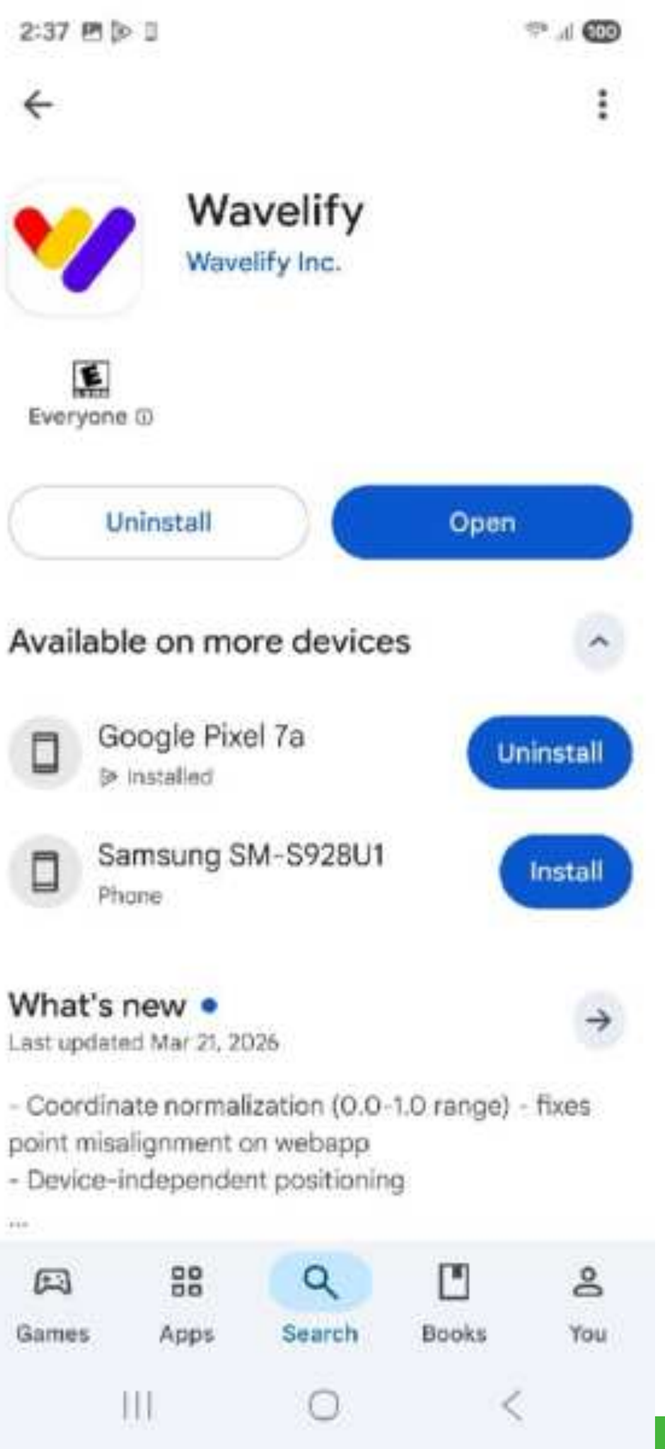
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avelify


wavelify.com/halow/site-maps/?mode=survey&folder=Nq6X5zJHCrxR5YTZTtAx&name=Enteranet%2520-%2520HaLow%2520Drone
HaLow Site
2 Maps
Create HaLow Site Map
Name *
Toronto
Address *
Enercare, 50, Frank Nighbor Place, Kana
Description
Wi-Fi HaLow Survey
Site Map Type *
Upload Floor Plan
For indoor coverage
Use Google Maps
For outdoor/campus
Click on map to set center location *
Map
Satellite
Google
Enter address above, then click to adjust location

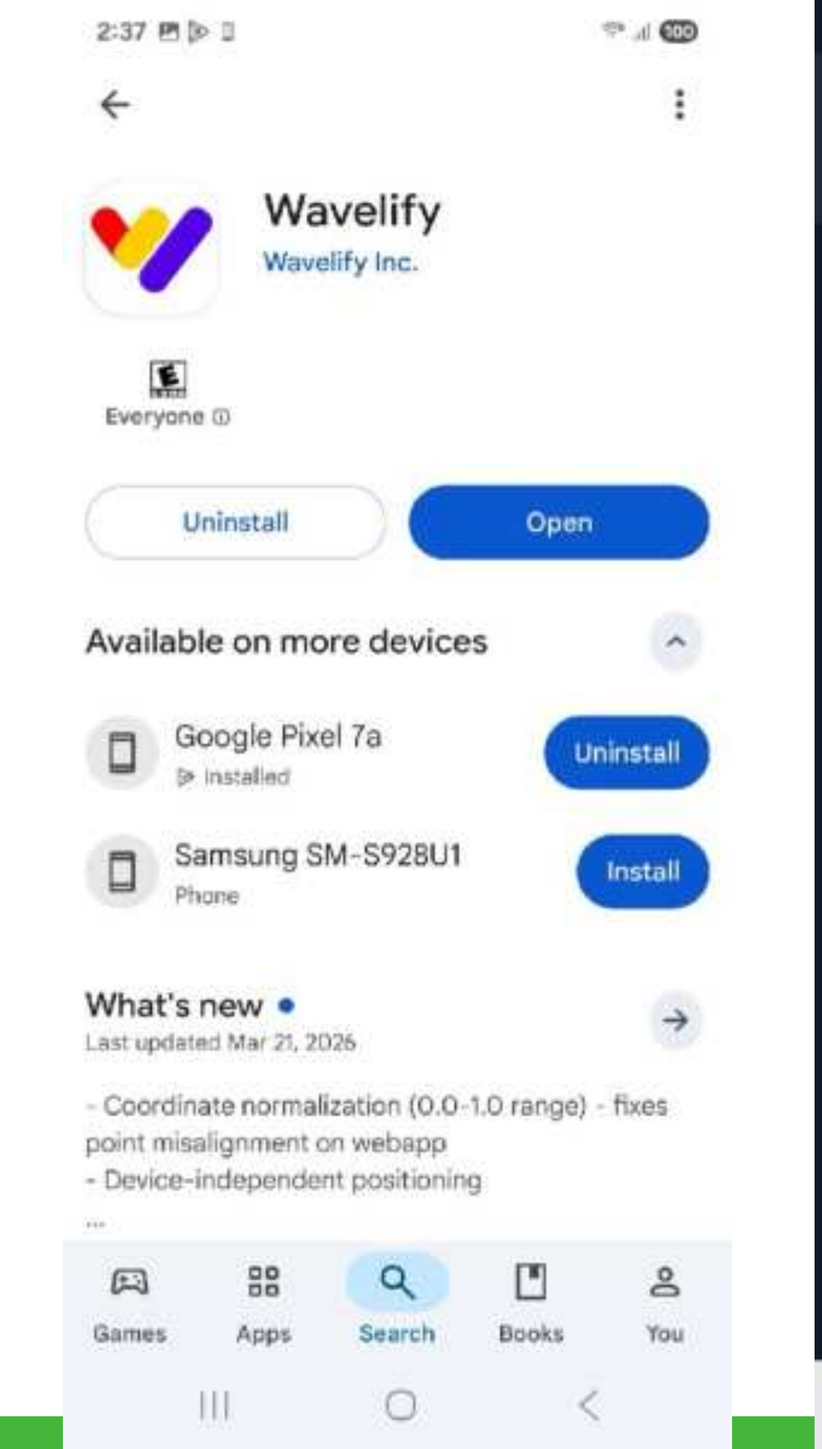


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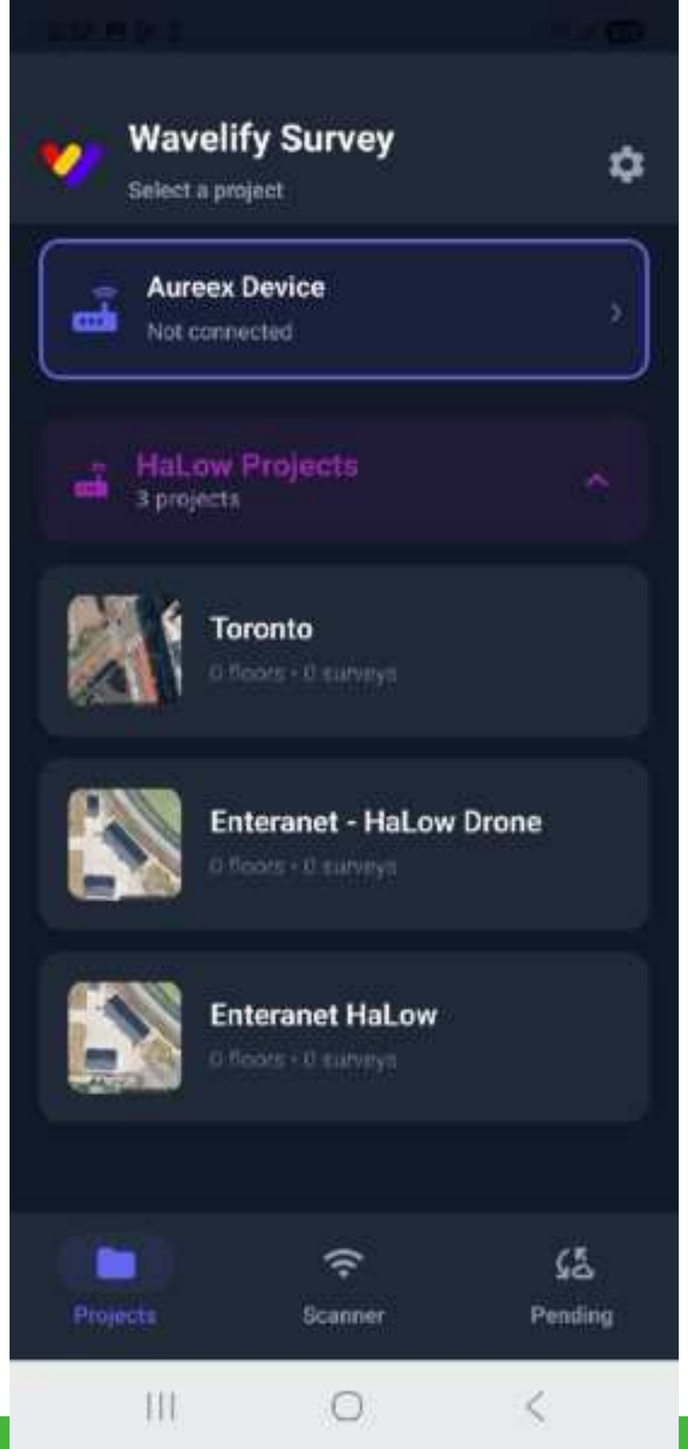

2:37
Wavelify
Wavelify Inc.
Everyone
Uninstall
Open
Available on more devices
Google Pixel 7a
Installed
Uninstall
Samsung SM-S928U1
Phone
Install
What's new
Last updated Mar 21, 2026
- Coordinate normalization (0.0-1.0 range) - fixes point misalignment on webapp
- Device-independent positioning
...
Games
Apps
Search
Books
You


Select Device
Select Device
Android
WLANPi
Aureex
Using Phone Scanner
Your phone's built-in WiFi scanner is ready to use for surveys.


Select Device
Select Device
Android
WLANPi
Aureex
WLANPi Connection
Connection Type
Bluetooth
Ethernet/USB
WLANPi via Bluetooth
Not connected
Connect via Bluetooth

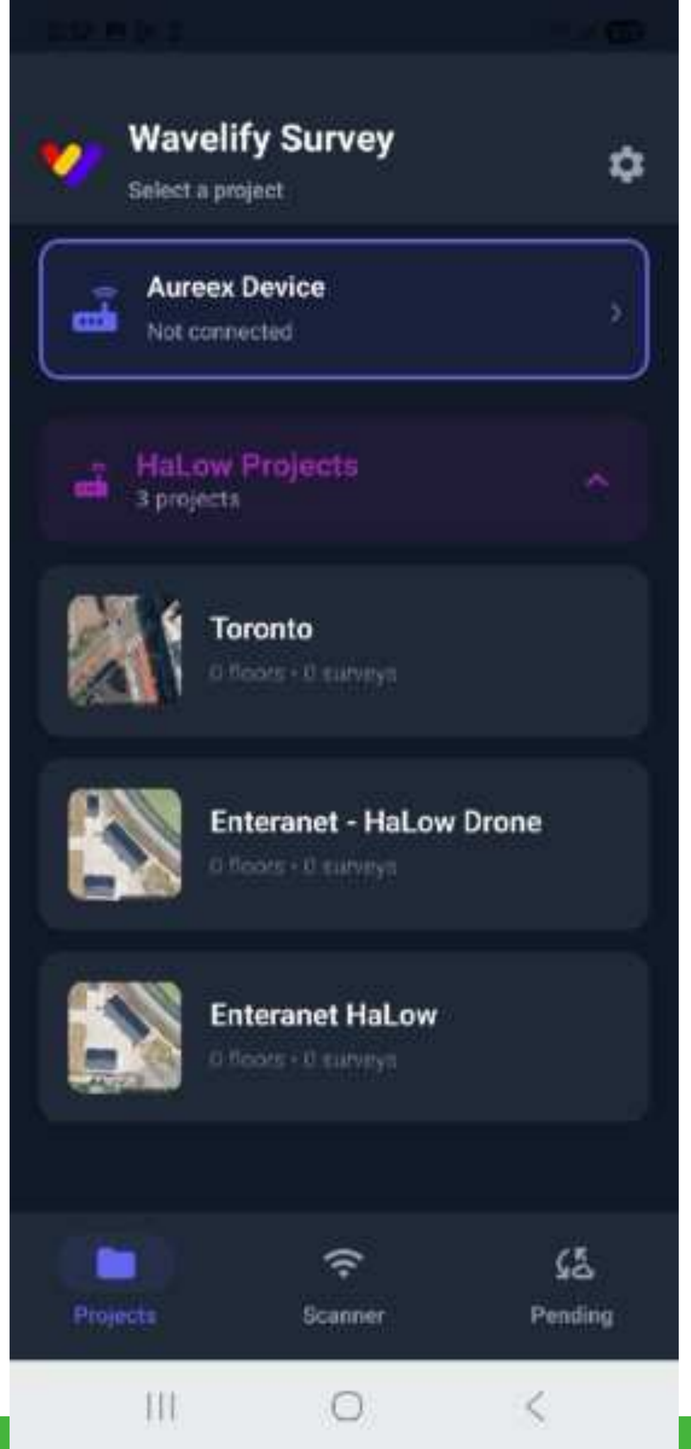

Select Device
Select Device
Android
WLANPi
Aureex
Aureex Connection
Connection Type
Bluetooth
Ethernet
Aureex via Ethernet
Not connected
IP Address
192.168.2.2
Port
8080
Connect via Ethernet

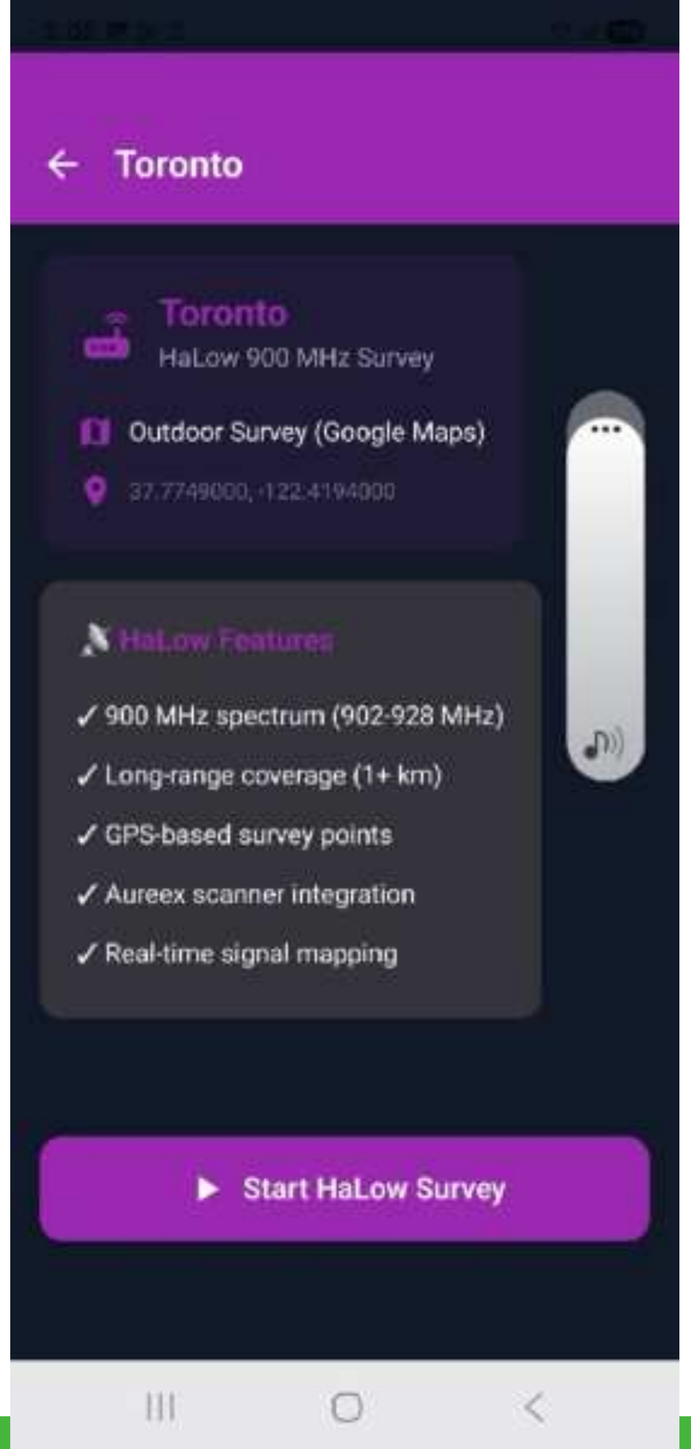




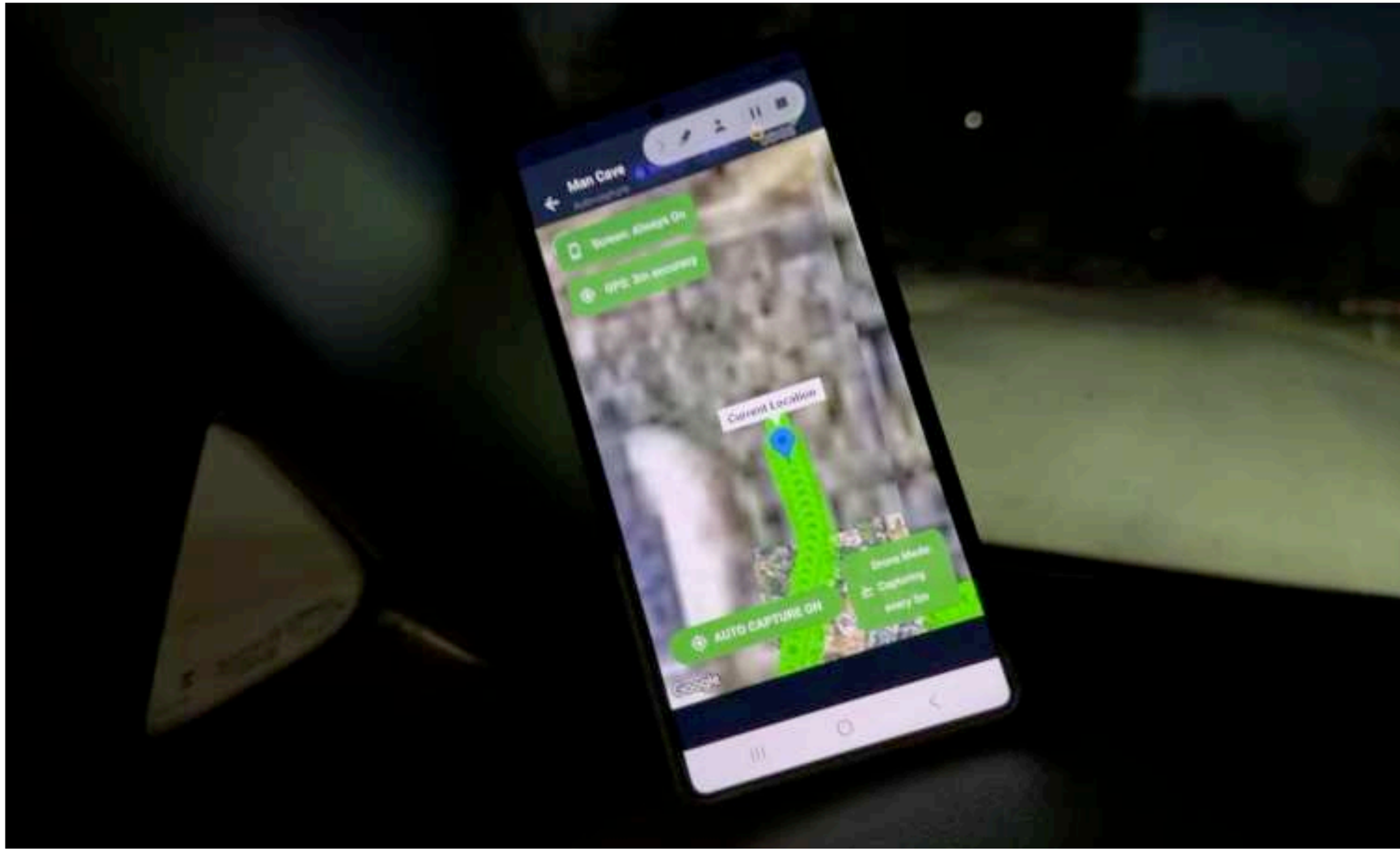
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Select Device
Select Device
Android
WLANPi
Aureex
Aureex Connection
Connection Type
Bluetooth
Ethernet
Aureex via Ethernet
Not connected
IP Address
192.168.2.2
Port
8080
Connect via Ethernet

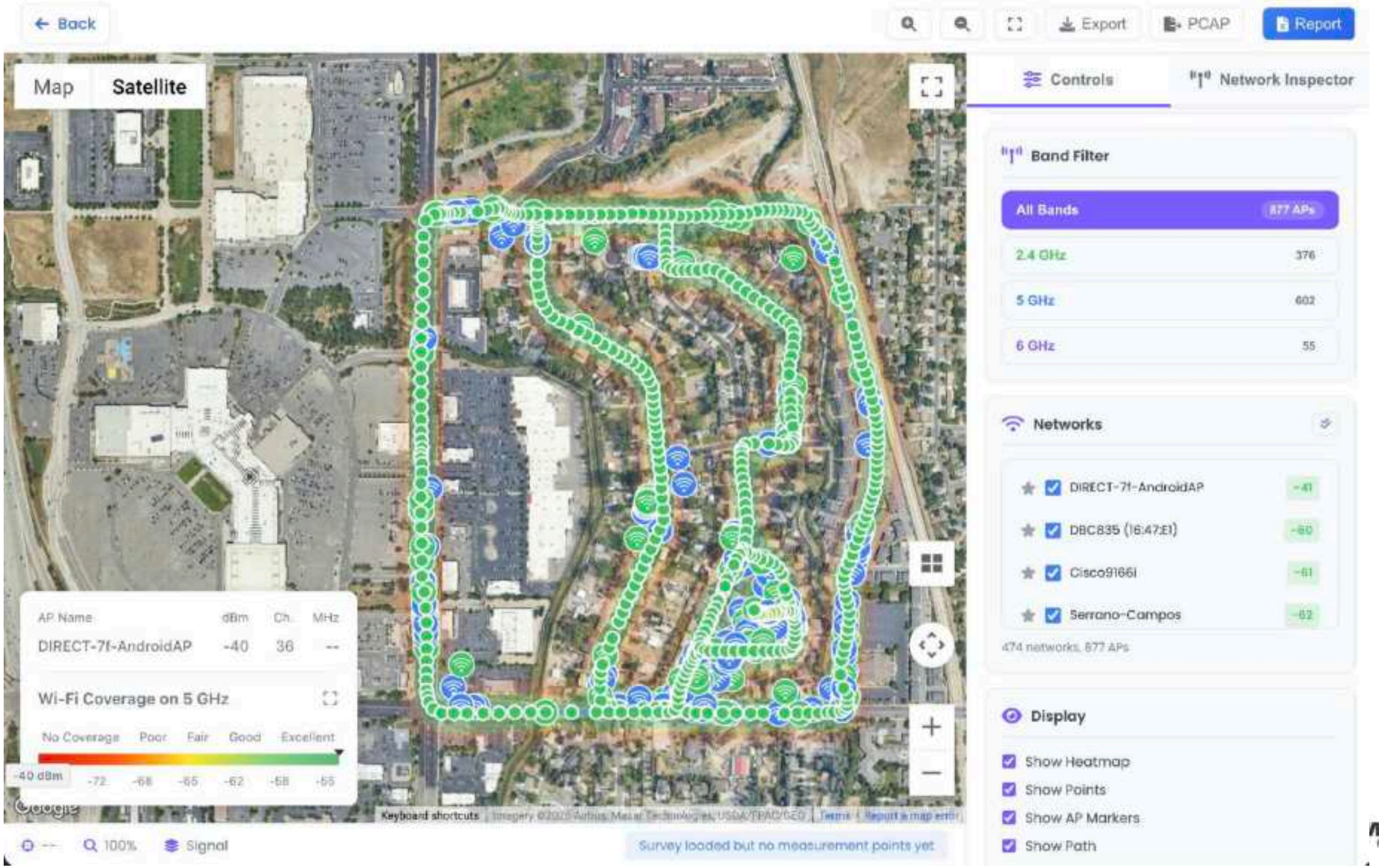

Wavelify Survey
Select a project
Aureex Device
Not connected
HaLow Projects
3 projects
Toronto
Enteranet - HaLow Drone
Enteranet HaLow
Projects
Scanner
Pending


Toronto
Toronto
HaLow 900 MHz Survey
Outdoor Survey (Google Maps)
37.7749000, -122.4194000
HaLow Features
✓ 900 MHz spectrum (902-928 MHz)
✓ Long-range coverage (1+ km)
✓ GPS-based survey points
✓ Aureex scanner integration
✓ Real-time signal mapping
Start HaLow Survey





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Back
Export
PCAP
Report
Map
Satellite
Controls
Network Inspector
Band Filter
All Bands
877 APs
2.4 GHz
376
5 GHz
602
6 GHz
55
Networks
DIRECT-7f-AndroidAP
DBC835 (16:47:E1)
Cisco9166I
Serrano-Campos
474 networks, 877 APs
Display
Show Heatmap
Show Points
Show AP Markers
Show Path
AP Name
dBm
Ch
MHz
DIRECT-7f-AndroidAP
-40
36
--
Wi-Fi Coverage on 5 GHz
No Coverage
Poor
Fair
Good
Excellent
-40 dBm
-72
-68
-65
-62
-58
-55
Keyboard shortcuts
Terms
Report a map error
100%
Signal
Survey loaded but no measurement points yet



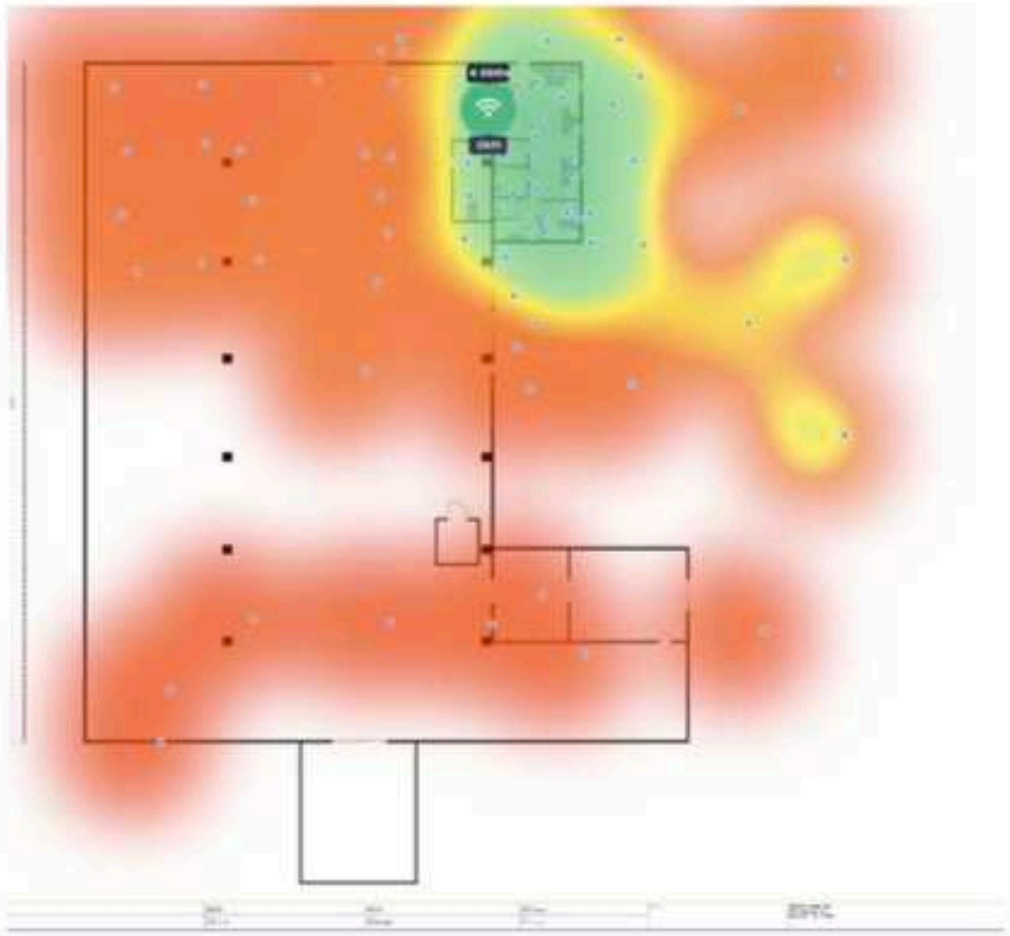
AP01 2.4 GHz

Channel: 11, Tx Power: N/A



-85 dBm | -65 | ≥ -30 dBm

AP01 2.4 GHz Android client view [Wavelify]

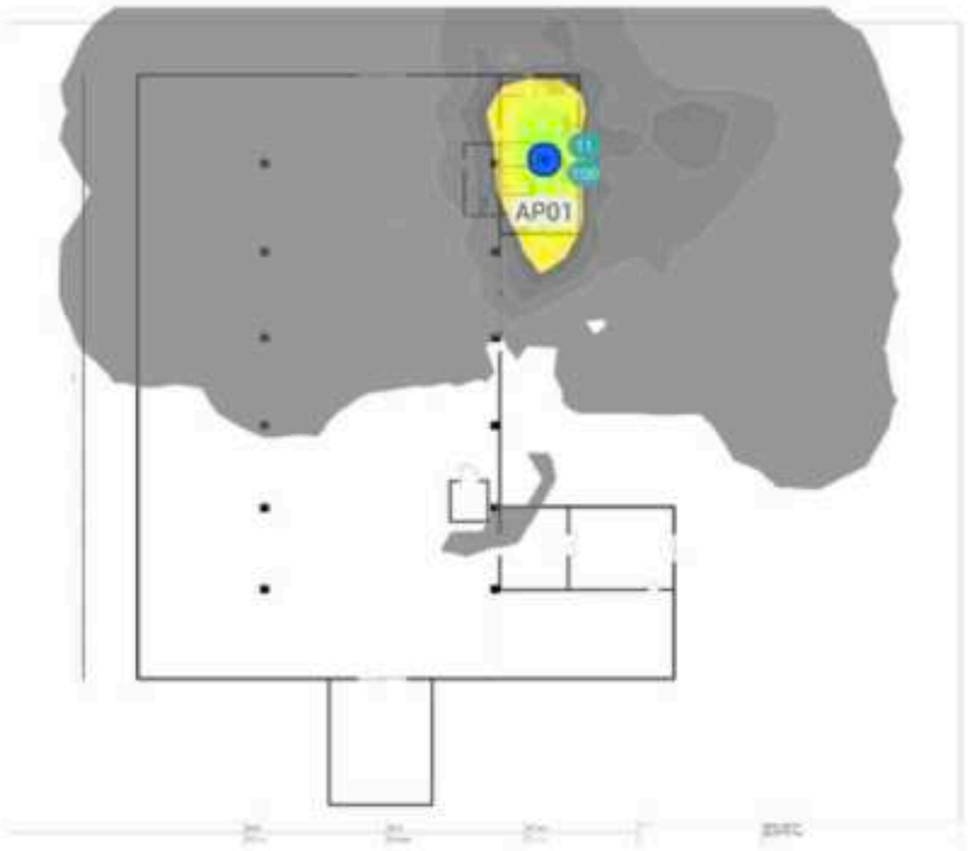


No Coverage Poor Fair Good Excellent

-75 -72 -68 -65 -62 -58 -55

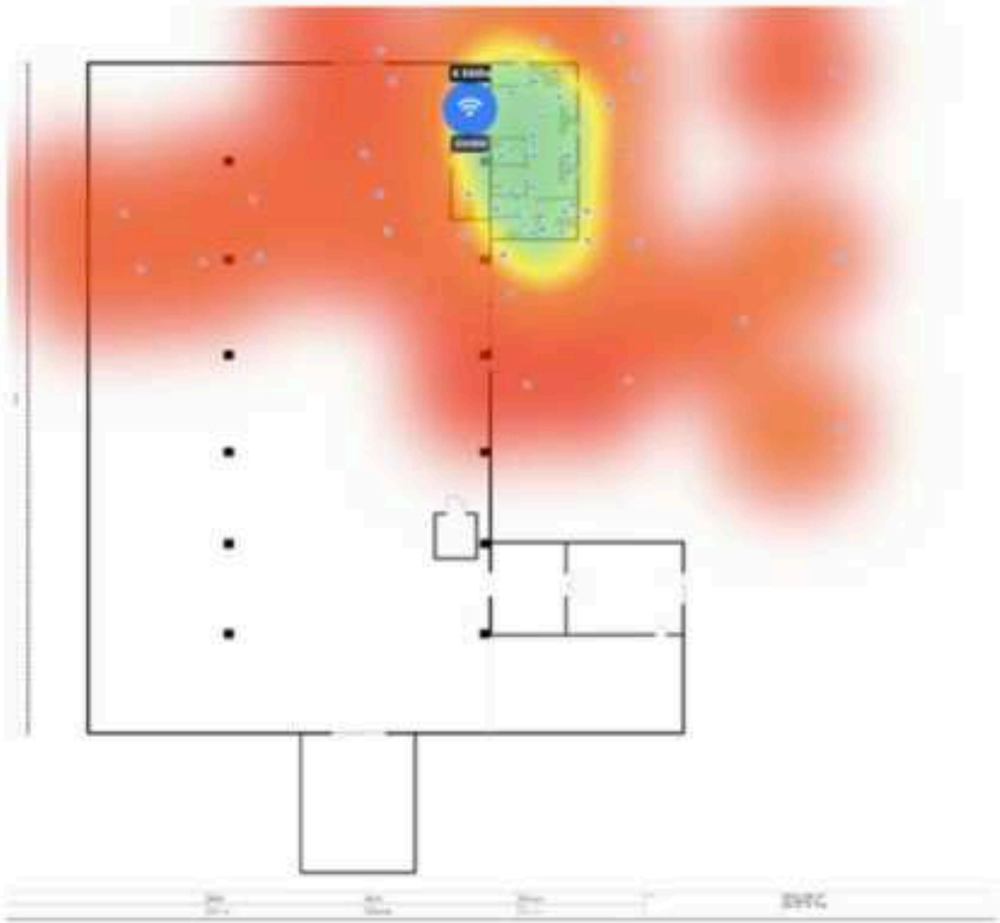
AP01 5 GHz

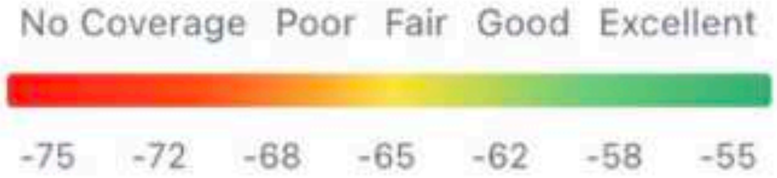
Channel: 100, Tx Power: N/A





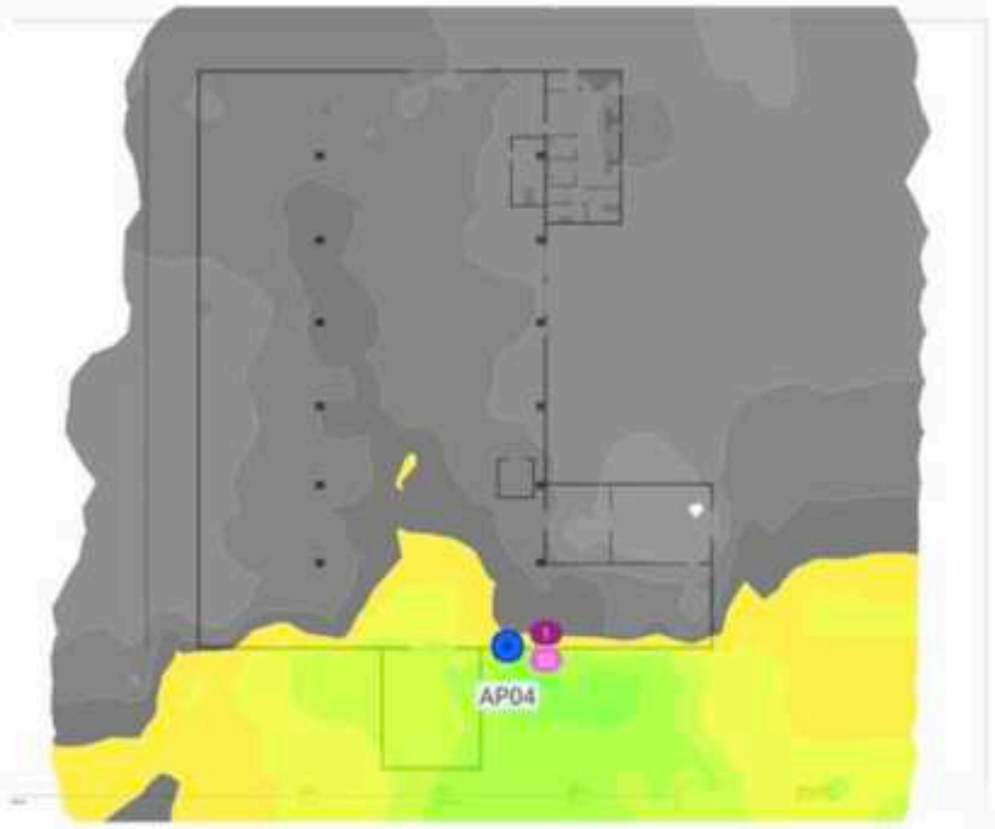
AP01 5 GHz Android client view [Wavelify]





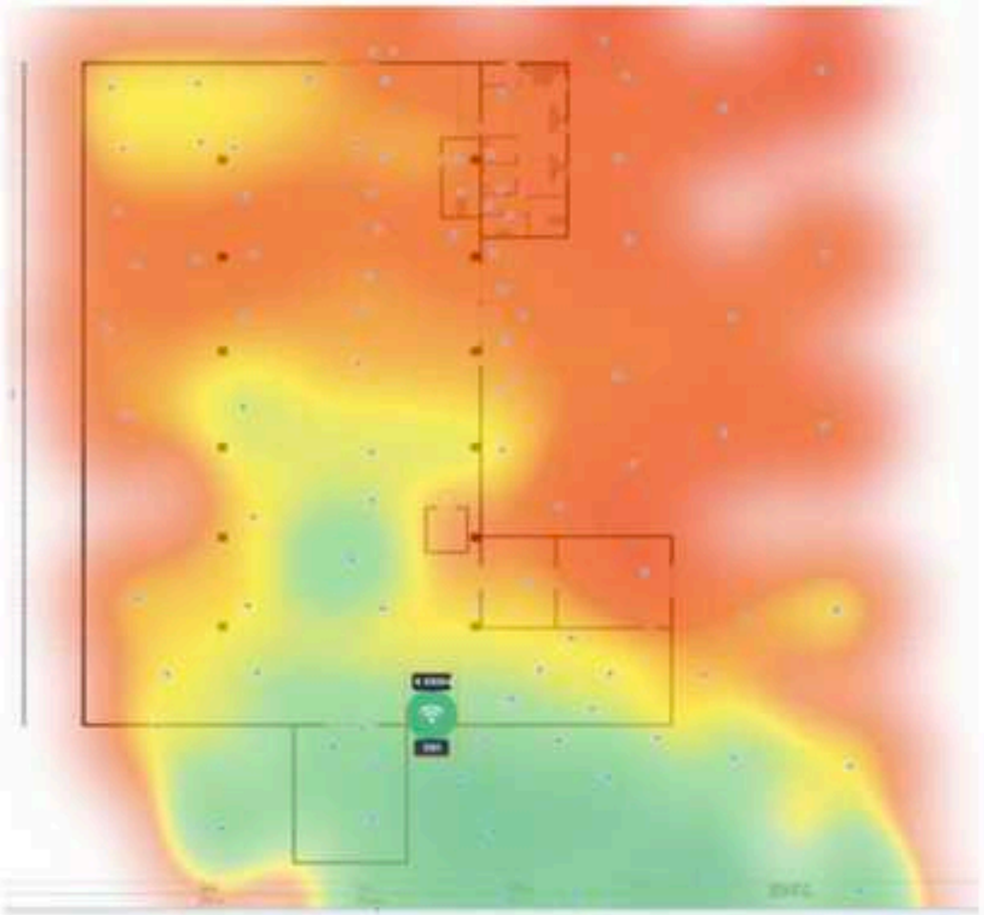
AP04 2.4 GHz

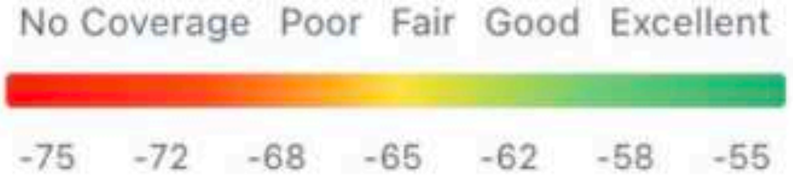
Channel: 1, Tx Power: N/A





AP04 2.4 GHz Android client view [Wavelify]





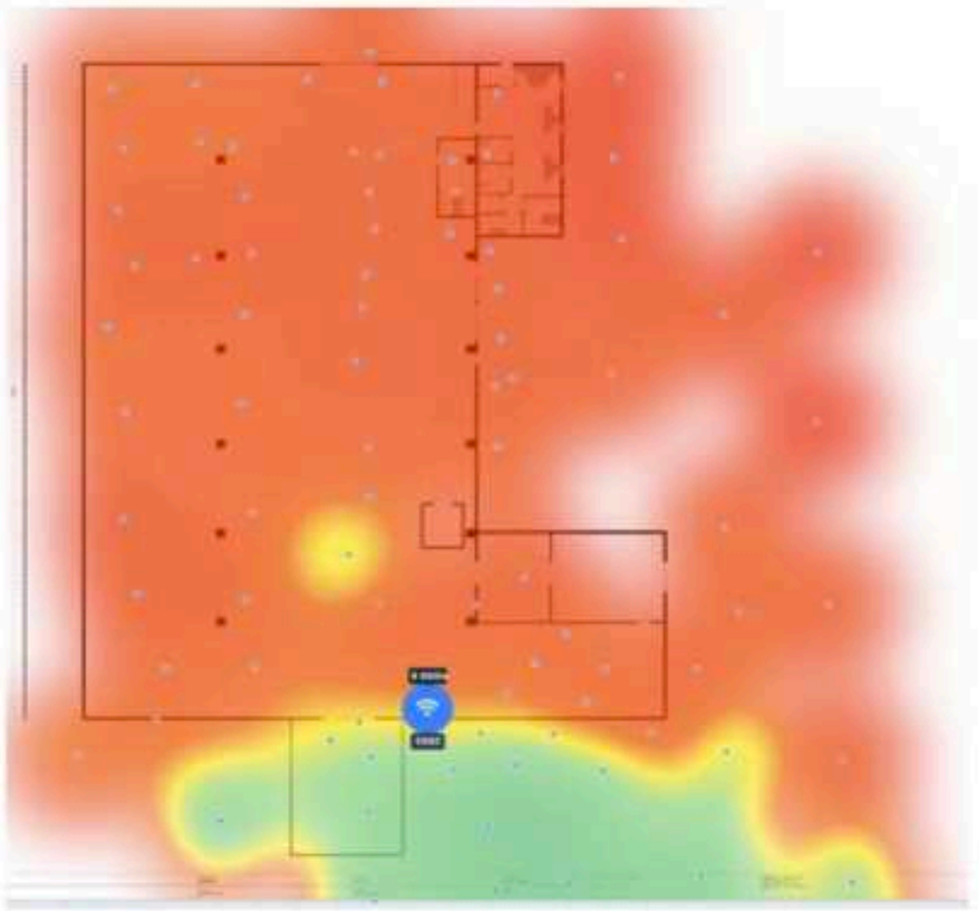
AP04 5 GHz

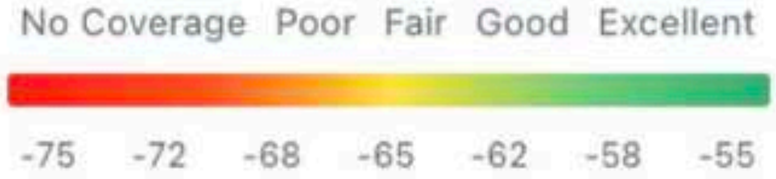
Channel: 52, Tx Power: N/A





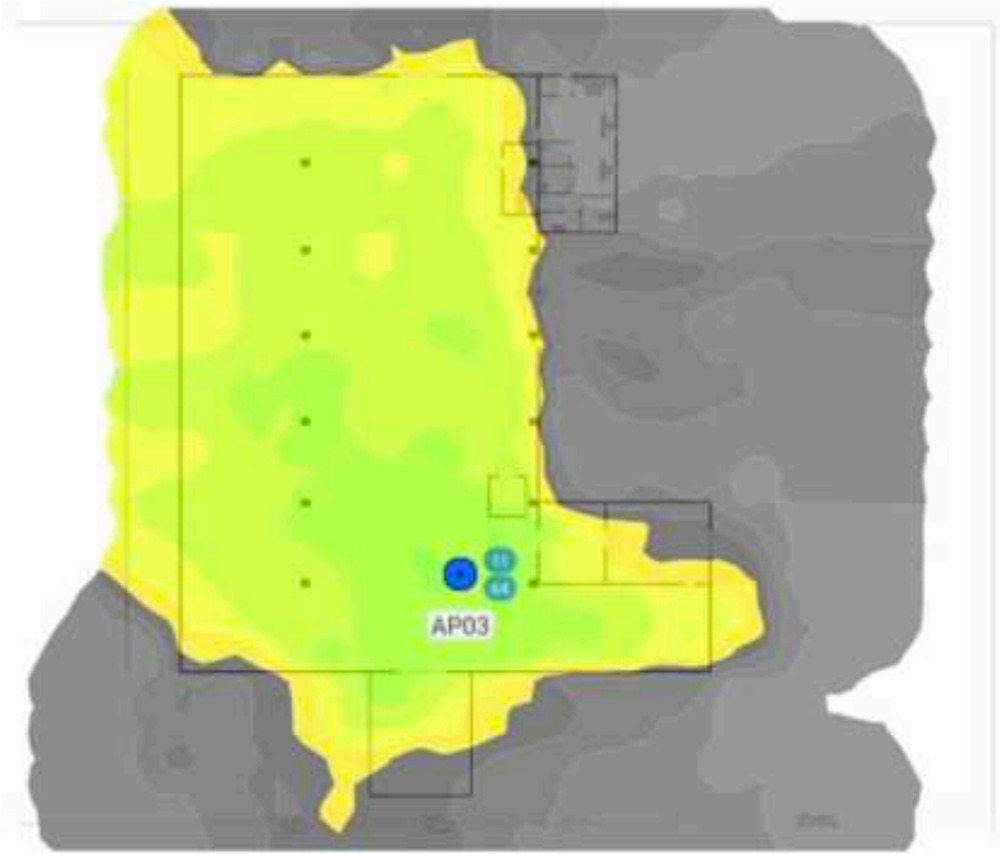
AP04 5 GHz Android client view [Wavelify]



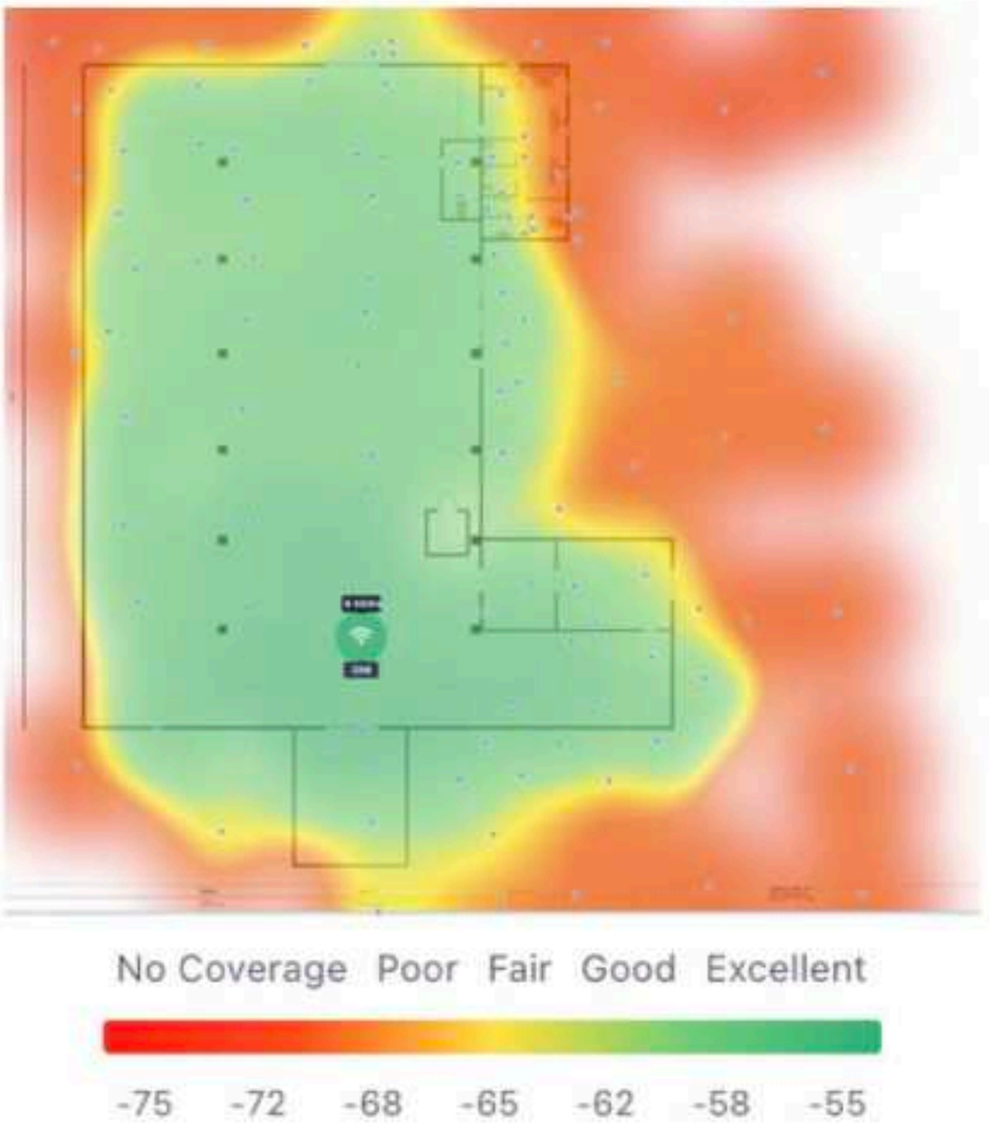


AP03 2.4 GHz

Channel: 11, Tx Power: N/A

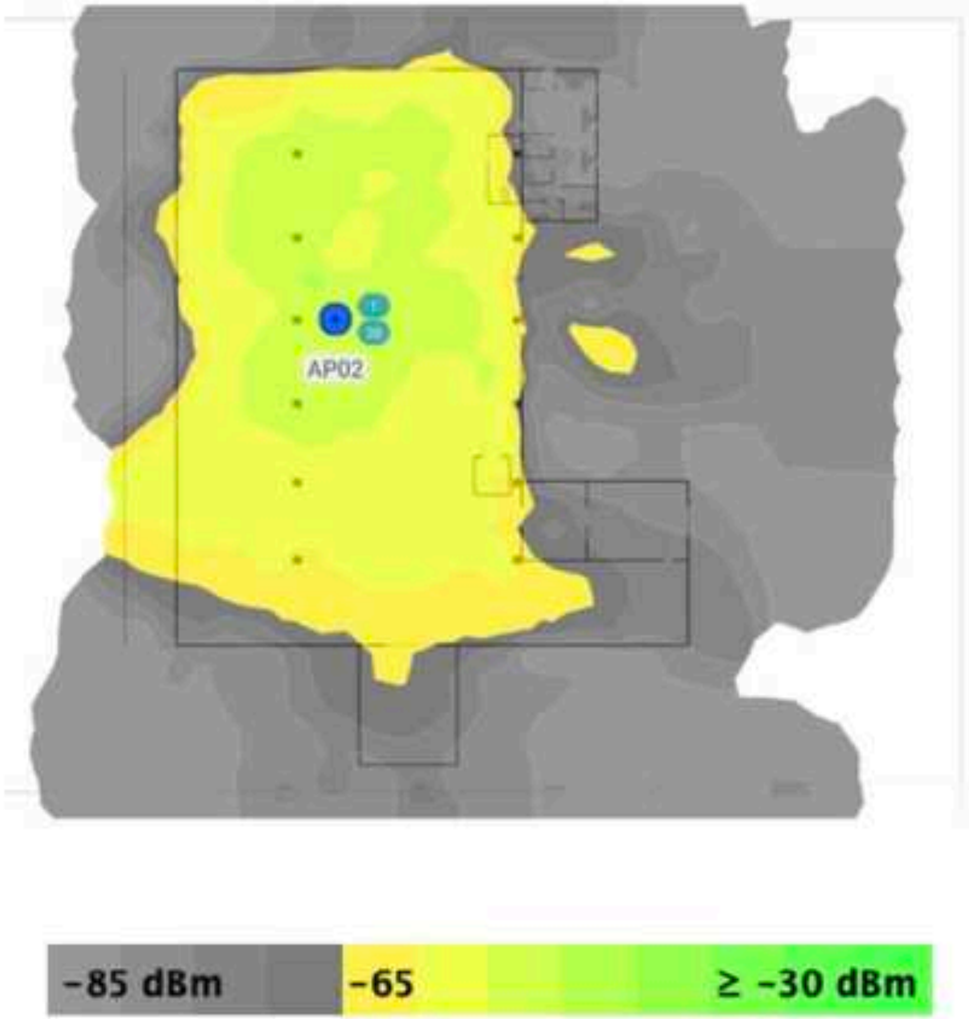


AP03 2.4 GHz Android client view [Wavelify]

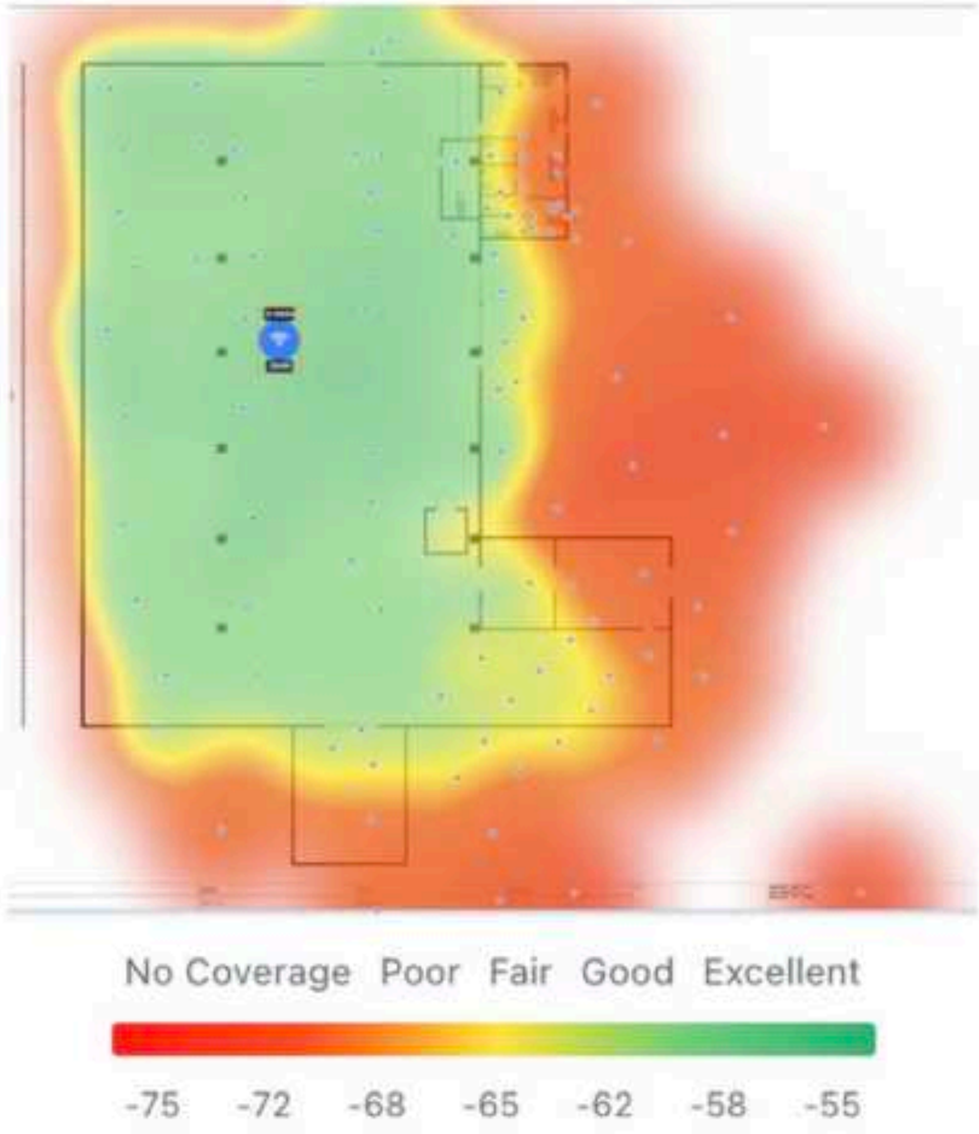


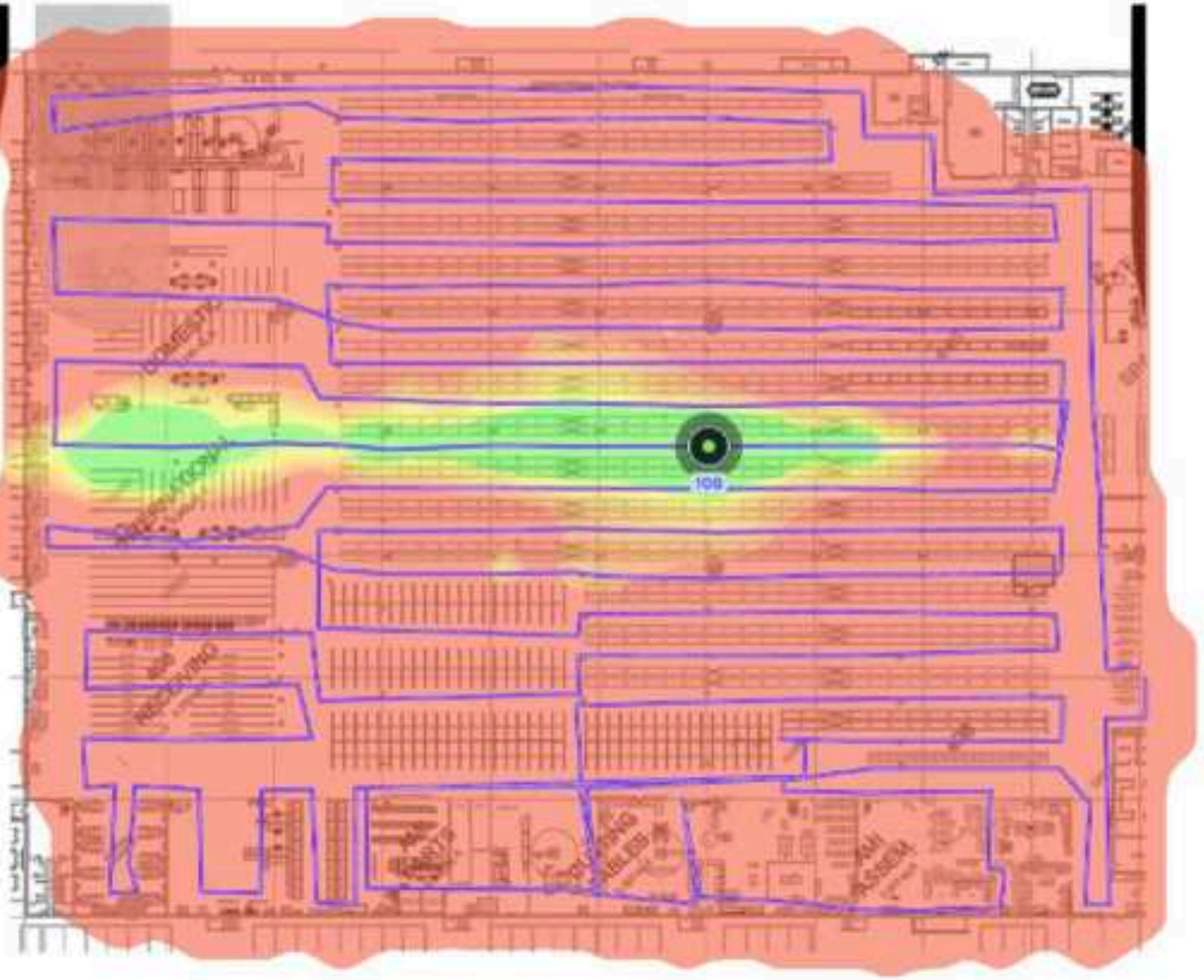
AP02 5 GHz

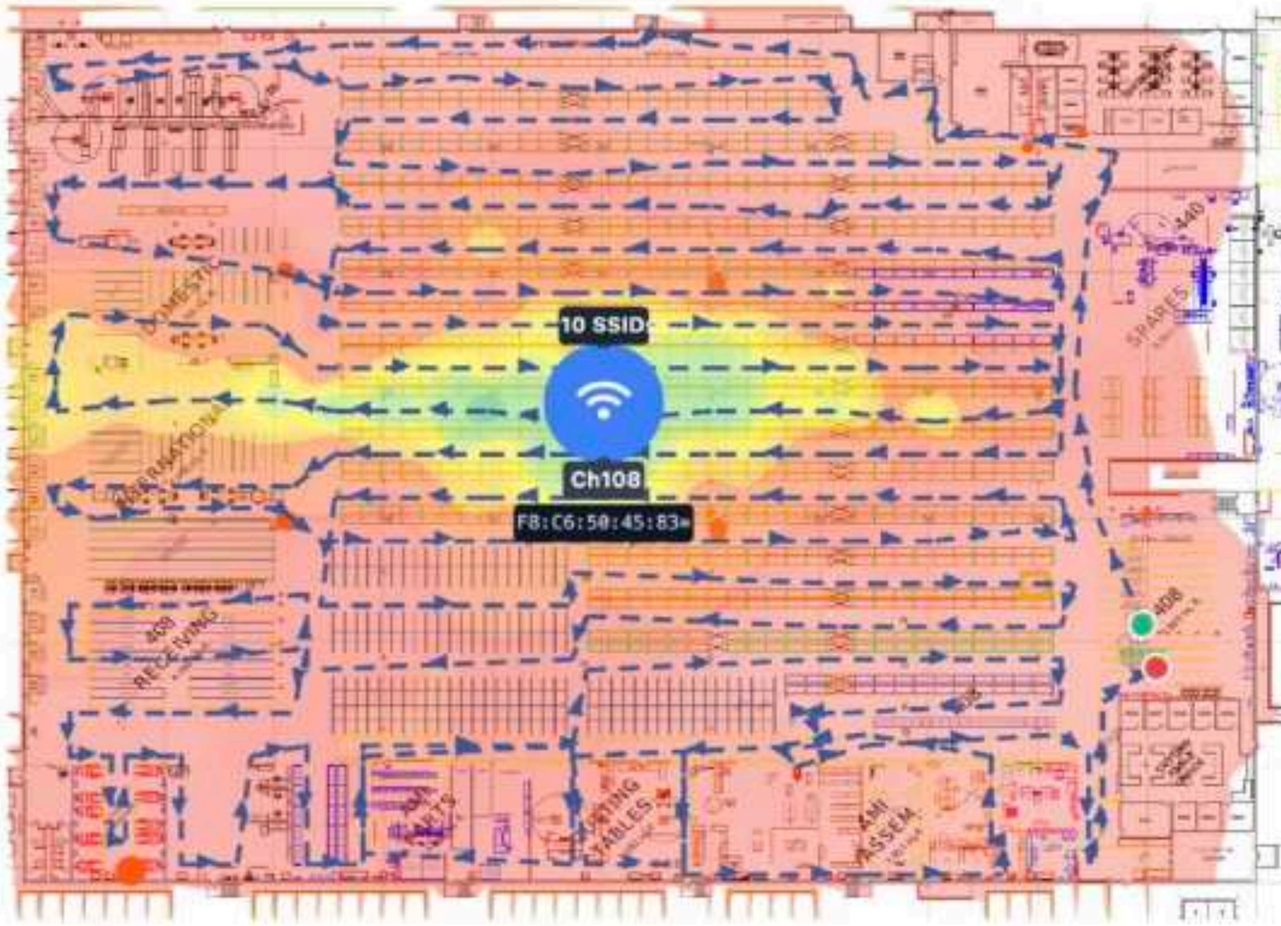
Channel: 36, Tx Power: N/A



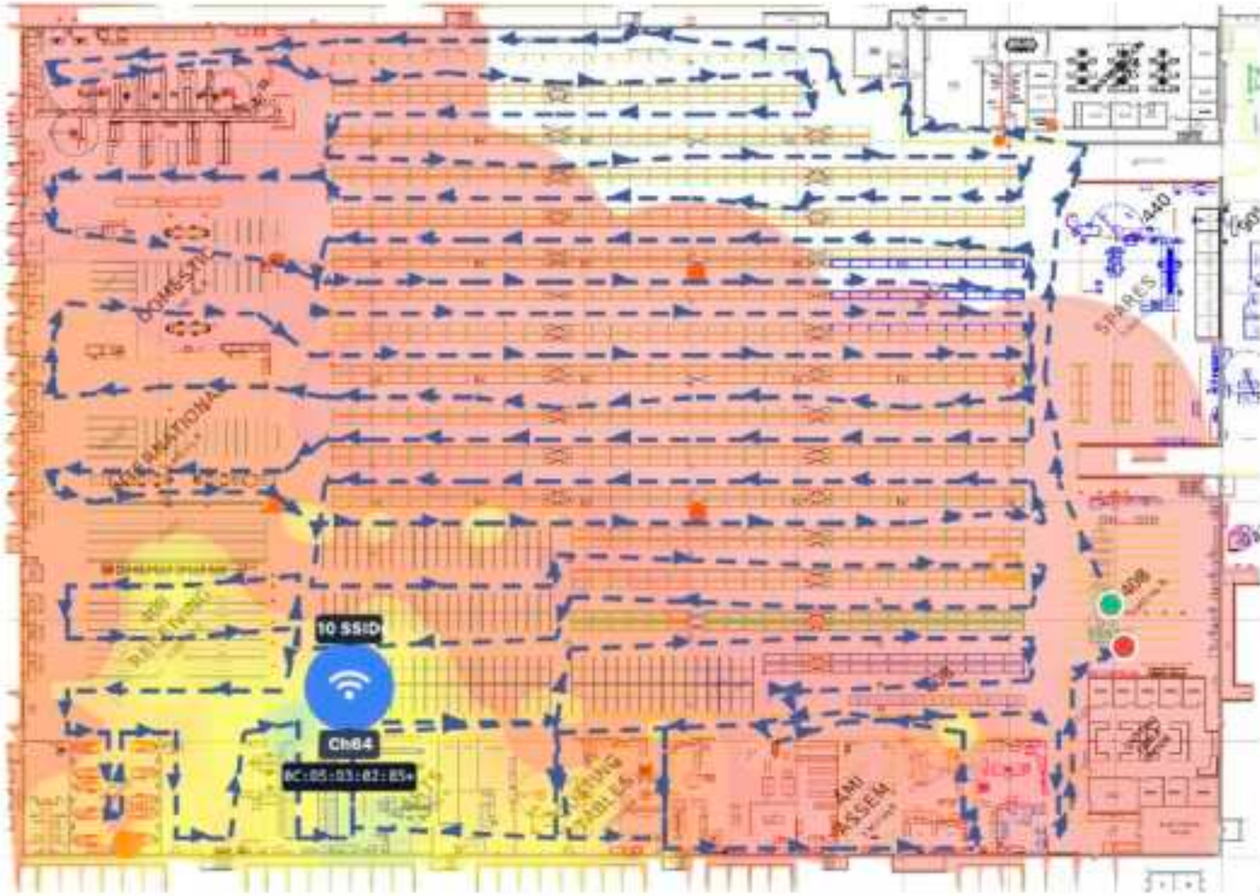
AP02 5 GHz Android client view [Wavelify]




108

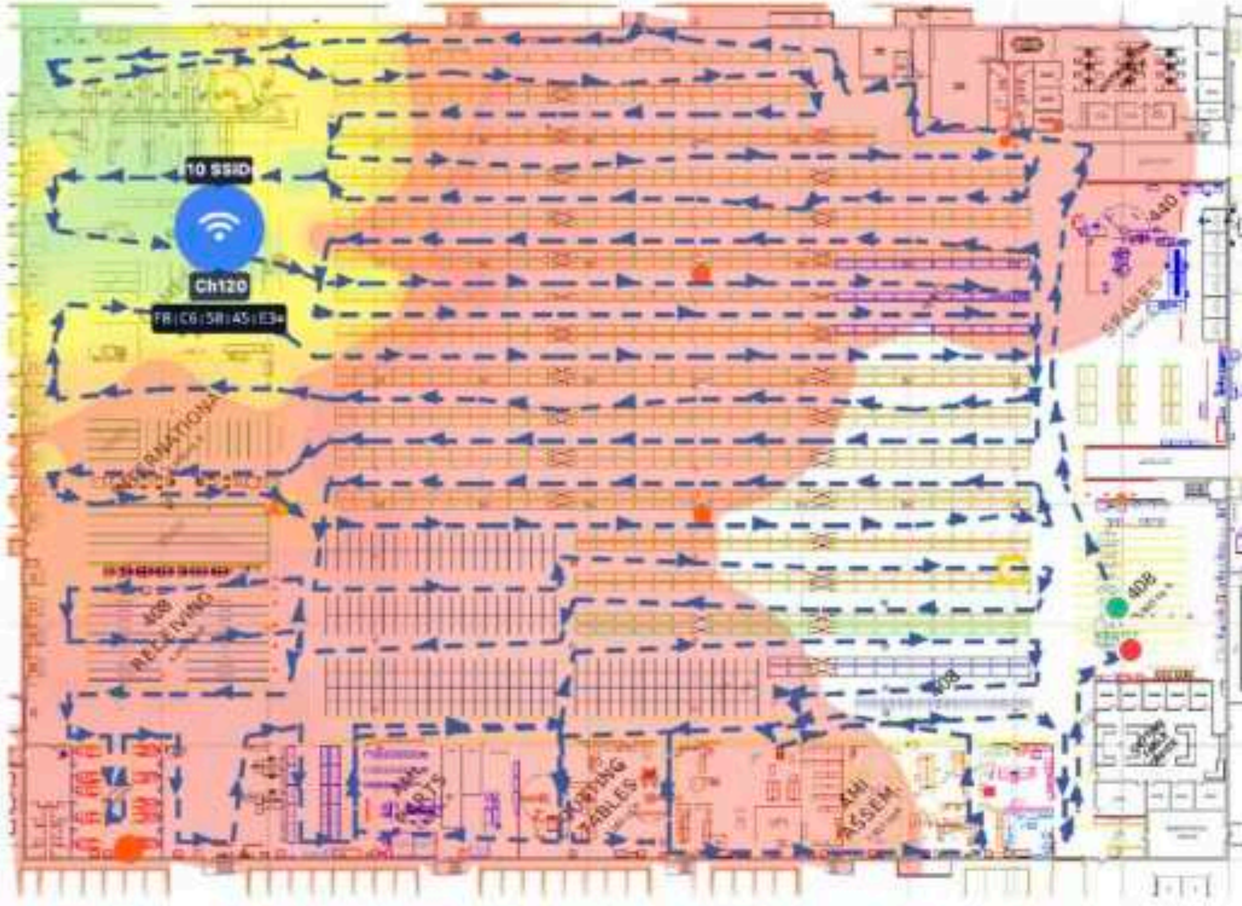

10 SSID
Ch108
F8:C6:50:45:83:



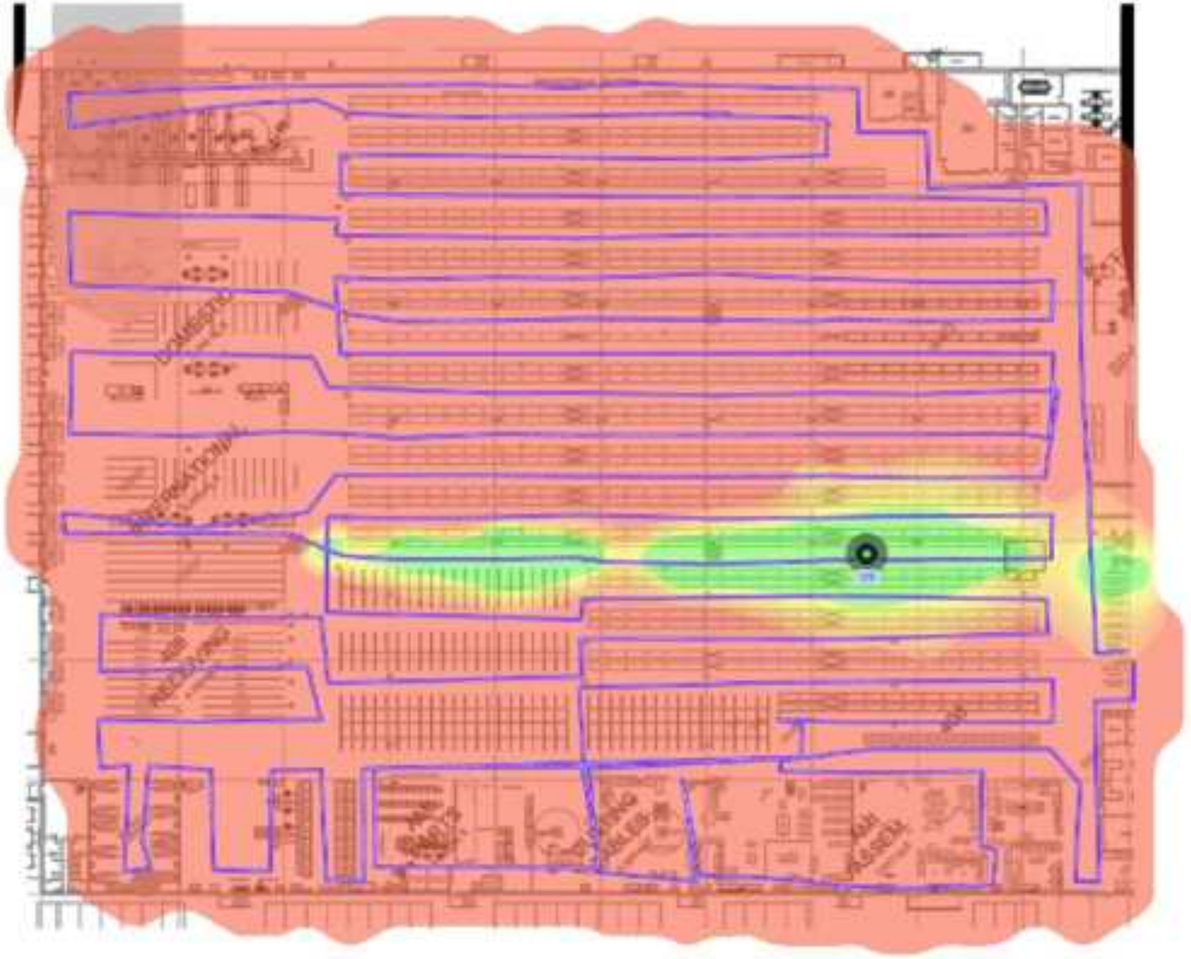


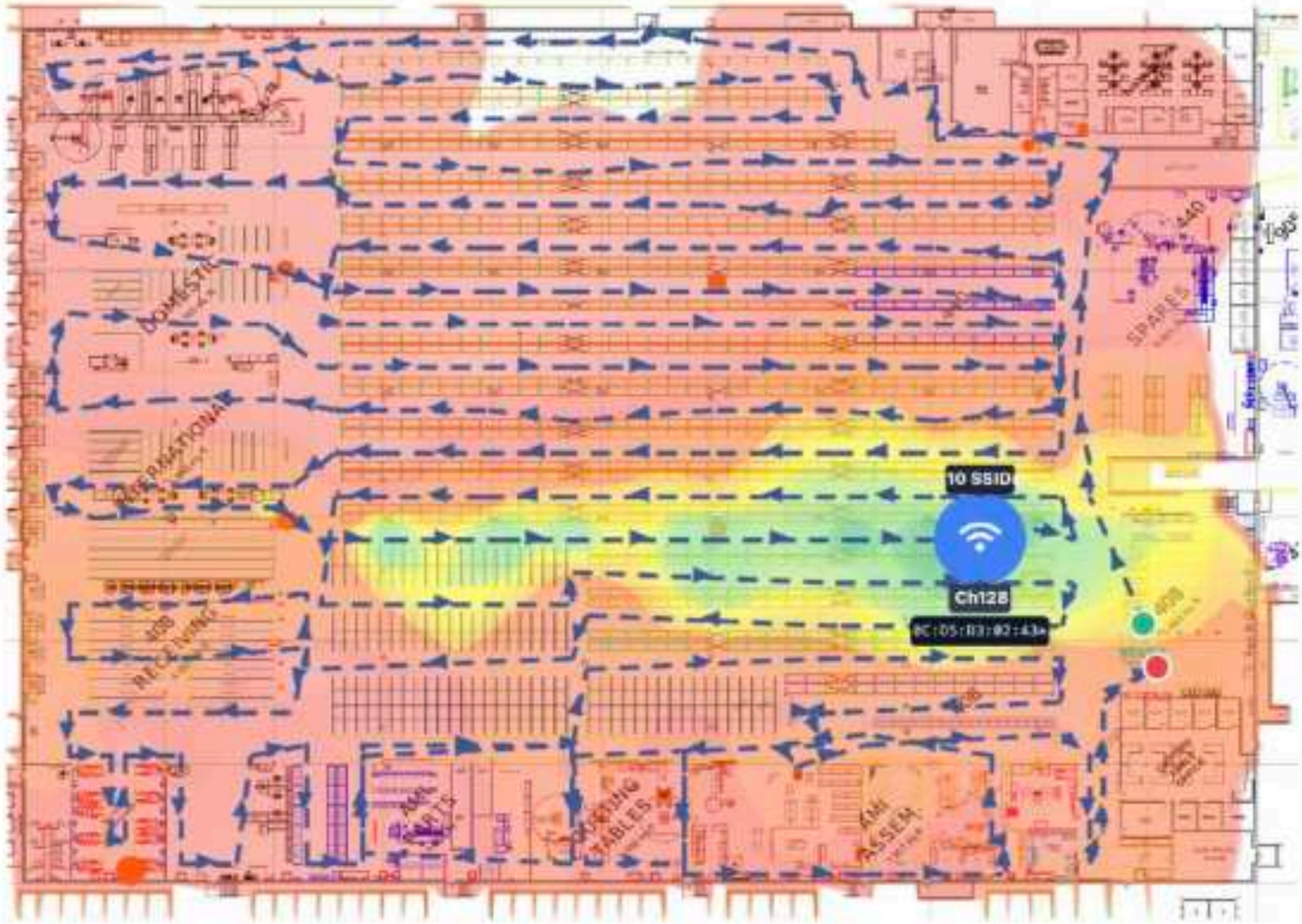
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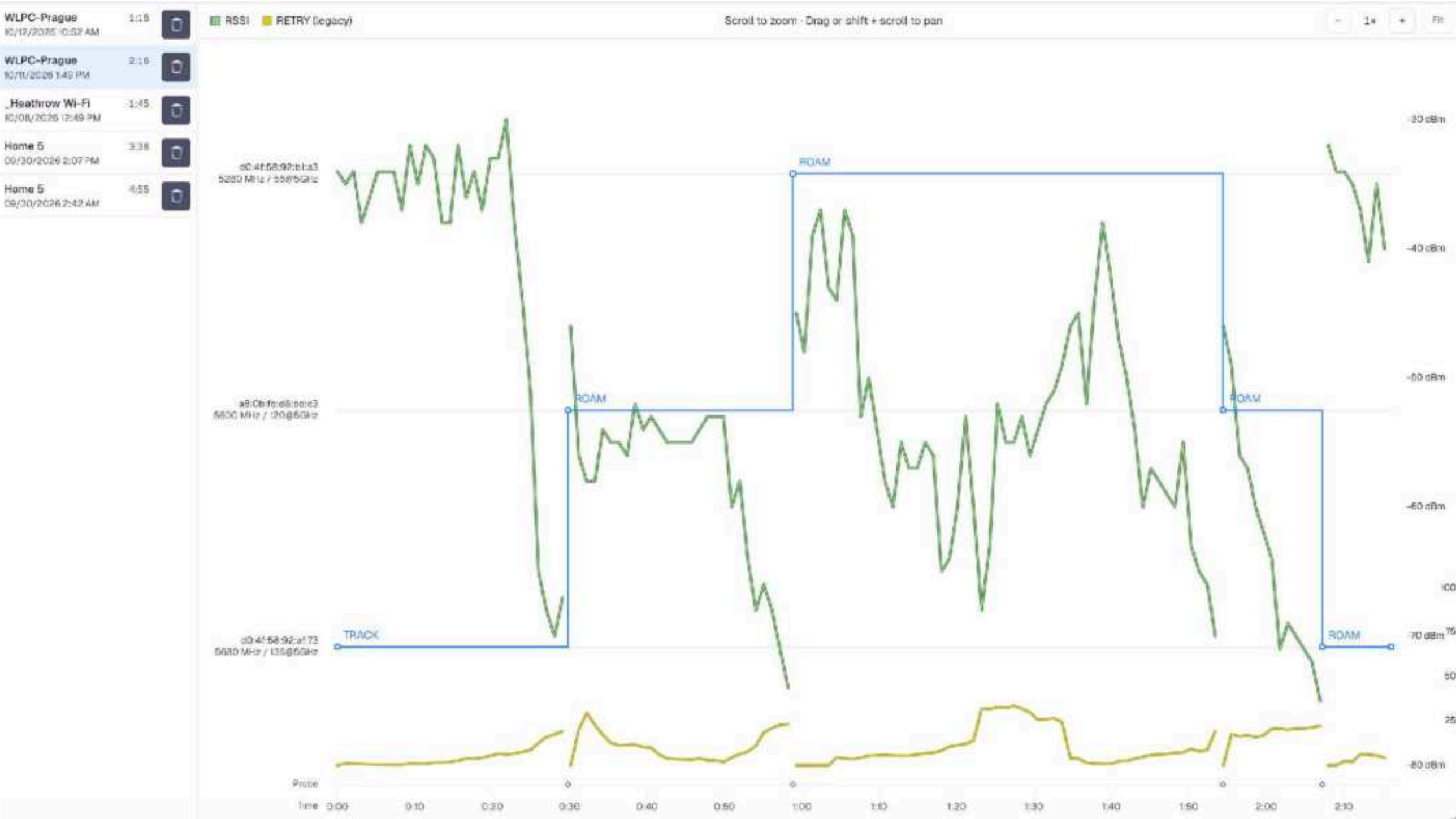


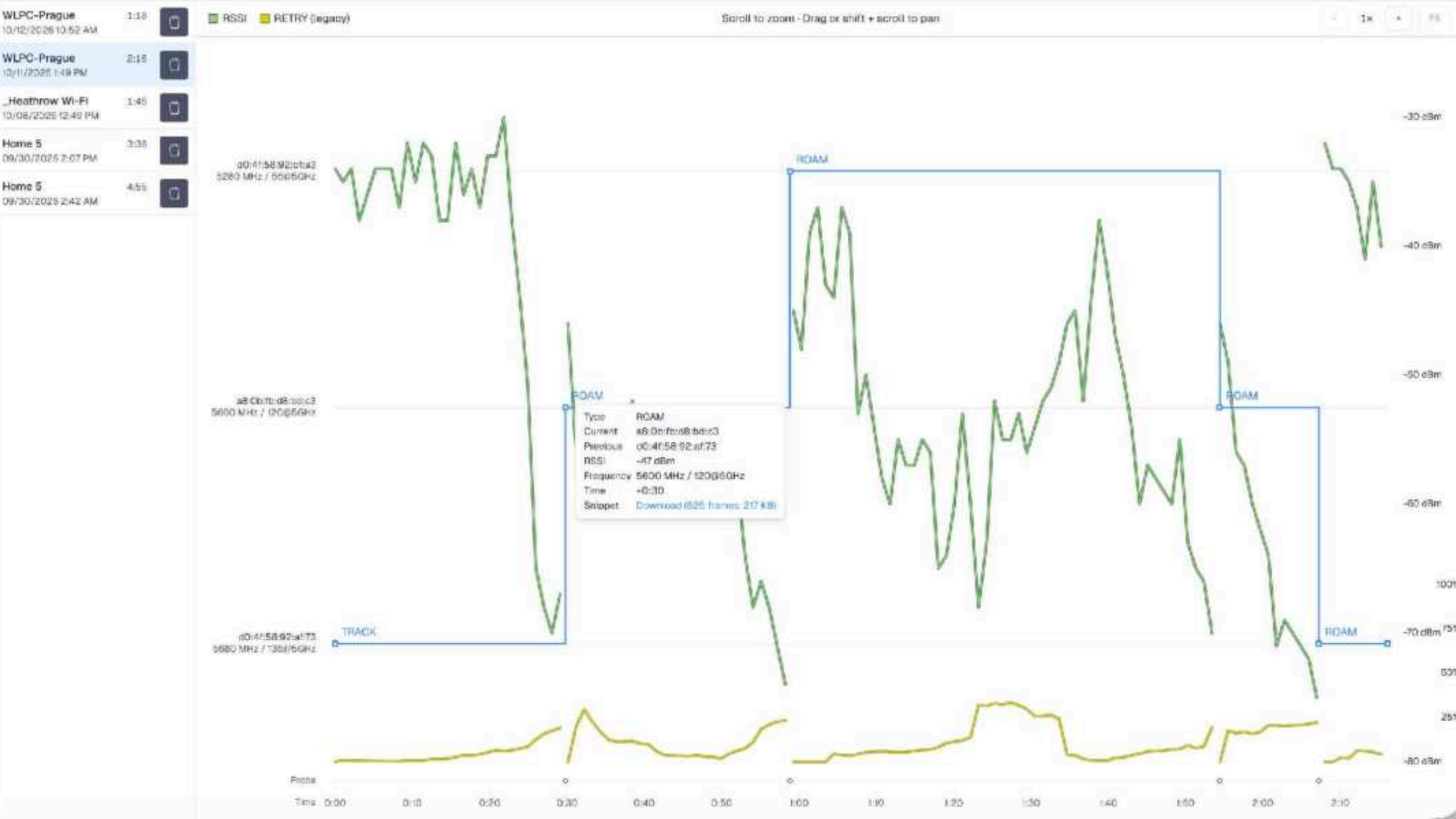







#4 00:04
CAPTURED 60 ms
#3 00:17
sidos
Netflix Streaming
No Rate Limit down to 2.5 Mbps
None
25 Mbps
10 Mbps
2.5 Mbps


WLPC-Prague
1:16
WLPC-Prague
2:16
_Heathrow Wi-Fi
1:45
Home 5
3:38
Home 5
4:55
RSSI
RETRY (legacy)
Scroll to zoom · Drag or shift + scroll to pan
1×
Fit
ROAM
ROAM
ROAM
TRACK
ROAM
-40 dBm
-50 dBm
-60 dBm
-70 dBm
-80 dBm
Probe
Time
0:00
0:10
0:20
0:30
0:40
0:50
1:00
1:10
1:20
1:30
1:40
1:50
2:00
2:10


WLPC-Prague
10/12/2026 10:52 AM
1:18
WLPC-Prague
10/1/2026 1:49 PM
2:18
_Heathrow Wi-Fi
10/08/2026 12:49 PM
1:46
Home 5
09/30/2026 2:07 PM
3:38
Home 5
09/30/2026 2:42 AM
4:55
RSSI
RETRY (legacy)
Scroll to zoom · Drag or shift + scroll to pan
1×
-30 dBm
-40 dBm
-50 dBm
-60 dBm
-70 dBm
-80 dBm
ROAM
ROAM
ROAM
ROAM
TRACK
Type ROAM
Current a8:0b:fb:d8:bd:c3
Previous d0:4f:58:92:af:73
RSSI -47 dBm
Frequency 5600 MHz / 120@6GHz
Time +0:30
Snippet Download (625 frames, 217 kB)
Probe
Time
0:00
0:10
0:20
0:30
0:40
0:50
1:00
1:10
1:20
1:30
1:40
1:50
2:00
2:10

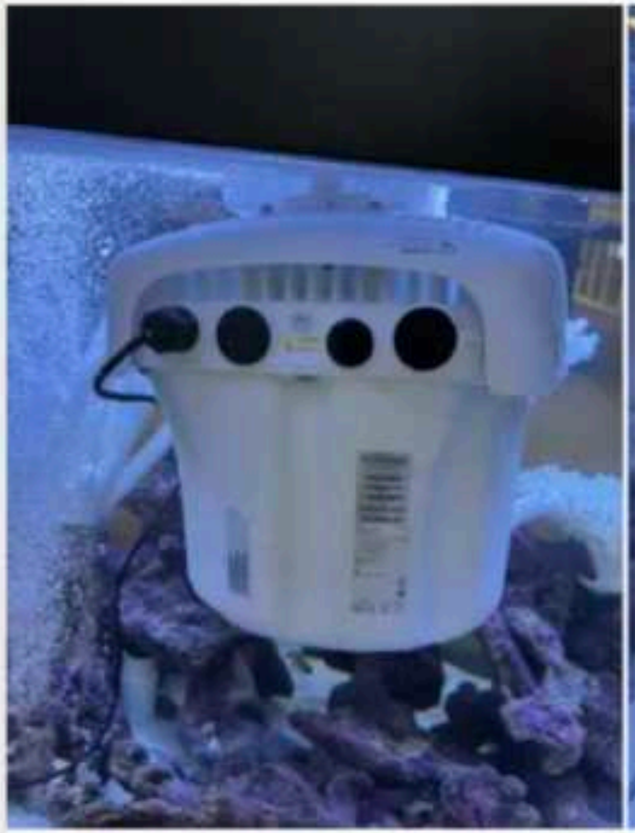


Part 107

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Scott Lester · Following
Enablement Technical Marketing Engineer @ HPE Aruba Networking / C...
7h ·

Developing Wi-Fi access points that work in the harshest conditions. That's what HPE Aruba Networking does. Check out our fully functional, completely submersed AP-675 on the Hewlett Packard Enterprise Discover show floor this week!





You and 100 others

6 comments · 9 reposts

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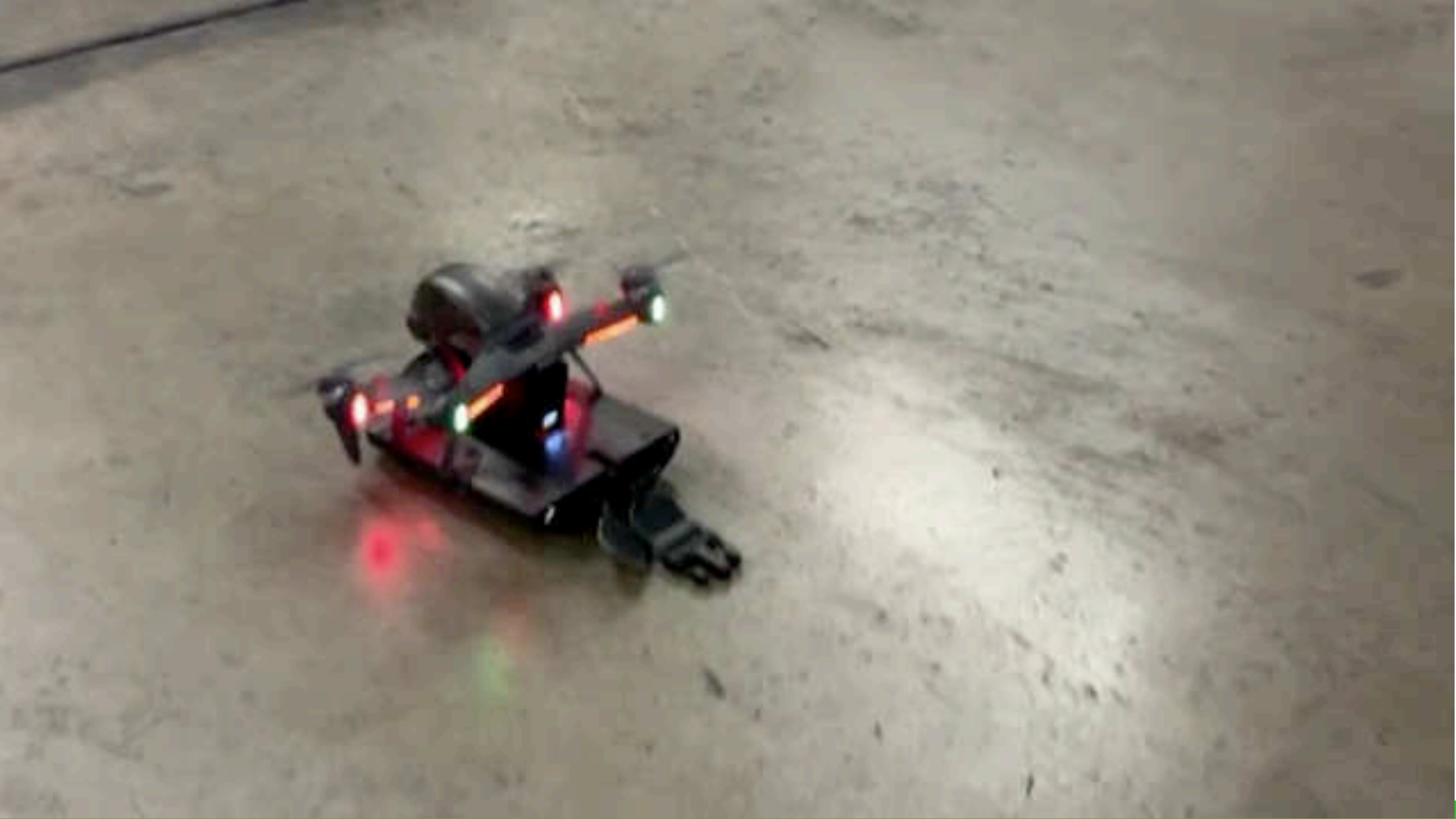




Sidekick


8:41PM Wed May 20
83%
SPECTRUM
2.4
5
6
Spectrum Live
5 GHz
Spectrum History
1 minute





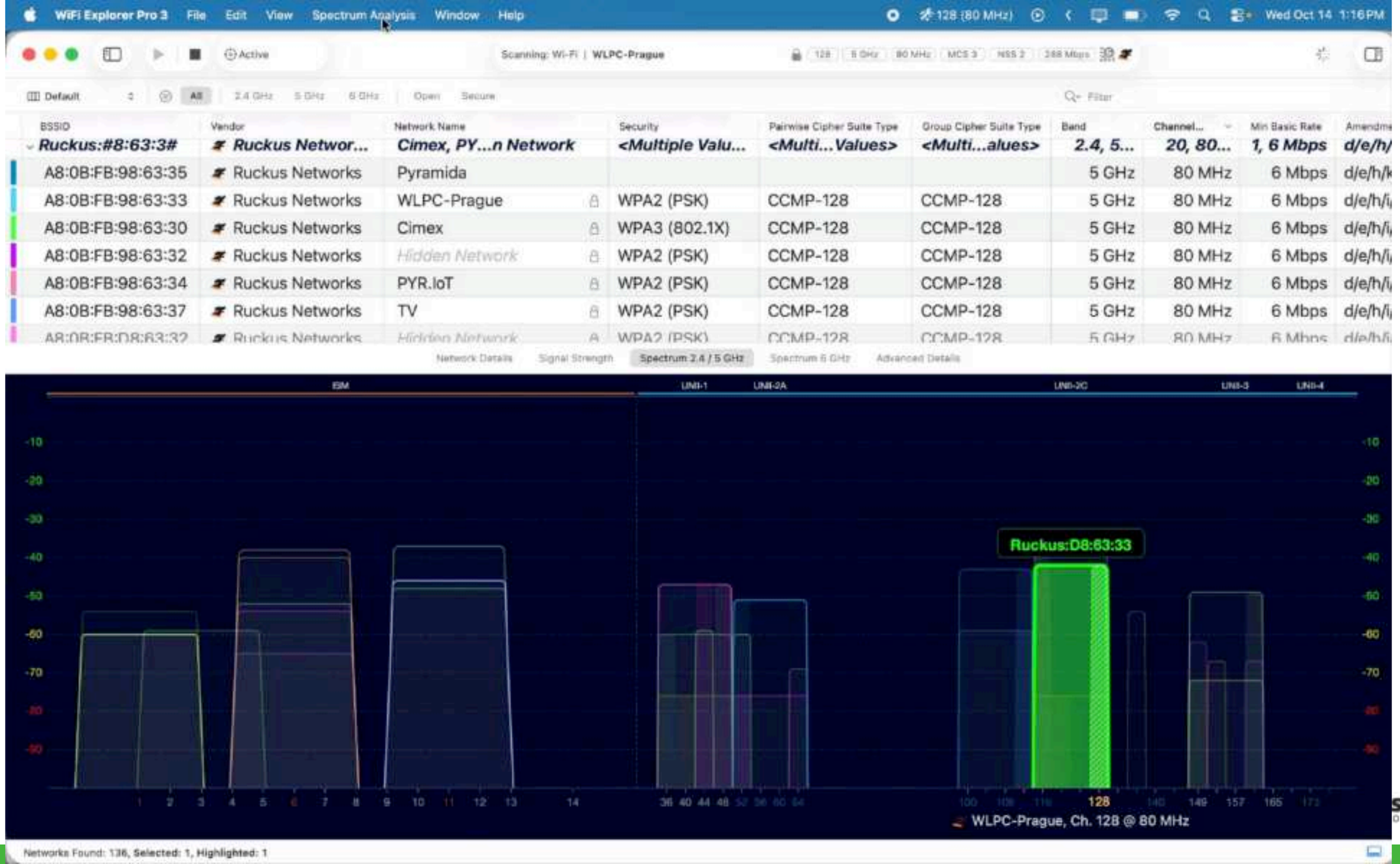
Sidekick

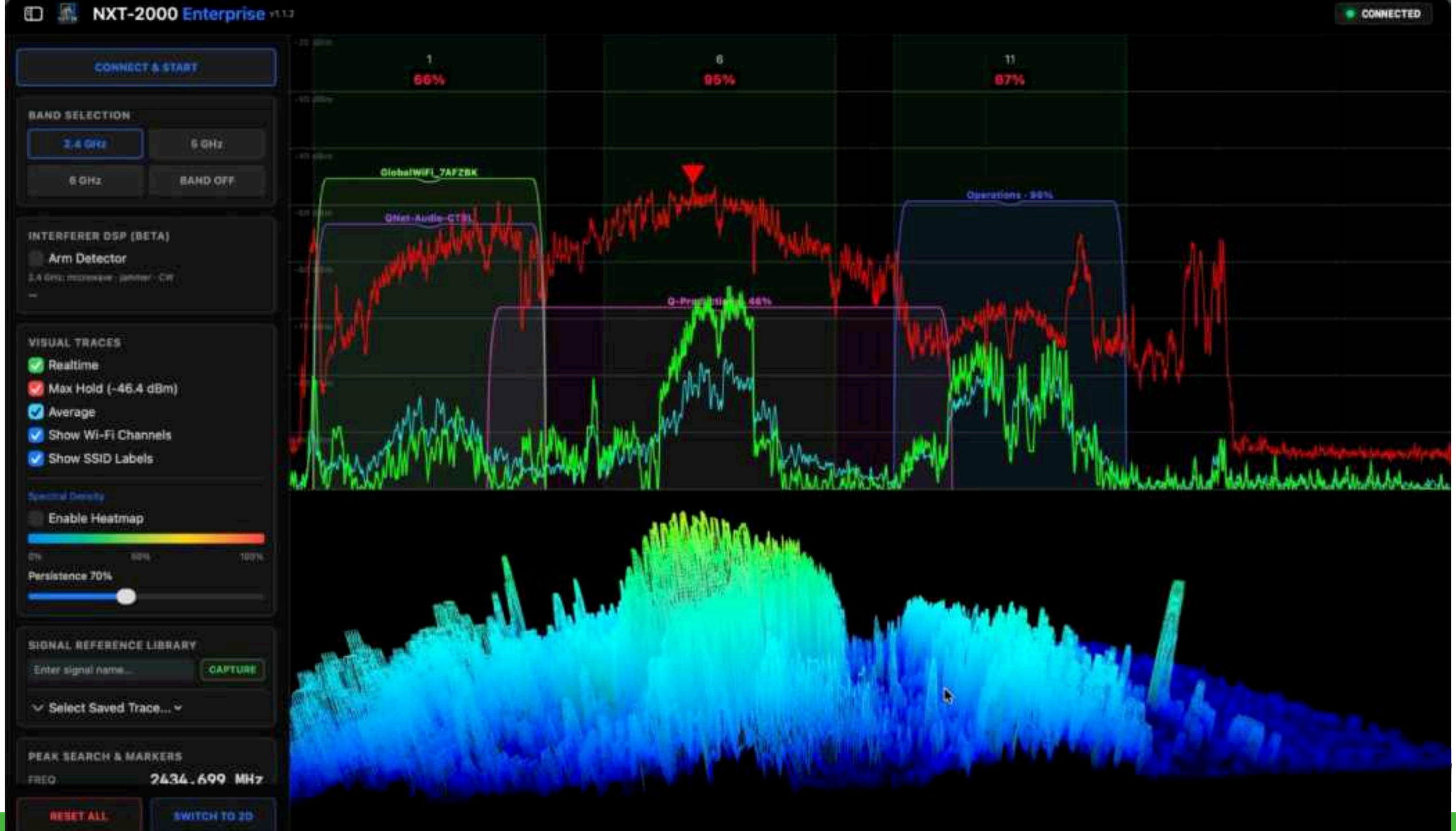
brick





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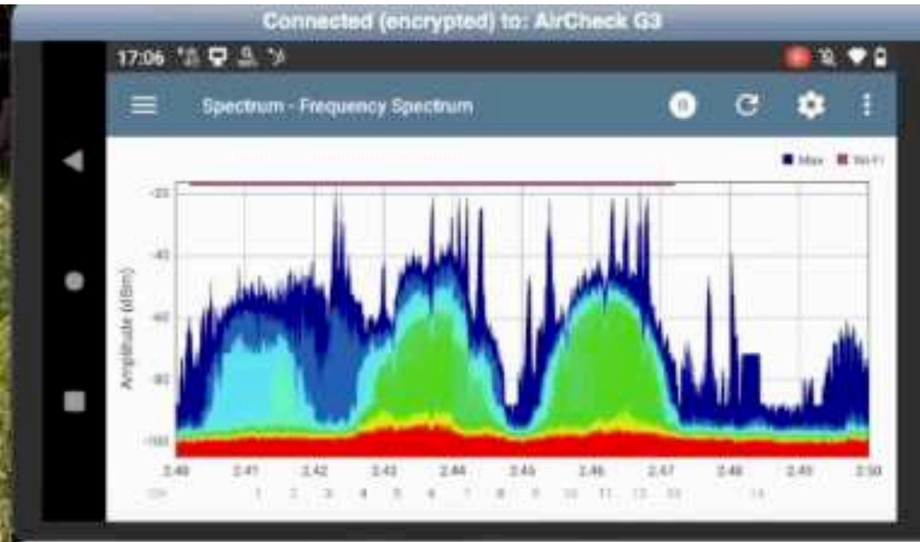

WiFi Explorer Pro 3 File Edit View Spectrum Analysis Window Help
Wed Oct 14 1:16 PM
Active
Scanning: Wi-Fi | WLPC-Prague
Default
All
2.4 GHz
5 GHz
6 GHz
Open
Secure
Filter
BSSID
Vendor
Network Name
Security
Pairwise Cipher Suite Type
Group Cipher Suite Type
Band
Channel...
Min Basic Rate
Ruckus:#8:63:3#
Ruckus Networ...
Cimex, PY...n Network
<Multiple Valu...
<Multi...Values>
<Multi...alues>
2.4, 5...
20, 80...
1, 6 Mbps
A8:0B:FB:98:63:35 Ruckus Networks Pyramida 5 GHz 80 MHz 6 Mbps
A8:0B:FB:98:63:33 Ruckus Networks WLPC-Prague WPA2 (PSK) CCMP-128 CCMP-128 5 GHz 80 MHz 6 Mbps
A8:0B:FB:98:63:30 Ruckus Networks Cimex WPA3 (802.1X) CCMP-128 CCMP-128 5 GHz 80 MHz 6 Mbps
A8:0B:FB:98:63:32 Ruckus Networks Hidden Network WPA2 (PSK) CCMP-128 CCMP-128 5 GHz 80 MHz 6 Mbps
A8:0B:FB:98:63:34 Ruckus Networks PYR.IoT WPA2 (PSK) CCMP-128 CCMP-128 5 GHz 80 MHz 6 Mbps
A8:0B:FB:98:63:37 Ruckus Networks TV WPA2 (PSK) CCMP-128 CCMP-128 5 GHz 80 MHz 6 Mbps
Network Details
Signal Strength
Spectrum 2.4 / 5 GHz
Spectrum 6 GHz
Advanced Details
ISM
UNII-1
UNII-2A
UNII-2C
UNII-3
UNII-4
-10
-20
-30
-40
-50
-60
-70
-80
-90
Ruckus:D8:63:33
1 2 3 4 5 6 7 8 9 10 11 12 13 14
36 40 44 48 52 56 60 64
100 108 116 128 140 149 157 165 173
WLPC-Prague, Ch. 128 @ 80 MHz
Networks Found: 136, Selected: 1, Highlighted: 1

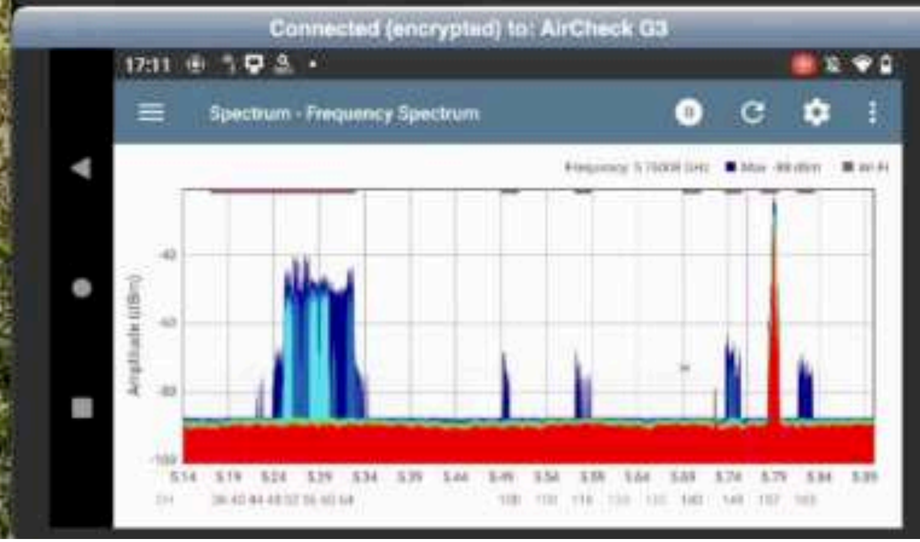

NXT-2000 Enterprise v1.1.2
CONNECTED
CONNECT & START
BAND SELECTION
2.4 GHz
5 GHz
6 GHz
BAND OFF
INTERFERER DSP (BETA)
Arm Detector
VISUAL TRACES
Realtime
Max Hold (-46.4 dBm)
Average
Show Wi-Fi Channels
Show SSID Labels
Enable Heatmap
Persistence 70%
SIGNAL REFERENCE LIBRARY
Enter signal name...
CAPTURE
Select Saved Trace...
PEAK SEARCH & MARKERS
FREQ
2434.699 MHz
RESET ALL
SWITCH TO 2D
1
66%
6
95%
11
87%
GlobalWiFi_7AFZBK
Operations - 98%




link-live.com/units/6799d307b47954224ec1d24d
Connected (encrypted) to: AirCheck G3
17:09
Spectrum - Frequency Spectrum
SunellPro







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Surveying Wi-Fi HaLow

By: Ferney Muñoz Vergara

CWNE #187, CWNT, Ekahau Master #56

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100% Free WLANPros Wi-Fi Design

Hamina

sidos

Wavelify

ekahau

https://thewifiofthings.com/